

«APEC PETROTECHNIC HIGHER COLLEGE» LLP

«APPROVED»

Deputy director
for academic work

_____ A.E. Shakenova

« ____ » _____ 2025

Educational-methodological complex of the module

Module: PM 01 «Maintenance and configuration of computer hardware and software»

Specialty: 06130100 – «Software Engineering (by types)»

Qualification: 4S06130105 – «Information Systems Technician»

Atyrau, 2025

«APEC PETROTECHNIC HIGHER COLLEGE» LLP

APPROVAL SHEET

Reviewed and approved at a meeting of the Subject-Cycle Methodological Commission «Software».

Protocol № ____ of « ____ » ____ 2026.

Chair of the SCMC: _____ A.E. Beket

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INTRODUCTION

The educational-methodological complex (EMC) of the module PM 01 «Maintenance and configuration of computer hardware and software» is intended for students of the specialty 06130100 – «Software Engineering (by types)», qualification 4S06130105 – «Information Systems Technician». It is developed in accordance with the working curriculum of the module and the methodical recommendations of the Technical and Professional Education Institute (TIPO) of 2025.

The module is aimed at studying the methods and means of maintaining, configuring and supporting the hardware and software of computer systems. Students master the principles of installation, configuration, diagnostics, technical maintenance and ensuring the stable operation of computer equipment and software tools.

The main goal of the module is to form the professional competencies that a future information systems technician needs to perform work on maintaining, configuring, diagnosing and supporting the hardware and software of computer systems.

The objectives of the module are: to study the basic principles of operation of the hardware and software of computer systems; to master the methods of installing, configuring and updating operating systems and application software; to learn to perform diagnostics, preventive maintenance and troubleshooting of computer equipment; to master the methods of configuring peripheral devices, drivers and system parameters; and to form the skills of ensuring the reliable, safe and stable operation of computer systems.

Upon completion of the module the student must know the occupational-safety requirements when operating computer systems; the architecture and operating principles of hardware and software of the computer infrastructure; the purpose and operating principles of peripheral devices and operating systems; and the technologies of deployment, cloning and support of computer systems.

The student must be able to perform diagnostics and minor repair of the hardware components of a personal computer; install, configure and update operating systems and specialized software; connect, configure and maintain peripheral devices; and create and apply master images of operating systems for deploying workplaces.

The total volume of the module is 336 academic hours. The module is built around seven learning outcomes (LO). Each learning outcome of this EMC contains an academic calendar, a glossary, lecture materials, practical (laboratory or industrial-training) materials, a list of software and multimedia support, and a list of literature. The closing sections contain self-study topics and exam questions for the whole module.

Prerequisites of the module: BM 02 «Design and analysis of electrical circuits and electronic devices in accordance with standards» and BM 03 «Application of the fundamentals of engineering and computer graphics». Post-requisites: PM 02 «Installation and configuration of personal-computer operating systems» and PM 03 «Installation, processing and configuration of application software packages».

I. ACADEMIC CALENDAR OF MODULE

PM 01	Maintenance and configuration of computer hardware and software	Weight, %
LO 1.1	Comply with the requirements and norms of occupational safety, industrial and ecological safety at information-service enterprises	6%
LO 1.2	Determine the operability of computer systems at the circuit-engineering level	11%
LO 1.3	Operate and control the work of computers and peripheral devices	7%
LO 1.4	Perform diagnostics and eliminate malfunctions and failures of the hardware of computing equipment	7%
LO 1.5	Restore equipment after hardware and software failures and prevent them	15%
LO 1.6	Replace the consumables used in computing and office equipment	7%
LO 1.7	Manage the internal and external devices of a personal computer	7%
	Intermediate certification (exam)	40%

II. LO 1.1 Comply with the requirements and norms of occupational safety, industrial and ecological safety at information-service enterprises

1. ACADEMIC CALENDAR

LO 1.1 Comply with the requirements and norms of occupational safety, industrial and ecological safety at information-service enterprises

Instructor: D.M. Yermekov

Specialty: 06130100 – «Software Engineering (by types)»

Total hours: 30 hours (15 lessons of 2 academic hours each)

№	Name of topics	Date of lesson	Type of activity
1	Fundamentals of electrical safety		Study of new material
2	The role and importance of occupational safety in the field of information services		Combined lesson
3	Features of organizing safe workplaces at information enterprises		Study of new material
4	Practical aspects of fire safety at information-sector enterprises		Study of new material
5	Prevention and treatment of occupational diseases among IT specialists		Skills-improvement lesson
6	Risk assessment and development of measures for their reduction at information-service enterprises		Study of new material
7	Analysis of occupational risks at the workplace of an IT specialist		Application and consolidation of knowledge
8	The role of training and briefings on safety techniques for staff at information enterprises		Skills-improvement lesson
9	The importance of the correct approach to connecting and using electronics and electrical equipment		Study of new material
10	Prevention of occupational diseases of IT specialists		Application and consolidation of knowledge
11	Compliance with norms and rules for ensuring safety when working with information and data on computers and servers		Skills-improvement lesson
12	Protective measures during the operation of electronic computing equipment. Individual electrical-protective equipment and tools, their classification and design		Study of new material
13	Means of electrical and fire safety in the IT field		Application and consolidation of knowledge
14	Fundamentals of fire safety		Combined lesson

15	Protective measures during the operation of electrical installations and electrical equipment		Comprehensive assessment of knowledge
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2. GLOSSARY

The glossary contains the main definitions, terms and concepts that reflect the core content of this learning outcome. The terms are arranged in alphabetical order.

Term	Definition
Earthing (grounding)	Intentional electrical connection of equipment housings to the ground, used to protect people from electric shock if the insulation fails.
Electrical safety	A system of organizational and technical measures and means that protect people from the harmful and dangerous effects of electric current, electric arc and static electricity.
ESD (electrostatic discharge)	A sudden flow of electricity between two charged objects; it is one of the main causes of latent damage to electronic components.
Fire safety	The state of an object at which the possibility of a fire is prevented and, if it occurs, the impact of dangerous fire factors on people and property is excluded or limited.
Occupational safety (labour protection)	A system of preserving the life and health of employees during their work, including legal, socio-economic, organizational, technical, sanitary and other measures.
Occupational disease	A chronic or acute disease of an employee that results from exposure to harmful production factors and entails a temporary or permanent loss of professional working capacity.
PPE (personal protective equipment)	Means worn or used by an employee to prevent or reduce the effect of harmful and dangerous production factors (anti-static straps, dielectric gloves, goggles, etc.).
Residual current device (RCD)	A protective device that quickly disconnects the supply when it detects a leakage (residual) current, protecting a person from electric shock.
Risk assessment	The process of identifying hazards, analysing the probability and severity of harm, and ranking risks to plan measures for their reduction.
Safety briefing	A form of training in which an employee is informed about safe methods of work; types include introductory, primary, repeated, unscheduled and targeted briefings.
Static electricity	Electric charges that accumulate on the surface of dielectrics and isolated conductors and may damage microelectronics through discharge.

3. LECTURE MATERIALS

LO 1.1 Comply with the requirements and norms of occupational safety, industrial and ecological safety at information-service enterprises

Lecture 1.1.1 Fundamentals of electrical safety

Content

The concept of electrical safety and the effect of electric current on the human body.

Factors that determine the severity of electric shock.

Classification of premises by the degree of electric-shock hazard.

Basic protective measures: earthing, insulation, RCD and personal protective equipment.

Electrical safety is a system of organizational and technical measures and means that protect people from the harmful and dangerous effects of electric current, electric arc, electromagnetic field and static electricity. For an information-systems technician, electrical safety is a basic professional competency, because all work with computer and server equipment is connected with the operation of devices powered by mains voltage.

Electric current passing through the human body produces thermal, electrolytic, mechanical and biological effects. The thermal effect causes burns of the skin and internal tissues; the electrolytic effect decomposes blood and other body fluids; the mechanical effect leads to ruptures of tissues; and the biological effect disturbs the work of the nervous system, the heart and the respiratory organs.

The severity of an electric shock depends on a number of factors: the magnitude of the current, the type and frequency of the current, the duration of its action, the path of the current through the body, and the individual physiological state of the person. The current value is considered the main factor. A current of about 1 mA is the threshold of perception; a current of 10–15 mA is the threshold of non-let-go, at which a person can no longer release the current-carrying part on their own; a current of 100 mA and above is, as a rule, fatal.

The voltage and the resistance of the human body determine the value of the current according to Ohm's law. The resistance of the body is not constant: it depends on the condition of the skin (dry or wet), the area and tightness of contact, and the duration of the current. Damp skin, sweat and metal jewellery sharply reduce resistance and increase the danger.

Alternating current of industrial frequency (50 Hz) is more dangerous for a person than direct current of the same value, because it causes more severe muscle contractions and is more likely to cause fibrillation of the heart. With increasing frequency the danger of electric shock decreases, but the danger of burns remains.

The path of the current through the body is of great importance. The most dangerous paths are those in which the current passes through the heart and respiratory organs — for example, 'hand–hand' and 'hand–legs'. Such loops cover the vital organs and most often lead to a fatal outcome.

According to the degree of danger of electric shock, all premises are divided into three classes. Premises without increased danger have no factors that increase the risk. Premises with increased danger are characterized by one of the following factors: dampness, conductive dust, conductive floors, high temperature, or the possibility of simultaneous contact with the earthed metal parts of equipment and the metal housings of electrical devices. Especially dangerous premises are characterized by two or more such factors at the same time, or by a chemically active environment.

Most computer classrooms and server rooms belong to the category of premises without increased danger or with increased danger, depending on the conditions. The technician must be able to determine the class of a room, because the permissible voltages of portable lamps and tools and the requirements for protective measures depend on it.

The main collective protective measures against electric shock include protective earthing, protective neutral bonding (zeroing), protective disconnection (RCD), the use of low (safe) voltage, electrical separation of networks, insulation of current-carrying parts, fencing and signalling, and safety signs.

Protective earthing is an intentional electrical connection of the metal non-current-carrying parts of equipment, which may become energized, to the earth or its equivalent. In the event of a breakdown of

the insulation, the earthing reduces the touch voltage to a safe value and ensures that the protection operates.

A residual current device (RCD) continuously compares the current flowing into a load with the current returning from it. If a leakage (residual) current appears — for example, because a person touched a faulty housing — the RCD detects the difference and disconnects the supply within tens of milliseconds, preventing dangerous consequences.

Personal protective equipment is used when collective measures cannot fully eliminate the danger. Basic electrical-protective equipment (dielectric gloves, insulating tools, voltage indicators) can withstand the operating voltage of the installation and allows work on live parts. Additional equipment (dielectric galoshes, mats, insulating stands) by itself does not provide protection but reinforces the action of the basic equipment.

When servicing computer equipment, the technician must always de-energize the device before opening the case, check the absence of voltage, discharge capacitors of power supplies and monitors, use a serviceable, earthed tool, and never work on a live installation alone. Compliance with these rules is the foundation of safe professional activity.

Thus, electrical safety combines knowledge of the physical action of current on the body, the ability to assess the danger of a particular room and operation, and the practical mastery of protective measures and equipment. These skills are mandatory for everyone who maintains and repairs computer and office equipment.

Control Questions

1. What types of effects does electric current have on the human body?
2. Which factors determine the severity of an electric shock, and why is the magnitude of the current considered the main one?
3. Into which classes are premises divided according to the degree of electric-shock danger, and by what features?
4. How do protective earthing and a residual current device (RCD) protect a person?
5. What is the difference between basic and additional personal electrical-protective equipment?

Lecture 1.1.2 The role and importance of occupational safety in the field of information services

Content

The concept, goals and tasks of occupational safety (labour protection).

The legal and regulatory basis of occupational safety.

Specific features of occupational safety at information-service enterprises.

Responsibilities of the employer and the employee in the field of labour protection.

Occupational safety, or labour protection, is a system of preserving the life and health of employees in the course of their work activity. It includes legal, socio-economic, organizational and technical, sanitary and hygienic, treatment-and-prevention, rehabilitation and other measures. The purpose of labour protection is to minimize the probability of injury or illness of the employee while maintaining their working capacity.

In the field of information services, occupational safety acquires its own specific features. The main production factors here are not heavy machinery, but prolonged work at a computer, electrical equipment, electromagnetic fields, an unfavourable microclimate, insufficient lighting, visual and neuro-emotional loads, and a forced working posture. Therefore labour protection in IT focuses on ergonomics, electrical safety, fire safety and the prevention of occupational diseases.

The legal basis of labour protection is formed by the Labour Code, state standards (GOST/ST RK) of the occupational-safety standards system, sanitary norms and rules (SanPiN), fire-safety rules, and

the rules for the technical operation of electrical installations of consumers. These documents establish the requirements that both the employer and the employee are obliged to fulfil.

The goals of occupational safety are achieved through a set of tasks: identifying and assessing harmful and dangerous production factors; bringing workplaces into conformity with the requirements of standards; training and briefing personnel; providing employees with personal protective equipment; organizing medical examinations; and investigating and recording accidents and occupational diseases.

A harmful production factor is a factor whose impact on an employee can lead to an illness. A dangerous production factor is a factor whose impact can lead to an injury. At an information enterprise, the increased level of noise from cooling systems, the electromagnetic field of equipment, and the eye strain from monitors are harmful factors, while electric current and the possibility of a fire are dangerous factors.

The employer bears the main responsibility for the state of labour protection. The employer is obliged to ensure safe conditions and labour protection at every workplace, to organize briefings and the verification of knowledge of labour-protection requirements, to provide personal protective equipment, to carry out a special assessment of working conditions, to organize medical examinations and to investigate accidents.

The employee, in turn, is obliged to comply with labour-protection requirements, to correctly apply the personal and collective protective equipment provided, to undergo training, briefing and verification of knowledge, to immediately notify the manager of any situation that threatens the life and health of people, and to undergo compulsory medical examinations.

An important element of the labour-protection system is the management of occupational risks. Risk management is a set of interrelated measures, including the identification of hazards, the assessment of occupational risks and the planning of measures to reduce them. A modern enterprise builds its labour-protection system precisely on the basis of risk assessment.

Briefings on labour protection occupy a special place in the field of information services. The introductory briefing is conducted with all newly hired employees; the primary briefing is conducted directly at the workplace; repeated, unscheduled and targeted briefings maintain and update the employee's knowledge. Without passing a briefing, an employee is not allowed to work.

The economic importance of labour protection is also significant. Accidents, illnesses and equipment downtime caused by violations of safety requirements lead to direct and indirect losses for the enterprise. Investments in labour protection — modern equipment, ergonomic furniture, training — reduce these losses and increase the productivity and motivation of the staff.

The social importance of labour protection lies in preserving the health and working capacity of people, in reducing injuries and occupational diseases, and in creating a comfortable and safe working environment. For the IT sphere, where qualified specialists are a key resource, this is especially important.

Thus, occupational safety in the field of information services is a comprehensive system that combines legal requirements, technical measures and the personal responsibility of every employee. Mastering its principles is a necessary condition for the professional activity of an information-systems technician.

Control Questions

1. What is occupational safety (labour protection) and what does this system include?
2. What specific features does labour protection have at information-service enterprises?
3. What is the difference between a harmful and a dangerous production factor? Give examples for the IT sphere.
4. What are the main responsibilities of the employer and of the employee in the field of labour protection?
5. What types of safety briefings exist and what is their purpose?

Lecture 1.1.3 Features of organizing safe workplaces at information enterprises

Content

The concept of a workplace and the requirements for its organization.

Ergonomic requirements for a computer workstation.

Microclimate, lighting and noise at the workplace of an IT specialist.

Placement of equipment and electrical wiring at the workplace.

A workplace is the part of the production area equipped with the necessary means in which an employee performs their job duties. The correct organization of the workplace of an IT specialist directly affects their health, working capacity and labour productivity, and also reduces the probability of injuries and occupational diseases.

Ergonomics is the science of adapting the working environment to the capabilities and characteristics of the human body. Ergonomic requirements for a computer workstation regulate the dimensions of the desk and chair, the height of the monitor, the distance to the screen, the position of the keyboard and the mouse, and the placement of documents and auxiliary devices.

The work chair must be lifting-and-swivelling and adjustable in height and in the angles of inclination of the seat and the backrest. This allows the posture to be changed during work and reduces the static load on the muscles of the back and neck. The feet should rest on the floor or on a footrest, and the angle at the knees should be close to a right angle.

The monitor should be placed so that the upper edge of the screen is at or slightly below eye level, and the distance from the eyes to the screen is 600–700 mm. The screen should be positioned to exclude glare from windows and lamps. Correct placement of the monitor reduces visual strain and the load on the cervical spine.

The microclimate of the premises is characterized by air temperature, relative humidity and air velocity. For premises with computer equipment, an optimal temperature of about 22–24 °C in the cold season and 23–25 °C in the warm season, a relative humidity of 40–60 %, and a low air velocity are recommended. Deviations from these values reduce the comfort and working capacity of the staff.

Lighting at the workplace must be sufficient and uniform. Both natural and artificial lighting are used. The illumination of the working surface for working with documents should be about 300–500 lux. It is important to avoid sharp contrasts between the brightness of the screen and the surrounding space and to exclude direct and reflected glare.

The level of noise at the workplace of an IT specialist is created by the cooling fans of computers and servers, printers, air-conditioning systems and other people. Increased noise tires the nervous system and reduces concentration. To reduce noise, low-noise equipment is used, server equipment is placed in separate rooms, and sound-absorbing materials are applied.

The placement of equipment at the workplace must ensure free access to all devices for maintenance and a convenient working posture. The distance between the side surfaces of monitors should be at least 1.2 m, and the distance from the rear surface of one monitor to the screen of another should be at least 2 m. This reduces the mutual influence of the electromagnetic fields of devices.

Particular attention is paid to electrical wiring and the connection of equipment. All cables must be laid in such a way as to exclude their mechanical damage and tripping of people. The use of damaged sockets, plugs and extension cords is not allowed. The number of devices connected to one socket group must not exceed its rated load.

It is recommended to connect computer equipment through sources of uninterruptible power supply and surge protectors. This protects the equipment from voltage surges and short-term power outages, and also reduces the risk of fire and loss of data. Earthing of the equipment housings is mandatory.

The organization of a safe workplace also includes the rational arrangement of furniture and passages, compliance with fire-safety requirements, the availability of primary fire-extinguishing means, and free access to the means of disconnecting electricity. All these elements together form a safe and productive working environment.

Thus, the safe organization of the workplace of an IT specialist is based on ergonomic principles, on compliance with the requirements for the microclimate, lighting and noise, and on the correct and safe placement of equipment and electrical wiring.

Control Questions

1. What is a workplace, and why is its correct organization important for an IT specialist?
2. What ergonomic requirements apply to a computer workstation?
3. What microclimate and lighting parameters are recommended for premises with computer equipment?
4. What requirements apply to the placement of equipment and the distances between workstations?
5. What requirements must be observed when connecting equipment and laying electrical wiring?

Lecture 1.1.4 Practical aspects of fire safety at information-sector enterprises

Content

Causes of fires at information enterprises.

Classes of fire and the choice of extinguishing agent.

Primary fire-extinguishing means for electrical equipment.

Actions of personnel in the event of a fire.

Fire safety is the state of an object at which the possibility of a fire is prevented, and in the event of its occurrence the impact of dangerous fire factors on people and property is excluded or limited. At information-sector enterprises, where a large amount of electrical equipment and combustible materials is concentrated, fire safety is of particular importance.

The main causes of fires at information enterprises are connected with the electrical part. These are short circuits in wiring and equipment, overloading of electrical networks, poor contact in connections leading to overheating, the use of faulty devices, and the violation of the rules for operating heating and electrical appliances. A significant proportion of fires is also caused by the human factor — careless handling of fire and smoking in unauthorized places.

Combustible materials at an information enterprise include the polymer cases of equipment, cable insulation, furniture, paper documents and finishing materials. When they burn, not only is heat released, but also toxic combustion products that pose a danger to people even before the fire reaches them.

Dangerous fire factors that affect people are open flame and sparks, an increased temperature of the environment, toxic combustion products, smoke, a reduced concentration of oxygen, and the destruction of building structures. The greatest number of victims in fires is caused precisely by the toxic combustion products and smoke.

Fires are classified into classes depending on the type of burning material. Class A is the burning of solid substances (paper, wood, plastics); class B is the burning of flammable liquids; class C is the burning of gases; class D is the burning of metals; and class E (in a number of classifications) is the burning of electrical installations under voltage. The choice of extinguishing agent depends on the class of the fire.

The extinguishing of electrical installations under voltage has important features. Water and foam extinguishers conduct electric current and cannot be used on live equipment, as this creates a danger of

electric shock. For extinguishing energized electrical equipment, carbon-dioxide and powder fire extinguishers are used, which do not conduct current.

A carbon-dioxide (CO₂) fire extinguisher is the preferred means for server rooms and computer classes, because carbon dioxide does not leave traces and does not damage the electronics. When using it, one must avoid contact of the skin with the bell of the extinguisher, the temperature of which drops sharply, and ensure ventilation of the room after extinguishing, since carbon dioxide displaces oxygen.

Powder fire extinguishers are universal and suitable for most classes of fire, including energized equipment up to a certain voltage. However, the powder contaminates the equipment and the room and may render the electronics inoperable, so for valuable equipment carbon-dioxide extinguishers are preferred.

Automatic fire-extinguishing and fire-alarm systems play a key role in protecting server rooms and data centres. Gas fire-extinguishing systems based on inert gases or chemical agents extinguish a fire without damaging the equipment and without harming people if the evacuation rules are observed. Fire detectors detect smoke or a temperature rise and trigger an alarm.

In the event of a fire, the personnel must act quickly and in a coordinated manner. The general sequence of actions is: immediately report the fire to the fire service by calling the emergency number; de-energize the equipment if it is safe to do so; begin evacuating people through the emergency exits; and, if possible and safe, begin extinguishing the fire with the available primary means.

Evacuation must be carried out according to the evacuation plan, which is posted in a visible place in each room. Employees must know the location of the emergency exits, fire extinguishers, fire hydrants and electrical-disconnection points. During evacuation it is forbidden to use lifts, and one should move along the evacuation routes, covering the respiratory organs from smoke.

The prevention of fires is based on compliance with the operating rules for electrical equipment, the timely inspection and repair of wiring, the prohibition of overloading the network and of using faulty devices, the maintenance of order at the workplace, and the regular training and briefing of personnel on fire safety.

Thus, fire safety at information-sector enterprises requires knowledge of the causes and classes of fires, the correct choice of extinguishing agents for electrical equipment, the availability and serviceability of fire-extinguishing means, and the clear and trained actions of personnel in the event of a fire.

Control Questions

1. What are the main causes of fires at information enterprises?
2. Into which classes are fires divided, and how does this affect the choice of extinguishing agent?
3. Why can water and foam extinguishers not be used on energized electrical equipment?
4. Which fire extinguishers are preferable for server rooms and computer classes, and why?
5. What is the sequence of actions of personnel in the event of a fire?

Lecture 1.1.6 Risk assessment and development of measures for their reduction at information-service enterprises

Content

The concept of occupational risk and the goals of risk management.

Identification of hazards at an information enterprise.

Methods of assessing occupational risks.

Development and implementation of risk-reduction measures.

Occupational risk is the probability of harm to the health of an employee as a result of exposure to harmful and (or) dangerous production factors in the performance of their job duties, taking into

account the severity of the possible consequences. Risk management is the basis of the modern system of labour protection and allows resources to be directed first of all to the most significant hazards.

The process of risk management includes several stages: the identification (revealing) of hazards, the assessment of the magnitude of the risk, the ranking of risks by significance, the development and implementation of measures to reduce them, and the monitoring of the effectiveness of these measures. This process is cyclical and is repeated regularly.

The identification of hazards is the first and key stage. At an information enterprise, the hazards include electric current, the possibility of a fire, an unfavourable microclimate, insufficient lighting, increased noise, an electromagnetic field, visual and neuro-emotional loads, a forced working posture, and the risk of injury when working with tools.

For the identification of hazards, the inspection of workplaces, the analysis of the technological process, the study of the statistics of accidents and illnesses, surveys of employees and the analysis of the requirements of regulatory documents are used. The result of identification is a list of hazards for each workplace.

The assessment of occupational risk can be carried out by qualitative, quantitative or combined (semi-quantitative) methods. The qualitative method describes the risk in words (low, medium, high). The quantitative method expresses the risk in numerical values. The combined method, the most widespread, uses a risk matrix.

The risk matrix is based on two parameters: the probability of the occurrence of a harmful event and the severity of its consequences. Each parameter is assessed on a scale (for example, from 1 to 5). The product or the combination of these estimates determines the level of risk, which is conveniently displayed in a coloured matrix from green (acceptable) to red (unacceptable).

After assessing and ranking the risks, measures for their reduction are developed. The hierarchy of risk-control measures provides for the priority of more reliable measures: first of all, the elimination of the hazard; then its replacement with a less dangerous one; then engineering (technical) controls; then administrative measures; and only at the last level — personal protective equipment.

For an information enterprise, engineering measures include the installation of RCDs and earthing, the replacement of faulty equipment, the improvement of ventilation and lighting, and the reduction of noise. Administrative measures include briefings, work-and-rest regimes, safe-operation instructions and access control. Personal protective equipment includes anti-static straps and dielectric tools.

An important principle of risk management is that the residual risk after the implementation of measures must be acceptable. If the residual risk remains high, additional measures are developed. All decisions and measures are documented, which allows their implementation and effectiveness to be monitored.

The monitoring of the effectiveness of measures is carried out through repeated inspections, the analysis of accident and illness statistics, and surveys of employees. If the measures have not produced the expected effect, the risk is reassessed and the program of measures is adjusted.

Risk management is closely connected with the special assessment of working conditions and with the system of labour-protection management of the enterprise. The results of risk assessment are used in planning the budget for labour protection, in training personnel and in the investigation of incidents.

Thus, the assessment of occupational risks and the development of measures for their reduction is a systematic, cyclical process that allows the enterprise to consciously and effectively manage the safety of its employees, directing efforts to the most significant hazards.

Control Questions

1. What is occupational risk and what is the purpose of risk management?
2. What stages does the process of occupational-risk management include?
3. What hazards are typical of an information-service enterprise?
4. How is a risk matrix constructed and how is the level of risk determined from it?

5. What is the hierarchy of risk-control measures, and which measures are considered the most reliable?

Lecture 1.1.9 The importance of the correct approach to connecting and using electronics and electrical equipment

Content

- Rules for connecting electrical equipment to the mains.
- Loading of electrical networks and protection against overloads.
- Earthing and protection against static electricity.
- Typical errors when connecting equipment and their consequences.

The correct connection and use of electronics and electrical equipment is a basic condition for the safe and reliable operation of computer systems. Errors at this stage lead to equipment failure, loss of data, electric shock to people and even to fires, therefore the technician must clearly know and follow the connection rules.

Before connecting any device, one must make sure that its rated supply voltage corresponds to the mains voltage, that the power cable and the plug are in good condition, and that the socket is serviceable and earthed. The use of devices with damaged cables, cracked plugs or traces of overheating is not allowed.

Each electrical network and each socket group is designed for a certain rated load. Connecting too many powerful devices to one socket through tees and extension cords leads to an overload, the heating of the wiring and the contacts, the melting of the insulation and the risk of fire. The total load must not exceed the permissible value.

To protect against overloads and short circuits, circuit breakers are used, which automatically disconnect the line when the current exceeds the permissible value. Fuses perform a similar function. The technician must not replace protective devices with 'jumpers' or devices with a higher rated current, since this disables the protection.

Sources of uninterruptible power supply (UPS) and surge protectors protect equipment from voltage surges, short-term outages and high-frequency interference. The UPS additionally provides power supply to the equipment from batteries for the time necessary for the correct shutdown of the system, which prevents loss of data.

Earthing of the equipment housings is a mandatory protective measure. It ensures the operation of the protective devices in the event of an insulation breakdown and removes the touch voltage from the housing. The connection of computer equipment to a socket without an earth contact deprives the protective measures of their effectiveness and is dangerous.

Static electricity poses a special danger to electronic components. Charges accumulating on a person's clothing and body can reach several kilovolts and, when touching a microchip, cause its damage — both obvious (immediate failure) and latent (a gradual degradation of parameters). Therefore, when working with components, anti-static measures are mandatory.

Anti-static measures include the use of an anti-static wrist strap connected to the earthed housing, anti-static mats, the discharge of static from oneself before starting work, the storage of components in anti-static packaging, and the maintenance of a sufficient humidity of the air in the room.

When connecting interface cables and peripheral devices, one must observe the polarity and orientation of the connectors, not apply excessive force, and, where required, connect and disconnect the devices with the power off. The 'hot' connection is permissible only for interfaces that explicitly support it (for example, USB).

Typical errors when connecting equipment include the connection to a socket without earthing, the overloading of the network, the use of faulty extension cords, the laying of cables in passages, the connection of equipment with a voltage that does not correspond to the rating, and the neglect of anti-static measures. Each of these errors can lead to serious consequences.

The correct organization of the power supply of computer equipment also includes the rational distribution of the load over the lines, the marking of cables and sockets, the maintenance of order in the cable economy, and the regular inspection of the condition of the wiring and connections.

Thus, the competent approach to connecting and using electrical equipment combines the verification of the conformity and serviceability of devices, the observance of the permissible load, the obligatory earthing, the protection against static electricity, and the avoidance of typical errors. This ensures the safety of people and the reliable operation of equipment.

Control Questions

1. What must be checked before connecting an electrical device to the mains?
2. What does the overloading of an electrical network lead to, and how is it prevented?
3. What functions do circuit breakers and sources of uninterruptible power supply perform?
4. Why is the earthing of equipment housings a mandatory protective measure?
5. What anti-static measures must be observed when working with electronic components?

Lecture 1.1.12 Protective measures during the operation of electronic computing equipment. Individual electrical-protective equipment and tools, their classification and design

Content

Protective measures during the operation of computing equipment.

Classification of electrical-protective equipment into basic and additional.

Design and purpose of the main types of electrical-protective equipment.

Rules for using and storing electrical-protective equipment.

The safe operation of electronic computing equipment is ensured by a combination of collective and individual protective measures. Collective measures (earthing, RCDs, insulation, low voltage) protect all employees, while individual measures protect a specific person performing work that involves the danger of electric shock.

Before any work on the maintenance of computing equipment, organizational and technical measures are carried out. Organizational measures include the issuance of a task, the admission to work and supervision. Technical measures include the disconnection of voltage, the prevention of erroneous switching-on, the verification of the absence of voltage, and the earthing and discharge of capacitors.

Individual electrical-protective equipment is divided into basic and additional. Basic equipment has insulation that reliably withstands the operating voltage of the electrical installation and allows work on live current-carrying parts to be performed. Additional equipment by itself does not provide protection at a given voltage but reinforces the action of the basic equipment.

For electrical installations up to 1000 V, the basic equipment includes dielectric gloves, insulating tools with handles up to a certain voltage, and voltage indicators. The additional equipment includes dielectric galoshes, dielectric mats, portable earths and insulating stands.

Dielectric gloves are made of special rubber and protect the hands from electric current. Before each use, they are checked for the absence of punctures and damage by rolling them towards the fingers and trapping air. Insulating tools (screwdrivers, pliers, wire-cutters) have handles covered with a dielectric that withstands the operating voltage.

A voltage indicator is a device for checking the presence or absence of voltage on current-carrying parts. Before and after the check, the serviceability of the indicator is verified on an obviously

energized source. The verification of the absence of voltage is a mandatory step before starting work on de-energized equipment.

Anti-static protective equipment occupies a special place when working with electronic components. An anti-static wrist strap is connected through a resistor to the earthed housing and continuously removes the static charge from the human body. An anti-static mat and anti-static packaging protect the components during repair and storage.

All electrical-protective equipment is subject to periodic testing with a high voltage to confirm its dielectric properties. On each item, a stamp is placed indicating the voltage for which it is intended and the date of the next test. The use of equipment with an expired test period or without a stamp is not allowed.

The rules for using electrical-protective equipment require checking its serviceability before each use, applying it strictly for its intended purpose and at the corresponding voltage, and observing the methods of working with it. Damaged or untested equipment is immediately removed from use.

The storage of electrical-protective equipment is organized in such a way as to protect it from mechanical damage, moisture, oils, the sun and high temperatures, which degrade the dielectric properties. Gloves and mats are stored separately from the tools, in dry, ventilated places.

For the maintenance of computing equipment, in addition to electrical-protective equipment, a special tool is used: insulated screwdrivers, a set of bits, pliers, an anti-static strap, a brush and a vacuum cleaner for cleaning, a multimeter for measurements, and a soldering station for repairs. The serviceability of the tool is a condition for safe work.

Thus, the protective measures during the operation of computing equipment combine organizational and technical measures, collective and individual protective equipment. Knowledge of the classification, design and rules for the use of electrical-protective equipment is a necessary part of the professional training of a technician.

Control Questions

1. What organizational and technical measures are carried out before working on computing equipment?
2. What is the difference between basic and additional electrical-protective equipment?
3. Which equipment is the basic and which the additional one for installations up to 1000 V?
4. How is the serviceability of dielectric gloves and a voltage indicator checked?
5. What rules must be observed when using and storing electrical-protective equipment?

Lecture 1.1.14 Fundamentals of fire safety

Content

The concept of combustion and the conditions for the occurrence of a fire.

The fire triangle and the methods of extinguishing a fire.

The fire-prevention regime at the enterprise.

Organization of evacuation and actions in the event of a fire.

Fire safety is based on an understanding of the nature of combustion. Combustion is a complex physico-chemical process of the oxidation of a substance, accompanied by the release of heat and light. For combustion to occur and be maintained, three conditions must be present simultaneously: a combustible substance, an oxidizer (usually the oxygen of the air) and an ignition source.

These three conditions are conventionally depicted as the 'fire triangle'. If at least one of the sides of the triangle is removed, combustion stops. This principle underlies all methods of extinguishing a fire: the removal or cooling of the combustible substance, the isolation of the source from the oxidizer, and the removal of the heat (the reduction of the temperature below the ignition point).

Accordingly, the main methods of extinguishing are cooling (with water), dilution and isolation from oxygen (with foam, carbon dioxide, powder or inert gases), and the chemical inhibition of the combustion reaction. The choice of method and agent depends on the class of the fire and the conditions in which it occurs.

A fire is an uncontrolled combustion that causes material damage, harm to the life and health of people, and the interests of society and the state. Unlike controlled combustion, a fire spreads spontaneously and is accompanied by dangerous factors: flame, high temperature, smoke, toxic products and a reduced oxygen concentration.

The fire-prevention regime at an enterprise is a set of rules of conduct of people and of the maintenance of the territory, buildings and premises that ensure the prevention of violations and the elimination of fires. It includes the procedure for maintaining evacuation routes, the rules for using electrical appliances, the designation of smoking places and the maintenance of fire-extinguishing means.

An important element of fire prevention is the maintenance of evacuation routes and exits in a free state. The cluttering of corridors, staircases and passages, the locking of emergency exits and the blocking of approaches to fire extinguishers are categorically forbidden, since this makes evacuation impossible in the event of a fire.

Fire alarm systems are designed for the early detection of a fire. Smoke, heat and flame detectors register the signs of a fire and transmit a signal to the control panel, which turns on sound and light annunciators and, if necessary, the automatic fire-extinguishing and smoke-removal systems.

Automatic fire-extinguishing systems are especially important for server rooms and data centres. Gas systems based on inert gases or chemical agents extinguish a fire without damaging the equipment. Water-based (sprinkler) systems are used in rooms where water will not damage the contents.

The organization of evacuation is based on the evacuation plan, which is developed for each floor and posted in visible places. The plan indicates the location of a person, the directions of movement to the exits, the location of fire extinguishers, fire hydrants and electrical-disconnection points. Employees must know the plan and the evacuation routes.

In the event of a fire, the personnel must immediately report it to the fire service, turn on the warning system, begin the evacuation of people, de-energize the equipment and, if it is safe, begin extinguishing the fire with primary means. During evacuation, it is forbidden to use lifts and to create panic.

The training of personnel in fire safety includes briefings, training in the fire-technical minimum, and practical evacuation exercises. Regular exercises develop in the staff a clear and confident sequence of actions, which in a real fire saves lives.

Thus, the fundamentals of fire safety include an understanding of the conditions of combustion and the methods of extinguishing, the observance of the fire-prevention regime, the maintenance of the means of detection and extinguishing in working order, and the readiness of the personnel for a competent evacuation and the elimination of an incipient fire.

Control Questions

1. What conditions are necessary for the occurrence and maintenance of combustion?
2. What is the 'fire triangle' and how is it connected with the methods of extinguishing?
3. What is the fire-prevention regime at an enterprise, and what does it include?
4. What requirements apply to the maintenance of evacuation routes and exits?
5. What is the sequence of actions of personnel in the event of a fire?

4. PRACTICAL CLASS MATERIALS

LO 1.1 Comply with the requirements and norms of occupational safety, industrial and ecological safety at information-service enterprises

Practical work 1.1.5 Prevention and treatment of occupational diseases among IT specialists

Objective: To form practical skills in identifying the risk factors of occupational diseases of IT specialists and in developing a set of preventive measures for a specific workplace.

Theoretical background

Prolonged work at a computer is associated with a number of harmful factors that can lead to occupational and work-related diseases. The main groups of such diseases are visual disorders (computer vision syndrome), diseases of the musculoskeletal system (osteocondrosis, carpal tunnel syndrome), and neuro-emotional and cardiovascular disorders caused by stress and a sedentary lifestyle.

Computer vision syndrome manifests itself as eye fatigue, dryness, redness, blurred vision and headache. It is caused by the prolonged focusing of the eyes on the screen, insufficient blinking, glare and incorrect placement of the monitor. Carpal tunnel syndrome is caused by the monotonous load on the hand when working with the keyboard and the mouse.

Prevention of occupational diseases is based on the organization of the workplace according to ergonomic requirements, on the observance of the work-and-rest regime with regulated breaks, on performing gymnastics for the eyes and the body, and on maintaining physical activity outside working hours.

Equipment and materials

- A computer workstation (desk, chair, monitor, keyboard, mouse).
- A measuring tape and a level for assessing the geometry of the workplace.
- A lux meter (or a smartphone application) for assessing illumination.
- An assessment checklist and report form.

Procedure

1. Study the main occupational diseases of IT specialists and the factors that cause them; fill in a table 'factor – disease – preventive measure'.
2. Examine the provided (or your own) workstation and assess its conformity with ergonomic requirements: the height of the desk and chair, the placement of the monitor, the position of the keyboard and the mouse.
3. Measure the illumination of the working surface with a lux meter and compare the result with the recommended value of 300–500 lux.
4. Perform a set of exercises for the eyes and for the musculoskeletal system, and describe in the report how often and at what intervals they should be carried out.
5. Develop a regulated work-and-rest regime for an IT specialist, indicating the duration of continuous work and the frequency and duration of breaks.
6. Formulate a list of recommendations for the prevention of occupational diseases for the examined workplace.

Tasks for independent completion

1. Compile a table of at least five occupational diseases of IT specialists with their causes and methods of prevention.
2. Prepare a memo 'Workplace gymnastics for an IT specialist' with a description of exercises for the eyes, hands, neck and back.
3. Propose measures to bring the examined workplace into conformity with ergonomic requirements.

Conclusion: As a result of the work, the student has formed the skills of identifying the risk factors of occupational diseases and of developing a comprehensive program for their prevention at the workplace of an IT specialist.

Control Questions

1. What are the main occupational diseases that arise from prolonged work at a computer?
2. What causes computer vision syndrome and how can it be prevented?
3. What ergonomic measures help to prevent diseases of the musculoskeletal system?
4. What work-and-rest regime is recommended when working at a computer?
5. What exercises are included in workplace gymnastics for an IT specialist?

Practical work 1.1.7 Analysis of occupational risks at the workplace of an IT specialist

Objective: To form practical skills in identifying hazards and in assessing occupational risks at a particular workplace using a risk matrix, and in developing measures for their reduction.

Theoretical background

The assessment of occupational risks at a particular workplace is carried out in several steps: the identification of hazards, the assessment of the probability and severity of each hazard, the determination of the level of risk using a matrix, and the planning of risk-reduction measures.

A risk matrix of dimension 5×5 is conveniently used, where the rows correspond to the probability of an event (from 'very unlikely' to 'almost certain'), and the columns correspond to the severity of the consequences (from 'negligible' to 'catastrophic'). The cell at the intersection determines the level of risk.

Equipment and materials

- A description (or a real example) of the workplace of an IT specialist.
- A risk-matrix template (5×5).
- A hazard-identification checklist.
- A report form for the results of the risk assessment.

Procedure

1. Choose a workplace for analysis (a computer class, a server room, or a workplace for repairing equipment).
2. Identify the hazards of this workplace and enter them in the table; for the IT sphere, take into account electric current, fire, microclimate, lighting, noise, electromagnetic field, ergonomic and psycho-emotional factors.
3. For each hazard, assess the probability of a harmful event on a scale from 1 to 5 and justify the assessment.
4. For each hazard, assess the severity of the possible consequences on a scale from 1 to 5 and justify the assessment.
5. Using the risk matrix, determine the level of risk for each hazard (acceptable, moderate, significant, unacceptable).
6. For the risks of a significant and unacceptable level, develop measures for their reduction in accordance with the hierarchy of controls and assess the residual risk.

Tasks for independent completion

1. Prepare a risk-assessment map for the chosen workplace with at least eight identified hazards.
2. Build a risk matrix and place the assessed hazards on it.
3. Develop a plan of measures to reduce the two most significant risks, indicating the type of measure and the expected result.

Conclusion: As a result of the work, the student has mastered the methodology of identifying hazards and assessing occupational risks using a matrix, and has learned to plan reasonable measures for their reduction at a specific workplace.

Control Questions

1. What steps does the assessment of occupational risks at a workplace consist of?
2. How are the probability and the severity of the consequences of a hazard assessed?
3. How is the level of risk determined using a risk matrix?
4. Which risk levels require the mandatory development of reduction measures?
5. What is residual risk and why must it be assessed after implementing the measures?

Practical work 1.1.8 The role of training and briefings on safety techniques for staff at information enterprises

Objective: To form practical skills in organizing and conducting safety briefings and in drawing up the corresponding documents (instructions and briefing-registration logs).

Theoretical background

Training and briefing personnel on labour protection is one of the most important organizational measures of the safety-management system. Without passing the established briefings and verifying their knowledge, employees are not allowed to work independently.

There are several types of briefings. The introductory briefing is conducted with all newly hired employees regardless of their profession. The primary briefing at the workplace is conducted before the start of independent work. The repeated briefing is conducted periodically to consolidate knowledge. The unscheduled briefing is conducted when the rules change, when equipment changes, or after an accident. The targeted briefing is conducted before performing one-time work.

All briefings, except the introductory one in some organizations, are registered in special logs indicating the date, the surname of the person being briefed and the person briefing, and their signatures. The instructions on labour protection for professions and types of work are the basis for conducting briefings.

Equipment and materials

- Samples of labour-protection instructions for an IT specialist.
- A template of a briefing-registration log.
- Regulatory requirements for the content of briefings.
- A computer for preparing documents.

Procedure

1. Study the types of briefings on labour protection and the conditions for conducting each of them; fill in a comparative table.
2. Develop the structure of a labour-protection instruction for the workplace of an IT specialist (general requirements, requirements before work, during work, in emergency situations, and after work).
3. Prepare the text of a primary briefing at the workplace for a newly hired technician for maintaining computer equipment.
4. Fill in the briefing-registration log for a conditional group of employees who have passed the briefing.
5. Conduct a role-play briefing in pairs, in which one student plays the role of the instructor and the other the role of the employee, and evaluate its completeness.
6. Formulate conclusions about the importance of briefings for ensuring safety at the enterprise.

Tasks for independent completion

1. Draw up a labour-protection instruction for the workplace of an IT specialist (in abbreviated form, by sections).
2. Prepare and fill in a briefing-registration log for at least five employees.
3. Make a list of situations in which an unscheduled briefing must be conducted.

Conclusion: As a result of the work, the student has formed the skills of organizing and conducting briefings on labour protection and of drawing up the accompanying documentation, which is an important element of the safety-management system at an information enterprise.

Control Questions

1. What types of briefings on labour protection exist and when is each of them conducted?
2. Who conducts the introductory briefing and who conducts the primary briefing at the workplace?
3. In which cases is an unscheduled briefing conducted?
4. What sections does a labour-protection instruction contain?
5. Why is the registration of briefings in a log necessary?

Practical work 1.1.10 Prevention of occupational diseases of IT specialists

Objective: To form practical skills in developing a comprehensive program for the prevention of occupational diseases of an IT specialist and in assessing the conformity of the working conditions with sanitary-hygienic requirements.

Theoretical background

The prevention of occupational diseases of IT specialists is built on three groups of measures: technical and ergonomic (the correct organization of the workplace), organizational (the work-and-rest regime, gymnastics, medical examinations), and individual (a healthy lifestyle and physical activity).

Regular medical examinations make it possible to detect the initial signs of diseases at an early stage and to take timely measures. For workers at a computer, periodic examinations by an ophthalmologist and a neurologist are especially important.

Equipment and materials

- A description of the workplace and the working conditions of an IT specialist.
- Sanitary-hygienic norms for premises with computer equipment.
- A checklist for assessing working conditions.
- A report form.

Procedure

1. Assess the working conditions of the chosen workplace by sanitary-hygienic parameters: microclimate, lighting, noise, ergonomics.
2. Identify the deviations of the parameters from the recommended values and determine which occupational diseases these deviations may cause.
3. Develop a set of technical and ergonomic measures to eliminate the identified deviations.
4. Make a regulated work-and-rest regime and a set of gymnastic exercises for the prevention of the diseases identified at the workplace.
5. Formulate recommendations for the individual prevention of occupational diseases (physical activity, nutrition, sleep regime).
6. Combine all the measures into a single prevention program and present it in the report.

Tasks for independent completion

1. Compile a comprehensive program for the prevention of occupational diseases for the workplace of an IT specialist.
2. Prepare a memo for the employee 'How to preserve health when working at a computer'.

3. Propose a list of medical specialists whose examinations are recommended for an IT specialist, and the frequency of the examinations.

Conclusion: As a result of the work, the student has formed the skills of comprehensive assessment of the working conditions and of developing a program for the prevention of occupational diseases that combines technical, organizational and individual measures.

Control Questions

1. What three groups of measures form the basis of the prevention of occupational diseases of IT specialists?
2. What sanitary-hygienic parameters of the workplace must be assessed?
3. What technical and ergonomic measures help to preserve the health of an employee?
4. Why are regular medical examinations important for workers at a computer?
5. What individual measures help to prevent occupational diseases?

Practical work 1.1.11 Compliance with norms and rules for ensuring safety when working with information and data on computers and servers

Objective: To form practical skills in observing the rules of safe work with information and data, including data protection, access control and safe operation of computer and server equipment.

Theoretical background

Safety when working with information and data combines two aspects: the physical and electrical safety of working with equipment, and information security — the protection of data from loss, unauthorized access and damage. The technician must take both aspects into account.

The basic principles of information security are confidentiality (access to data only for authorized persons), integrity (protection of data from unauthorized modification) and availability (data accessible to authorized users when needed). They are ensured by organizational and technical measures: access control, backup, antivirus protection and encryption.

Equipment and materials

- A computer (workstation) with an operating system installed.
- Rules and instructions for safe work with information.
- Software for backing up and protecting data.
- A report form.

Procedure

1. Study the rules of safe work with computer and server equipment and the requirements for the protection of information; fill in a checklist.
2. Configure user accounts with differentiation of access rights and a password policy on the workstation (or describe the procedure for doing so).
3. Perform a backup of important data to an external medium and check the possibility of restoring it.
4. Check the operation of the antivirus protection and update its databases; describe the procedure for scanning the system.
5. Develop a set of rules for safe handling of confidential information for an IT specialist.
6. Describe the procedure for safely de-energizing and maintaining server equipment without losing data.

Tasks for independent completion

1. Make a list of organizational and technical measures for protecting information at the workplace.
2. Prepare an instruction 'Rules of safe work with confidential data'.
3. Draw up a data-backup schedule for a small enterprise with a justification for the chosen frequency.

Conclusion: As a result of the work, the student has mastered the practical skills of observing the norms and rules of safe work with information and data, combining the requirements of physical safety and information security.

Control Questions

1. What two aspects does safety when working with information and data combine?
2. What are the basic principles of information security?
3. What measures ensure the confidentiality, integrity and availability of data?
4. Why is regular data backup necessary, and how is its frequency chosen?
5. What rules must be observed for the safe maintenance of server equipment without loss of data?

Practical work 1.1.13 Means of electrical and fire safety in the IT field

Objective: To form practical skills in selecting and using the means of electrical and fire safety for premises with computer equipment and in checking their serviceability.

Theoretical background

The means of electrical safety in the IT field include earthing and neutral-bonding devices, residual current devices (RCDs), circuit breakers, sources of uninterruptible power supply, voltage indicators and individual electrical-protective equipment.

The means of fire safety include fire alarms and detectors, automatic fire-extinguishing systems, primary fire-extinguishing means (carbon-dioxide and powder fire extinguishers), fire hydrants, and evacuation plans and signs. For computer rooms, the means that do not damage the electronics are especially important.

Equipment and materials

- Carbon-dioxide and powder fire extinguishers (or their mock-ups).
- A residual current device and a circuit breaker for examination.
- A voltage indicator and individual electrical-protective equipment.
- An evacuation plan of the room and a report form.

Procedure

1. Examine the means of electrical safety of the room (earthing, RCD, circuit breakers, UPS) and describe their purpose and operating principle.
2. Examine the available fire extinguishers, determine their type and the classes of fire for which they are intended, and check the pressure indicator and the date of the next inspection.
3. Determine which fire extinguishers may be used to extinguish energized computer equipment and justify the choice.
4. Study the evacuation plan of the room, find the location of the exits, the fire extinguishers and the electrical-disconnection points.
5. Develop an action algorithm for an employee in the event of an electrical fire in a computer class.
6. Compile a checklist for checking the serviceability of the means of electrical and fire safety of the room.

Tasks for independent completion

1. Make a table of the means of electrical and fire safety of the room indicating their purpose.
 2. Prepare an action algorithm for an employee in the event of a fire of energized equipment.
 3. Develop a checklist for the periodic inspection of fire extinguishers and electrical-safety means.
- Conclusion:** As a result of the work, the student has formed the skills of selecting, using and checking the serviceability of the means of electrical and fire safety for premises with computer equipment.

Control Questions

1. What means of electrical safety are used in premises with computer equipment?
2. What means of fire safety are used in the IT field?
3. Which fire extinguishers may be used to extinguish energized equipment, and why?
4. What information must be checked on a fire extinguisher before use?
5. What does the action algorithm of an employee in the event of an electrical fire include?

Practical work 1.1.15 Protective measures during the operation of electrical installations and electrical equipment

Objective: To check and consolidate, in a comprehensive form, the knowledge and skills of occupational, electrical and fire safety acquired in the learning outcome, by solving situational problems and developing a set of protective measures for a given object.

Theoretical background

This lesson generalizes the material of the learning outcome 'Comply with the requirements and norms of occupational safety'. The student must demonstrate the ability to apply, in a comprehensive way, knowledge of electrical safety, ergonomics, risk assessment, briefings and fire safety to a specific object.

Protective measures during the operation of electrical installations are divided into organizational (admission to work, briefings, supervision) and technical (de-energizing, earthing, verification of the absence of voltage, fencing and signs). Their combination ensures the safety of work.

Equipment and materials

- A description of a model object (a computer class or a server room).
- Sets of situational problems on occupational, electrical and fire safety.
- Templates of a risk-assessment map and a labour-protection instruction.
- A final-assessment form.

Procedure

1. Study the description of the model object and determine the class of the room by the degree of electric-shock danger.
2. Identify the hazards of the object and assess the occupational risks using a matrix.
3. Develop a set of collective and individual protective measures for the safe operation of the electrical equipment of the object.
4. Determine the means of fire safety necessary for the object and the procedure for actions in the event of a fire.
5. Solve the proposed situational problems on the provision of first aid in the event of an electric shock and on the actions in emergency situations.
6. Combine all the decisions into a single 'safety passport' of the object and present it to the group.

Tasks for independent completion

1. Prepare a comprehensive 'safety passport' of the model object, including risk assessment and a set of protective measures.
2. Solve a situational problem on the sequence of actions in the event of an electric shock to a person.
3. Draw up a list of protective equipment and fire-extinguishing means necessary for the safe operation of the object.

Conclusion: As a result of the work, the student has demonstrated comprehensive knowledge and skills in occupational, electrical and fire safety and the ability to develop a holistic set of protective measures for the safe operation of electrical installations and computer equipment.

Control Questions

1. What organizational and technical protective measures are applied during the operation of electrical installations?
2. How is the class of a room determined by the degree of electric-shock danger?
3. What is the sequence of providing first aid in the event of an electric shock?
4. What set of protective measures ensures the safe operation of a server room?
5. How are the requirements of electrical and fire safety combined in the operation of computer equipment?

5. LIST OF SOFTWARE AND MULTIMEDIA SUPPORT FOR TRAINING SESSIONS

- Multimedia projector and interactive (smart) board; digital podium; white marker board.
- PowerPoint presentations for every topic of the module.
- Personal computers with Internet access and access to local networks.
- Disassembled training PC stands; sets of motherboards, power supplies, RAM modules, storage drives and graphics adapters.
- Peripheral and office equipment: printers, scanners, multifunction devices (MFPs), plotters.
- Measuring instruments: multimeters, POST-cards, power-supply and cable testers.
- Anti-static wrist straps and mats; PC assembly and repair tool kits; soldering stations.
- Hardware-diagnostic software: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo, HWMonitor, OCCT.
- Programming and scripting tools: Python 3 (with psutil, WMI, pywin32), PowerShell, Windows Subsystem for Linux.
- Consumables of office equipment (cartridges, toner, ink, ribbons) for industrial training.
- Learning Management System (LMS) for hosting materials and assignments; video lectures on PC maintenance and repair.

6. LIST OF LITERATURE

Core (fundamental) literature:

1. Phil Hughes MBE, Ed Ferrett. Introduction to Health and Safety at Work. ISBN: 978-0-415-72308-4.
2. Mark A. Friend, James P. Kohn. Fundamentals of Occupational Safety and Health. ISBN-13: 978-0-86587-171-7.
3. Charles K. Alexander, Matthew N. O. Sadiku. Fundamentals of Electric Circuits. ISBN: 978-0-07-352955-4.
4. John L. Hennessy, David A. Patterson. Computer Architecture: A Quantitative Approach (5th Edition). ISBN-13: 978-8178672663.
5. Jim Ledin. Modern Computer Architecture and Organization (2nd Edition). ISBN-13: 978-1803234519.

Additional literature:

1. Saurabh Shrivastava, Neelanjali Srivastav. Solutions Architect's Handbook (2nd Edition). ISBN-13: 978-1801816618.
2. Joseph Ingeno. Software Architect's Handbook. ISBN-13: 978-1788624060.
3. Robert C. Martin. Clean Architecture: A Craftsman's Guide to Software Structure and Design. ISBN-13: 978-0134494166.

Internet resources:

1. Computer Architecture Topics — https://en.wikipedia.org/wiki/Computer_architecture
2. Fire Protection and Prevention (OSHA) — <https://www.osha.gov/sites/default/files/2019-03/fireprotection.pdf>
3. <https://learn.microsoft.com> — official Microsoft documentation on Windows and administration.
4. <https://support.microsoft.com> — technical support and instructions for configuring hardware and software.
5. <https://docs.python.org> — documentation on automation and system administration in Python.

III. LO 1.2 Determine the operability of computer systems at the circuit-engineering level

1. ACADEMIC CALENDAR

LO 1.2 Determine the operability of computer systems at the circuit-engineering level

Instructor: D.M. Yermekov

Specialty: 06130100 – «Software Engineering (by types)»

Total hours: 60 hours (30 lessons of 2 academic hours each)

№	Name of topics	Date of lesson	Type of activity
1	Fundamentals of circuit engineering		Study of new material
2	Capacitors. Types of capacitors and their connections		Combined lesson
3	Analysis of the types of capacitors and their application in computer devices		Application and consolidation of knowledge
4	Detecting faults in the power supplies of computer systems		Skills-improvement lesson
5	Investigation of the causes of faults of the power supplies of computer systems		Application and consolidation of knowledge
6	Conductors and dielectrics in an electric field		Study of new material
7	The fundamentals of circuit engineering and its role in determining the operability of computer systems		Combined lesson
8	Methods of analysing circuits for faults in computer systems		Study of new material
9	Practical skills in measuring and analysing electrical circuits in computers		Combined lesson
10	Testing and debugging of central processors in computer systems		Study of new material
11	Working with memory and the analysis of memory circuits in computers		Combined lesson
12	Analysis and testing of the circuits of the motherboards of computers		Study of new material
13	Investigation of the methods of diagnosing graphics adapters and monitors		Application and consolidation of knowledge
14	Electromagnetism		Combined lesson
15	Analysis of conductors, dielectrics and electromagnetic phenomena in electronic circuits		Application and consolidation of knowledge
16	Analysis and repair of graphics cards and monitors		Study of new material
17	The role of the BIOS and the methods of its diagnostics in		Skills-

	computers		improvement lesson
18	Analysis of the methods of diagnosing the BIOS and the network interfaces of a computer		Application and consolidation of knowledge
19	Testing and repair of network interfaces and network cards		Skills-improvement lesson
20	Analysis and repair of the circuits of data-storage devices (hard disk drives and SSDs)		Study of new material
21	Investigation of faults of data-storage devices and cooling systems of computers		Application and consolidation of knowledge
22	The principles of operation and the analysis of the circuits of cooling systems of computers		Skills-improvement lesson
23	Methods of diagnosing and measuring the parameters of the electrical circuits of a computer		Application and consolidation of knowledge
24	Detecting faults in the circuits of the sound subsystem of computers		Combined lesson
25	Testing and analysis of the circuits of USB and other input/output ports		Skills-improvement lesson
26	Analysis of the circuits of temperature monitoring and control in computers		Study of new material
27	Analysis of the typical faults of the central processor, the random-access memory and the motherboard		Application and consolidation of knowledge
28	The role of circuit engineering in protecting computer systems from overheating and overvoltage		Skills-improvement lesson
29	Modern methods of diagnosing integrated circuits, microcontrollers and monitoring systems of computers		Application and consolidation of knowledge
30	Diagnostics and repair of integrated circuits and microcontrollers in computers		Consolidation and assessment of knowledge

2. GLOSSARY

The glossary contains the main definitions, terms and concepts that reflect the core content of this learning outcome. The terms are arranged in alphabetical order.

Term	Definition
Capacitor	A passive two-terminal component that stores energy in an electric field; characterized by its capacitance, rated voltage and type of dielectric.

Circuit engineering (circuitry)	The branch of engineering that deals with the principles of building and analysing electronic circuits from discrete and integrated components.
Conductor	A material with a large number of free charge carriers that easily conducts electric current (for example, copper or aluminium).
Dielectric	A material that practically does not conduct electric current and is used for insulation and in capacitors.
Electromagnetism	The branch of physics that studies the interconnection of electric and magnetic phenomena; the basis for the operation of transformers, motors and inductive components.
Integrated circuit (IC)	A miniature electronic circuit manufactured on a single semiconductor crystal that contains many interconnected components.
Multimeter	A universal measuring instrument for measuring voltage, current, resistance and other parameters of electrical circuits.
Oscilloscope	A measuring instrument that displays the dependence of a voltage on time, used to observe the shape of signals.
POST card	A diagnostic board that displays the codes of the power-on self-test (POST) of a motherboard, helping to localize faults.
Power supply unit (PSU)	A device that converts the alternating mains voltage into the stabilized direct voltages required by the components of a computer.
Resistor	A passive component that limits the current and creates a voltage drop in a circuit; characterized by its resistance and dissipated power.
Short circuit	An abnormal connection of points of a circuit with a very small resistance, leading to a sharp increase in current.
Voltage	The potential difference between two points of an electric circuit, measured in volts (V).

3. LECTURE MATERIALS

LO 1.2 Determine the operability of computer systems at the circuit-engineering level

Lecture 1.2.1 Fundamentals of circuit engineering

Content

The subject of circuit engineering and its role in computing technology.
 Passive and active components of electronic circuits.
 Basic laws of electric circuits (Ohm's law, Kirchhoff's laws).
 Types of circuit diagrams and conventional graphic symbols.

Circuit engineering is the branch of engineering that studies the principles of building, analysing and calculating electronic circuits made up of discrete and integrated components. For an information-systems technician, the fundamentals of circuit engineering are necessary in order to understand how computer devices work and to be able to determine their operability at the level of components and circuits.

All elements of electronic circuits are divided into passive and active. Passive elements (resistors, capacitors, inductors) do not amplify a signal and do not require an external power source for their operation. Active elements (diodes, transistors, integrated circuits) can control the current and amplify signals, and they require power to be supplied.

A resistor is a passive element that limits the current and creates a voltage drop. Its main parameters are the resistance, measured in ohms, and the dissipated power, measured in watts. Resistors are used to set the operating modes of other components, to divide voltage and to limit currents.

A capacitor is a passive element that stores energy in an electric field between two plates separated by a dielectric. Its main parameter is the capacitance, measured in farads. Capacitors are used for filtering, decoupling power circuits, storing charge and forming time delays.

An inductor is a passive element that stores energy in a magnetic field. Its parameter is the inductance, measured in henries. Inductors are used in filters, power-supply circuits and signal circuits, where their property of resisting a change in current is important.

The operation of electric circuits is described by fundamental laws. Ohm's law establishes the relationship between voltage, current and resistance: the current in a section of a circuit is directly proportional to the voltage and inversely proportional to the resistance. This is the basic relationship for any calculation of a circuit.

Kirchhoff's laws describe the behaviour of currents and voltages in branched circuits. The first law (the current law) states that the sum of the currents flowing into a node is equal to the sum of the currents flowing out of it. The second law (the voltage law) states that the sum of the voltage drops in a closed loop is equal to the sum of the electromotive forces.

Circuit diagrams are used to represent electronic devices. A schematic (principal) diagram shows all the components and the electrical connections between them by means of conventional graphic symbols. A block diagram shows the device at the level of functional blocks. An assembly diagram shows the actual placement of the components on the board.

Conventional graphic symbols are a standardized 'language' of circuit engineering. Each type of component has its own symbol: a resistor is depicted as a rectangle, a capacitor as two parallel lines, a diode as a triangle with a line, and so on. The ability to read these symbols is the basis for understanding any circuit diagram.

In computer technology, circuit engineering is applied at all levels: from the power supply, which converts and stabilizes voltages, to the motherboard, the central processor and memory, which contain millions of components. Understanding the principles of circuit engineering allows the technician to localize a fault to a specific node or component.

Modern computer devices are built mainly on integrated circuits, in which a huge number of components are placed on a single crystal. Nevertheless, the discrete components of the strapping (resistors, capacitors, inductors) remain an important part of the circuits and are often the cause of faults.

Thus, the fundamentals of circuit engineering give the technician an understanding of the components of electronic circuits, the laws of their operation and the methods of representing circuits. This knowledge is the foundation for diagnosing and determining the operability of computer systems.

Control Questions

1. What does circuit engineering study, and why is it necessary for an information-systems technician?
2. What is the difference between passive and active components of electronic circuits?
3. Formulate Ohm's law and explain its meaning.
4. What do Kirchhoff's first and second laws state?
5. What types of circuit diagrams exist and what is the difference between them?

Lecture 1.2.2 Capacitors. Types of capacitors and their connections

Content

The design and operating principle of a capacitor.

The main parameters of capacitors.
Types of capacitors by the type of dielectric.
Series and parallel connection of capacitors.

A capacitor is one of the most widespread passive components of electronic circuits. It consists of two conductive plates separated by a dielectric. When a voltage is applied, charges of opposite signs accumulate on the plates, and energy is stored in the electric field between them.

The main parameter of a capacitor is its capacitance — the ability to accumulate a charge at a given voltage. Capacitance is measured in farads, but in practice fractions are used: microfarads, nanofarads and picofarads. The capacitance depends on the area of the plates, the distance between them and the properties of the dielectric.

The second most important parameter is the rated voltage — the maximum voltage at which the capacitor can operate for a long time without breakdown of the dielectric. Exceeding the rated voltage leads to the failure of the capacitor, often with a characteristic swelling or rupture of the case of electrolytic capacitors.

Other parameters include the tolerance on the capacitance, the temperature coefficient, the equivalent series resistance (ESR) and the leakage current. For the power circuits of a computer, a low ESR is especially important, since capacitors with an increased ESR cease to filter the ripple effectively and cause instability.

Capacitors are classified by the type of dielectric. Ceramic capacitors have a small capacitance, low losses and are used in high-frequency and decoupling circuits. Electrolytic (aluminium and tantalum) capacitors have a large capacitance and are used in power filters; they are polar and require the correct connection of the polarity.

Film capacitors have good stability and are used in signal and timing circuits. Tantalum capacitors are compact and stable but sensitive to exceeding the voltage. The choice of the type of capacitor is determined by the requirements of the circuit for capacitance, frequency, stability and dimensions.

Electrolytic capacitors of the power supplies and motherboards of computers are one of the most frequent causes of faults. Over time and under the influence of heat, the electrolyte dries out, the ESR increases, and the case swells. The replacement of swollen capacitors is a typical repair operation.

Capacitors can be connected in series and in parallel. With a parallel connection, the capacitances are added, and the total capacitance increases; the rated voltage remains equal to the smallest of the rated voltages. A parallel connection is used to obtain a large capacitance and to reduce the total ESR.

With a series connection, the total capacitance decreases (the reciprocal of the total capacitance is equal to the sum of the reciprocals of the capacitances), but the permissible voltage increases, since it is distributed among the capacitors. A series connection is used to operate at an increased voltage.

When working with capacitors, one must remember that charged capacitors of power supplies and monitors can store a dangerous charge even after the device is disconnected from the mains. Before servicing, such capacitors must be safely discharged through a resistor.

For checking capacitors, a multimeter with a capacitance-measurement function, an ESR meter and a visual inspection for swelling and leakage of electrolyte are used. The combination of these methods allows faulty capacitors to be reliably detected.

Thus, knowledge of the design, parameters, types and methods of connecting capacitors allows the technician to understand their role in the circuits of a computer, to correctly select replacements and to diagnose the typical faults associated with them.

Control Questions

1. What does a capacitor consist of and on what principle does it operate?
2. What are the main parameters of a capacitor, and why is the rated voltage important?
3. What types of capacitors are distinguished by the type of dielectric, and where are they applied?

4. How does the total capacitance change with a series and a parallel connection of capacitors?
5. Why are electrolytic capacitors a frequent cause of faults, and how are they checked?

Lecture 1.2.6 Conductors and dielectrics in an electric field

Content

- The structure of conductors and dielectrics.
- The behaviour of conductors in an electric field.
- Polarization of dielectrics and their breakdown.
- The application of conductors and dielectrics in computer devices.

All materials with respect to the conduction of electric current are divided into conductors, dielectrics and semiconductors. This division is based on the presence and mobility of free charge carriers. Understanding the behaviour of conductors and dielectrics in an electric field is the basis for understanding the operation of any electronic device.

Conductors are materials with a large number of free electrons that can move under the action of an electric field. Metals (copper, aluminium, gold, silver) are good conductors. In computer technology, conductors are used for the conductive tracks of boards, wires, contacts and connectors.

In an external electric field, the free electrons of a conductor are redistributed so that the field inside the conductor becomes equal to zero. This phenomenon — electrostatic induction — underlies the screening (shielding) of electronic devices from external electromagnetic interference.

Dielectrics are materials in which there are practically no free charge carriers, so they do not conduct electric current. Examples are the textolite of boards, the plastic of cases, ceramics, mica and air. Dielectrics are used for insulation and as the working medium of capacitors.

In an electric field, the dielectric is polarized: the bound charges are displaced, forming an internal field directed against the external one. The degree of polarization is characterized by the dielectric permittivity. The higher the permittivity, the greater the capacitance of a capacitor with such a dielectric.

If the field strength in a dielectric exceeds a certain limit, an electrical breakdown occurs — the dielectric loses its insulating properties and begins to conduct current. The value of the field strength at which breakdown occurs is called the electrical (dielectric) strength. Breakdown is one of the causes of the failure of capacitors and insulation.

Semiconductors occupy an intermediate position between conductors and dielectrics. Their conductivity depends strongly on temperature, illumination and the addition of impurities. It is on the basis of semiconductors (silicon, germanium) that diodes, transistors and integrated circuits — the foundation of all modern electronics — are built.

In a printed circuit board, conductors and dielectrics work together: the conductive copper tracks are placed on a dielectric base of textolite, and between the layers of a multilayer board there are dielectric layers. The correct combination of materials ensures the working of the circuit and the absence of unwanted leakages.

The properties of dielectrics depend on external conditions. Moisture, contamination and dust on the surface of a board can create conductive paths and cause leakage currents, faults and even fires. Therefore the cleanliness and dryness of boards is important for their reliable operation.

Knowledge of the properties of conductors and dielectrics is applied in the diagnostics of faults: a short circuit is a place where a dielectric has lost its properties or where conductors have closed; a break is a place where a conductor has lost its continuity. The technician must understand the physical nature of these faults.

The screening of cables and devices, the insulation of wires, the choice of materials for boards and the protection against static electricity — all this is based on an understanding of the behaviour of conductors and dielectrics in an electric field.

Thus, the division of materials into conductors, dielectrics and semiconductors and the understanding of their behaviour in an electric field is a fundamental basis for understanding the design, operation and faults of computer devices.

Control Questions

1. Into which groups are materials divided according to their ability to conduct current, and on what basis?
2. How does a conductor behave in an external electric field?
3. What is the polarization of a dielectric, and how is it connected with the dielectric permittivity?
4. What is the electrical breakdown of a dielectric, and why is it dangerous?
5. How are conductors and dielectrics used together in a printed circuit board?

Lecture 1.2.7 The fundamentals of circuit engineering and its role in determining the operability of computer systems

Content

Circuit engineering as a tool for assessing operability.

Functional nodes of a computer from the point of view of circuitry.

Signs of operability and inoperability of nodes.

The sequence of circuit-engineering diagnostics.

Determining the operability of a computer system at the circuit-engineering level means establishing whether each functional node performs its function and whether the electrical signals and voltages correspond to the nominal values. Circuit engineering provides the technician with the methods and criteria for such an assessment.

Any computer can be represented as a set of functional nodes connected by buses and power circuits: the power supply, the voltage-regulator modules (VRM), the central processor, the chipset, the memory, the graphics subsystem, the storage and input/output controllers. Each node has its own power supply and signal circuits.

The operability of a node is determined by several criteria: the presence of the correct supply voltage, the presence of a clock signal, the presence of a reset signal, and the correct exchange of signals over the buses. The absence or distortion of any of these signs indicates a fault.

The power supply of the nodes deserves special attention. Modern processors and memory require several supply voltages formed by the VRM directly on the motherboard. A fault of the VRM leads to the inoperability of the processor even with a serviceable power supply.

The clock signals are formed by quartz resonators and clock generators. Without a clock signal, the digital nodes cannot work. The reset signal puts the nodes into the initial state when switched on. The verification of the presence of these signals is an important step in diagnostics.

For determining operability, both passive and active methods are used. Passive methods include a visual inspection, the verification of voltages and signals without intervention in the operation of the circuit. Active methods include the substitution of nodes, the injection of test signals and the use of diagnostic equipment.

An important diagnostic tool is the power-on self-test (POST), which the motherboard performs at start-up. The POST checks the main nodes and signals the result with codes (on a POST card or indicators) and sound signals. The decoding of POST codes allows the faulty node to be localized.

The sequence of circuit-engineering diagnostics is usually built from the simple to the complex: a visual inspection, the verification of the power supply, the verification of the clock and reset signals, the analysis of the POST codes, and only then the in-depth verification of specific circuits and components.

The role of circuit engineering in diagnostics is that it allows one to move from the symptom of a fault (the computer does not turn on, hangs, restarts) to a specific cause at the level of a node or component. Without an understanding of circuitry, diagnostics is reduced to a blind replacement of parts.

The localization of a fault is based on the principle of dividing the system into parts and checking each of them. By measuring the voltages and signals at the control points and comparing them with the nominal values, the technician gradually narrows the search area to a specific faulty element.

Modern diagnostics combines circuit-engineering methods with software ones: diagnostic utilities read the data of the sensors of voltage, temperature and frequency, which complements the picture obtained by measurements. The combination of these approaches gives the most complete assessment of operability.

Thus, circuit engineering plays a key role in determining the operability of computer systems: it provides the criteria for the operability of nodes, the methods of measurement and the logic of localizing faults, allowing the technician to perform diagnostics meaningfully and effectively.

Control Questions

1. What does it mean to determine the operability of a computer at the circuit-engineering level?
2. Into which functional nodes can a computer be divided from the point of view of circuitry?
3. What are the criteria for the operability of a node?
4. What is the difference between passive and active methods of diagnostics?
5. How is the sequence of circuit-engineering diagnostics built?

Lecture 1.2.8 Methods of analysing circuits for faults in computer systems

Content

Visual and instrumental methods of circuit analysis.

Measurement of voltages, currents and resistances.

Use of an oscilloscope to analyse signals.

The method of comparison with a serviceable device.

The analysis of circuits for faults is the central task of the diagnostics of computer systems at the circuit-engineering level. It combines visual and instrumental methods, each of which provides its own part of the information about the state of the circuit.

The visual method is the first and simplest. An inspection reveals burnt and swollen components, traces of overheating and soot, cracks in the board, cold solder joints, broken tracks and foreign objects. A significant part of faults can be detected precisely by visual inspection, including with the use of a magnifying glass and a microscope.

The measurement of voltages with a multimeter or an oscilloscope at the control points allows one to check whether the supply voltages and signals correspond to the nominal values. The deviation of a voltage from the nominal value, its absence or instability indicate a fault in the corresponding circuit.

The measurement of resistance (in the power-off state) is used to detect short circuits and breaks. A resistance close to zero between a power circuit and ground indicates a short circuit; an infinitely large resistance where there should be a connection indicates a break.

The measurement of current consumption allows one to assess the general state of a node. An increased current consumption may indicate a short circuit or a faulty component, and the absence of consumption may indicate a break in the power circuit or the inoperability of the node.

The oscilloscope is an indispensable tool for analysing dynamic signals. It allows one to observe the shape, amplitude, frequency and timing relations of signals — the clock pulses, the data buses, the control signals. The distortion of the shape of a signal often indicates a fault that cannot be detected by a multimeter.

The method of comparison with a serviceable device is very effective. By measuring the parameters at the same points of a faulty and a serviceable device, the technician quickly detects discrepancies and localizes the fault. This method is especially useful for complex circuits whose nominal values are unknown.

The 'freezing' and 'heating' methods are used to detect faults that depend on temperature. Local cooling or heating of a component allows one to detect an element that loses its operability when the temperature changes, which is typical of 'floating' faults.

The method of signal injection (signal tracing) consists in supplying a test signal to the input of a circuit and tracing its passage through the nodes. The place where the signal disappears or is distorted indicates the faulty node. This method is applied in the diagnostics of signal paths.

The analysis of the consumed current by the thermal method uses a thermal imager or simply the touch of a finger (with the power off and observing safety) to detect overheating components. An overheating element is often the faulty one or the one operating in an abnormal mode.

Modern diagnostics is built on a combination of these methods. The technician begins with a visual inspection and the measurement of the power supply, then uses the oscilloscope and the method of comparison, and applies thermal and other methods to localize complex faults.

Thus, the methods of analysing circuits for faults form an arsenal of the technician, allowing one to reliably detect and localize faults at the level of nodes and components and to determine the operability of computer systems.

Control Questions

1. What does the visual method of circuit analysis reveal?
2. What faults are detected by the measurement of voltage and what by the measurement of resistance?
3. Why is an oscilloscope necessary, and what does it allow one to see?
4. What is the essence of the method of comparison with a serviceable device?
5. How do the 'freezing' and 'heating' methods help to detect temperature-dependent faults?

Lecture 1.2.9 Practical skills in measuring and analysing electrical circuits in computers

Content

Measuring instruments and the rules for their use.
Safety techniques when measuring electrical circuits.
Methods of measuring the main parameters.
Interpretation of the measurement results.

The practical mastery of the methods of measuring and analysing electrical circuits is a key skill of a technician. It includes knowledge of the measuring instruments, the rules for their use, the methods of measurement and the ability to interpret the results.

The main measuring instrument is the digital multimeter, which measures voltage, current, resistance, and also has functions for checking diodes and continuity, and sometimes for measuring capacitance and temperature. The multimeter must be serviceable, with serviceable probes and a charged battery.

For the measurement of voltage, the multimeter is connected in parallel to the section of the circuit, observing the polarity and choosing the correct measurement range. For the measurement of current, the multimeter is connected in series in the open circuit, which requires caution to avoid a short circuit.

The measurement of resistance and continuity is performed only with the power off, otherwise the readings will be incorrect and the instrument may be damaged. The continuity mode with a sound signal is convenient for quickly checking the integrity of conductors and the presence of short circuits.

The oscilloscope is used for the analysis of dynamic signals. When working with it, it is important to correctly choose the time and amplitude scan, to connect the ground of the probe to the ground of the circuit, and to take into account the input parameters of the probe. The oscilloscope allows one to see the shape of the signal that the multimeter cannot show.

Safety techniques when measuring are of paramount importance. The measurements on energized equipment are performed with caution, using insulated probes, without touching the bare current-carrying parts. In the power supply, dangerous voltages are present even after disconnection, on the charged capacitors.

When measuring, control (reference) points are chosen — the contacts of connectors, the terminals of components, the test points of the board. The measurement results are compared with the nominal values from the documentation or with the readings of a serviceable device.

The interpretation of the results requires an understanding of the operation of the circuit. A voltage below the nominal value may indicate an overload or a fault of the stabilizer; the absence of a clock signal indicates a faulty generator; an increased current consumption indicates a short circuit. The technician must be able to connect a measurement with a possible cause.

The systematic recording of the measurement results in a table or protocol allows one to track the dynamics, to compare the data and to substantiate the conclusion about the fault. The accuracy and completeness of the measurements directly affect the reliability of the diagnosis.

Typical errors when measuring include the wrong choice of the measurement range, a bad contact of the probes, the measurement of resistance on an energized circuit, and the incorrect connection of the ground of the oscilloscope. Knowledge of these errors helps to avoid them.

The combination of measurements with a multimeter and an oscilloscope, the comparison with the nominal values and with a serviceable device, and the competent interpretation of the results form a complete picture of the state of the circuit and allow the fault to be localized.

Thus, the practical skills of measuring and analysing electrical circuits — the correct use of instruments, the observance of safety, the methods of measurement and the interpretation of the results — are the basis of the practical diagnostics of computer systems.

Control Questions

1. How is the multimeter connected when measuring voltage and when measuring current?
2. Why is the measurement of resistance performed only with the power off?
3. What safety rules must be observed when measuring energized circuits?
4. How are the control points for measurement chosen?
5. How is the result of a measurement connected with a possible cause of a fault?

Lecture 1.2.10 Testing and debugging of central processors in computer systems

Content

The architecture and power supply of the central processor.

Signs of a faulty processor.

Methods of testing the processor.

Stress testing and stability check.

The central processor (CPU) is the main computing node of a computer, which executes the instructions of programs. Its testing and debugging require an understanding of the architecture of the processor, of its power supply and of the methods of checking its operability and stability.

A modern processor is a complex integrated circuit containing billions of transistors, several computing cores, cache memory and integrated controllers. The processor requires several supply voltages formed by the voltage-regulator module (VRM) of the motherboard, and a clock signal.

The signs of a faulty processor or of its strapping may be: the complete failure of the computer to start, a start without an image, restarts and freezes, errors during calculations, and overheating. It is important to distinguish a fault of the processor itself from a fault of its power supply, the socket or the motherboard.

Before concluding that the processor is faulty, one must check its power supply (the voltages of the VRM), the integrity of the contacts of the socket (the absence of bent or oxidized pins), the correct installation and the quality of the thermal contact with the cooler.

The testing of the processor begins with checking the start: the presence of the supply voltages, the clock signal and the passage of the POST. If the computer passes the POST and loads the operating system, the processor is generally operational, and further testing checks its stability and performance.

Stress testing is used to check the stability of the processor under maximum load. Programs such as Prime95, LinX, OCCT and AIDA64 load all the cores of the processor with calculations and monitor the temperature, the frequency and the absence of errors. Errors or shutdowns during stress testing indicate instability.

The temperature of the processor is a critical parameter. Under load, it must not exceed the limit value specified by the manufacturer (usually about 90–100 °C, with throttling beginning earlier). Overheating leads to a forced reduction of the frequency (thermal throttling) and to shutdowns.

Thermal throttling is a protective mechanism that reduces the frequency and voltage of the processor when the limit temperature is reached. If a processor under load constantly throttles, this indicates an insufficient efficiency of the cooling system: a poor thermal contact, dried-out thermal paste or a clogged radiator.

The monitoring of the parameters of the processor during testing is performed by utilities (HWMonitor, HWiNFO, Core Temp), which read the data of the built-in sensors: the temperature, the voltage, the frequency and the load of each core. The analysis of these data helps to assess the state of the processor and its power supply.

Debugging the operation of the processor includes checking and correcting the settings of the BIOS/UEFI: the frequencies, the voltages, the power-supply modes and the memory profiles. Incorrect settings (for example, an excessive overclocking) can cause instability that is eliminated by returning to the default settings.

When replacing or reinstalling a processor, it is important to observe the rules: discharge the static, do not touch the contacts, correctly orient the processor in the socket by the key, apply the thermal paste in the right amount, and evenly tighten the cooler. Violation of these rules leads to faults.

Thus, the testing and debugging of central processors combines checking the power supply and the contacts, analysing the start and the POST, stress testing for stability, monitoring the temperature and the parameters, and correcting the settings of the BIOS/UEFI.

Control Questions

1. What supply voltages and signals does a central processor require?
2. What signs may indicate a fault of the processor or of its strapping?
3. What must be checked before concluding that the processor is faulty?
4. What is stress testing, and what programs are used for it?
5. What is thermal throttling, and what does its constant occurrence indicate?

Lecture 1.2.11 Working with memory and the analysis of memory circuits in computers

Content

- Types and operating principles of computer memory.
- The power supply and signal circuits of the memory.
- Signs of memory faults.
- Testing and analysing memory circuits.

The random-access memory (RAM) is one of the most important nodes of a computer, which stores the data and the code of the running programs. Working with memory and analysing its circuits require an understanding of the types of memory, of its power supply and of the methods of detecting faults.

Modern computers use dynamic memory of the DDR standard (DDR3, DDR4, DDR5) in the form of DIMM and SO-DIMM modules. Dynamic memory stores a bit of information in a capacitor of a memory cell and requires constant regeneration (refresh) of the charge, which is performed by the memory controller.

The memory module contains memory chips, an SPD chip with the parameters of the module, power circuits and a connector. The power supply of the memory is formed by a separate stabilizer of the motherboard; for DDR4, the supply voltage is about 1.2 V, and there are also reference voltages and the voltage of the controller.

The signs of memory faults are varied: the failure of the computer to start with a characteristic sound signal, the 'blue screens' of the operating system, freezes, spontaneous restarts, errors of applications and damage to data. These signs may be caused both by the modules themselves and by their power supply or the controller.

Before testing the memory, one must check the correct installation of the modules in the slots, the cleanliness of the contacts and the compatibility of the modules with the motherboard and the processor. Often the problem is solved by reseating the module and cleaning the contacts.

The testing of memory is performed by software: the built-in Windows tool, and the specialized programs MemTest86 and TestMem5. These programs write test patterns into the memory and check that they are read correctly, detecting errors of individual cells.

The testing of memory is performed for a long time and in several passes, since some errors appear only after the memory warms up or under a certain load. The detection of even a single error indicates a problem that must be eliminated.

If errors are detected, the localization is performed by the method of exclusion: the modules are checked one by one in different slots. This allows one to determine whether the faulty element is a specific module, a slot or the memory controller of the processor.

The analysis of the memory circuits at the hardware level includes checking the supply and reference voltages, the integrity of the contacts of the slots and the absence of short circuits. A fault of the memory power stabilizer leads to the inoperability of all the modules.

The settings of the memory in the BIOS/UEFI affect its stability. Incorrect timings, an overestimated frequency or an aggressive XMP/EXPO profile can cause errors. Returning to the standard settings or manually adjusting the parameters often restores stability.

When working with memory, anti-static measures must be observed, the modules must be taken by the edges without touching the contacts and the chips, and they must be installed until the latches click. The cleaning of the contacts is performed with a soft eraser or alcohol.

Thus, working with memory and analysing memory circuits combines an understanding of the types and operation of memory, checking the installation and the power supply, software testing, the localization of faults by the method of exclusion and the correction of the settings of the BIOS/UEFI.

Control Questions

1. What type of memory is used in modern computers, and how does it work?
2. What does a memory module consist of, and what is the SPD chip for?
3. What signs may indicate a memory fault?
4. What programs are used to test memory, and why is testing performed in several passes?
5. How is a memory fault localized by the method of exclusion?

Lecture 1.2.12 Analysis and testing of the circuits of the motherboards of computers

Content

The structure and functional nodes of a motherboard.

The power circuits of a motherboard.

The sequence of the start of a motherboard.

Methods of testing and diagnosing a motherboard.

The motherboard is the basic board of a computer, which unites all its components: the processor, the memory, the expansion cards, the storage and the peripheral devices. The analysis and testing of its circuits is one of the most complex tasks of diagnostics, requiring an understanding of the structure and the start sequence of the board.

The main functional nodes of a motherboard are: the processor socket, the memory slots, the chipset, the VRM power modules, the clock generator, the BIOS/UEFI chip, the expansion slots, the storage and input/output controllers, and the connectors of the power supply and the front panel.

The chipset is a set of microcircuits that organizes the interaction of the processor with the memory, the buses and the peripheral devices. In modern platforms, part of the functions of the chipset is integrated into the processor, and the remaining part is performed by a single chip (PCH). The chipset is the centre of the connections of the board.

The power circuits of a motherboard are formed by several stabilizers (VRM) that produce the voltages for the processor, the memory, the chipset and other nodes. The start of the board begins with the appearance of the standby voltage, which is present immediately when the board is connected to the power supply.

The start sequence of a motherboard is strictly defined: the appearance of the standby voltage, the formation of the power-on signal, the start of the power supply, the appearance of all the supply voltages, the formation of the clock signals, the removal of the reset signal, and the start of the processor with the execution of the BIOS/UEFI code.

The diagnostics of a motherboard is built on checking each step of this sequence. The technician checks the presence of the standby voltage, the operation of the power-on, the appearance of the supply voltages, the clock signals and the reset, gradually localizing the place where the start sequence breaks down.

The POST card is an effective tool for diagnosing a motherboard. It is inserted into a slot and displays the codes of the passage of the POST. The last code displayed indicates the stage at which the start stopped, which helps to localize the faulty node.

The visual inspection of a motherboard reveals swollen capacitors, traces of overheating of the VRM, mechanical damage, cracks of the board, faults of the connectors and traces of moisture or oxidation. Many faults of motherboards are detected precisely visually.

The measurement of voltages at the control points (the chokes of the VRM, the connectors, the components) and the comparison with the nominal values allow one to detect faults of the power circuits. The measurement of resistance to ground at the power circuits reveals short circuits, which are a frequent cause of the failure of the board to start.

The faults of motherboards are diverse: faults of the VRM, of the BIOS/UEFI chip, of the clock generator, of the chipset, of the connectors and slots, breaks of tracks and short circuits. Some faults are eliminated by replacing components, while others require complex repair or the replacement of the board.

The testing of a serviceable motherboard includes assembling a minimal configuration (the board, the processor, one memory module, the power supply) and checking the start, the passage of the POST and the loading of the system. The gradual connection of the components helps to detect an incompatible or faulty device.

Thus, the analysis and testing of the circuits of a motherboard is based on the knowledge of its structure and start sequence, on the use of a POST card and measurements, on the visual inspection and on the method of the minimal configuration, which allows the most complex faults to be localized.

Control Questions

1. What are the main functional nodes of a motherboard?
2. What function does the chipset perform?
3. What is the start sequence of a motherboard?
4. How does a POST card help to diagnose a motherboard?
5. What faults of motherboards are detected by a visual inspection?

Lecture 1.2.14 Electromagnetism

Content

The magnetic field and its sources.

Electromagnetic induction.

Inductive components and transformers.

Electromagnetic interference and compatibility.

Electromagnetism is the branch of physics that studies the interconnection of electric and magnetic phenomena. The understanding of electromagnetism is necessary for understanding the operation of many nodes of a computer — power supplies, transformers, chokes, motors of disk drives and fans — and also the phenomena of interference.

A magnetic field arises around moving electric charges, that is, around any conductor carrying a current. The magnetic field acts on other currents and magnets and is characterized by the magnetic induction. The direction of the field around a conductor is determined by the right-hand rule.

A coil of wire (a solenoid) creates a magnetic field similar to that of a permanent magnet. By placing a ferromagnetic core inside the coil, the magnetic field is greatly amplified. This principle underlies the operation of electromagnets, relays, chokes and transformers.

The phenomenon of electromagnetic induction, discovered by Faraday, consists in the appearance of an electromotive force in a circuit when the magnetic flux through it changes. The faster the flux changes, the greater the induced EMF. This phenomenon is the basis for the operation of transformers and generators.

An inductor (a choke) is a component that, owing to self-induction, resists a change in the current through it. It stores energy in a magnetic field and is used in filters, in the power circuits of the VRM, and in pulse power supplies, where it participates in the conversion of energy.

A transformer converts an alternating voltage of one value into a voltage of another value owing to electromagnetic induction between two windings on a common core. The ratio of the voltages is determined by the ratio of the numbers of turns. Transformers are used in power supplies to isolate and convert voltage.

In pulse power supplies of computers, a high-frequency transformer operates, which, owing to the high frequency of conversion, has small dimensions with a large power. The choke and the transformer are key components that determine the operation of the power supply.

Electric motors of fans and disk drives also operate on the principle of the interaction of a magnetic field and a current. The understanding of electromagnetism helps in the diagnostics of these devices and of the circuits that control them.

Electromagnetic interference is the unwanted electromagnetic effect of some devices on others. The fast switching of currents in digital circuits and power supplies generates interference that can disrupt the operation of neighbouring nodes and devices.

Electromagnetic compatibility (EMC) is the ability of a device to operate normally in the presence of interference and not to create unacceptable interference for others. EMC is ensured by screening, filtering, the correct layout of boards and the earthing of cases and cables.

The screening of cables and cases, the use of ferrite filters, the correct earthing and the layout of the tracks of a board are practical applications of the knowledge of electromagnetism for ensuring the reliable operation of computer equipment.

Thus, electromagnetism explains the operation of the inductive components, transformers and motors of a computer, and also the phenomena of interference and electromagnetic compatibility, the knowledge of which is necessary for the diagnostics and reliable operation of equipment.

Control Questions

1. Around what does a magnetic field arise, and how is its direction determined?
2. What is the phenomenon of electromagnetic induction?
3. How do a choke and a transformer work, and where are they applied in a computer?
4. Why does a pulse power supply use a high-frequency transformer?
5. What is electromagnetic compatibility, and how is it ensured?

Lecture 1.2.16 Analysis and repair of graphics cards and monitors

Content

The structure of a graphics card and the typical faults of its nodes.

Diagnostics of the power circuits and the GPU of a video card.

The structure of a monitor and the typical faults.

Methods of repairing video cards and monitors.

Graphics cards and monitors are devices of the graphics subsystem of a computer that form and display the image. The analysis and repair of these devices require an understanding of their structure, of the typical faults and of the methods of their elimination.

A graphics card consists of a graphics processor (GPU), video memory, power circuits (VRM), a cooling system, a BIOS chip, and output connectors. The GPU is the most complex node, which performs the calculations of the image; it requires powerful power supply and intensive cooling.

The typical faults of video cards are connected with overheating and the power circuits. Overheating leads to the degradation of the GPU and of the soldering of the BGA chips, the appearance of artifacts on the image and shutdowns. The faults of the VRM lead to the inability to start or to the instability of the card.

The diagnostics of a video card begins with a visual inspection of the cooling system, the capacitors and the connectors. Then the power circuits are checked: the presence of the supply voltages of the GPU and the memory, the operation of the VRM. The measurement of the resistance to ground reveals short circuits in the power circuits.

Artifacts on the image (distortions, stripes, dots) often indicate problems of the video memory or the GPU, including a violation of the soldering of the BGA. In some cases, the situation is temporarily improved by warming up, but a reliable repair requires the resoldering or replacement of the chip.

The cooling system of a video card requires regular maintenance: the cleaning of the radiator from dust, the replacement of the dried-out thermal paste and the thermal pads. A significant part of the problems of video cards is caused precisely by the deterioration of the cooling and the overheating that results from it.

A monitor consists of a power supply, a backlight (LED), a matrix (the LCD panel), a control board (the scaler) and the input connectors. The faults of a monitor may be located in any of these nodes, and the diagnostics determines the faulty one.

The typical faults of monitors are: the absence of the image with a working indicator (a problem of the backlight or the power supply of the backlight), the complete absence of power (the power supply), stripes and spots on the image (the matrix or the loop), and the absence of a reaction to the signal (the scaler).

The diagnostics of the power supply of a monitor includes checking the output voltages and the electrolytic capacitors, which often swell. The replacement of swollen capacitors restores the operability of many monitors. The diagnostics of the backlight requires checking the LED strips and their power supply.

The repair of video cards and monitors requires a soldering tool, a soldering station, sometimes an infrared station for working with BGA chips, measuring instruments and replacement components. The repair of complex faults (the GPU, the matrix) is not always economically justified.

Safety techniques during repair require disconnecting the device from the mains, discharging the capacitors, observing anti-static measures and caution when working with a hot soldering tool. The high voltages in the power supply of a monitor are especially dangerous.

Thus, the analysis and repair of graphics cards and monitors is based on the knowledge of their structure and typical faults, on the diagnostics of the power circuits, the cooling and the nodes, and on the mastery of the methods of repair and the rules of safety.

Control Questions

1. What nodes does a graphics card consist of, and which of them is the most complex?
2. What faults of video cards are connected with overheating?
3. How is the diagnostics of the power circuits of a video card performed?
4. What nodes does a monitor consist of, and what are their typical faults?
5. Why does the replacement of swollen capacitors often restore a monitor?

Lecture 1.2.20 Analysis and repair of the circuits of data-storage devices (hard disk drives and SSDs)

Content

The structure and operating principle of HDDs and SSDs.
The power and interface circuits of the drives.
Typical faults of HDDs and SSDs.
Diagnostics and methods of restoring drives.

Data-storage devices store the information of a computer permanently. The two main types are hard disk drives (HDD) and solid-state drives (SSD). The analysis and repair of their circuits require an understanding of their structure, the operating principle and the typical faults.

A hard disk drive stores data on rotating magnetic plates, from which the information is read and written by movable heads. The HDD consists of a hermetic block (the plates, the heads, the motor) and

a control board with a controller, memory and power circuits. The mechanical part makes the HDD sensitive to shocks.

A solid-state drive stores data in flash memory and has no moving parts. The SSD consists of flash-memory chips, a controller, a buffer (DRAM) and power circuits. The absence of mechanics makes the SSD faster and more resistant to shocks, but the flash memory has a limited number of rewrite cycles.

The power and interface circuits of the drives are located on the control board. The drives receive the supply voltages +5 V and +12 V (for the HDD) through the SATA power connector and exchange data over the SATA or NVMe interface. The faults of the power circuits lead to the inoperability of the drive.

The typical faults of HDDs are connected with the mechanics and the electronics: the wear and sticking of the heads, the jamming of the motor, bad sectors on the plates, and the faults of the control board. A characteristic sign of mechanical faults is an extraneous sound (knocking, clicking) during operation.

The typical faults of SSDs are connected with the controller and the flash memory: the failure of the controller (the drive is not detected), the depletion of the resource of the memory cells, and faults of the firmware. SSDs more often fail suddenly and completely, which makes the backing-up of data especially important.

The S.M.A.R.T. technology (self-monitoring) is built into the drives and accumulates data on their state: the number of bad sectors, the temperature, the number of switch-ons, the wear of the resource. The analysis of the S.M.A.R.T. attributes allows one to assess the state of the drive and to predict its failure.

The diagnostics of drives is performed by utilities (CrystalDiskInfo, Victoria, the manufacturers' utilities), which read the S.M.A.R.T., test the surface and the speed of reading, and detect bad sectors and degraded blocks. The deterioration of the S.M.A.R.T. attributes is a signal for replacing the drive.

The restoration of the operability of a drive depends on the nature of the fault. Software faults (logical bad sectors, errors of the file system) can be eliminated by utilities. Hardware faults (the mechanics of the HDD, the controller of the SSD) require complex repair, which is most often not advisable.

The recovery of data from a faulty drive is a separate complex task. Logical recovery is performed by programs; physical recovery (in the event of a mechanical fault of the HDD) requires special conditions and equipment. The most reliable protection against the loss of data is a backup.

When working with drives, one must observe caution: not subject the HDD to shocks, observe anti-static measures, correctly connect the power and the interface. The data on the drives is often more valuable than the drives themselves, so caution is paramount.

Thus, the analysis and repair of the circuits of data-storage devices is based on the knowledge of the structure of HDDs and SSDs, the analysis of the S.M.A.R.T., the diagnostics by utilities and the understanding of the limits of the advisability of repair and the importance of backing up data.

Control Questions

1. What is the difference in the structure and operating principle of an HDD and an SSD?
2. What faults are typical of hard disk drives and what for SSDs?
3. What is the S.M.A.R.T. technology, and how is it used in diagnostics?
4. What utilities are used to diagnose drives?
5. Why is a backup the most reliable protection against the loss of data?

Lecture 1.2.24 Detecting faults in the circuits of the sound subsystem of computers

Content

The structure and operating principle of the sound subsystem.
The signal and power circuits of the sound path.
Typical faults of the sound subsystem.
Methods of diagnosing sound faults.

The sound subsystem of a computer provides the input and output of audio signals. It may be integrated into the motherboard (an audio codec) or made as a separate sound card. The understanding of its structure and circuits allows one to diagnose the typical faults of sound.

The basis of the sound subsystem is the audio codec — a microcircuit that converts the digital sound into an analog signal (a DAC) for output to speakers and headphones, and the analog signal into a digital one (an ADC) for input from a microphone. The codec is connected to the chipset over a special interface.

The sound path includes the codec, the analog amplification and filtering circuits, the output connectors (the jacks), the input circuits of the microphone and the line input. The quality of the sound depends on the quality of these circuits and on the screening from the interference of the digital part.

The power supply of the sound subsystem is performed from the supply voltages of the board, often through additional filters to reduce interference. The analog circuits are sensitive to interference, so they are separated from the digital part and screened.

The typical faults of the sound subsystem are: the complete absence of sound, the absence of sound in one channel, noise and crackling, a low volume, and the inoperability of the microphone or the line input. These faults may be caused by the hardware, the drivers or the settings.

Before searching for a hardware fault, one must exclude software causes: check the connection of the speakers, the volume settings, the choice of the playback device, and the presence and operation of the driver of the audio codec. A significant part of the problems of sound is solved at the software level.

The hardware diagnostics of the sound subsystem includes checking the output connectors (the jacks), which often wear out and oxidize, the integrity of the contacts, and the operability of the codec. The faults of the connectors are a frequent cause of the absence of sound in a channel.

The measurement of the signals in the sound path with an oscilloscope allows one to trace where the signal disappears or is distorted. The comparison with a serviceable device and the method of substitution help to localize the faulty node.

Noise and crackling in the sound are often caused by the interference of the digital part, a bad earthing or a fault of the filtering circuits. The improvement of the screening and the earthing, and the elimination of the sources of interference reduce the noise.

If the integrated sound subsystem is faulty and the repair is impossible, the simplest solution is to install a separate sound card or to use an external USB sound adapter, which completely replaces the faulty path.

The diagnostics of the microphone and the line input is performed similarly: the checking of the connectors, the settings, the driver and the signal circuits. The recording and playback of a test signal helps to assess the operability of the input path.

Thus, the detection of faults in the circuits of the sound subsystem combines the exclusion of software causes, the checking of the connectors and the codec, the measurement of the signals and the methods of substitution and comparison, which allows the typical faults of sound to be localized and eliminated.

Control Questions

1. What does the sound subsystem of a computer consist of, and what function does the audio codec perform?
2. What is included in the sound path, and why are the analog circuits screened?

3. What are the typical faults of the sound subsystem?
4. What software causes of the absence of sound must be excluded before searching for a hardware fault?
5. How can the operability of a faulty integrated sound subsystem be restored?

Lecture 1.2.26 Analysis of the circuits of temperature monitoring and control in computers

Content

- Temperature sensors and their placement in a computer.
- The circuits of monitoring and reading the sensors.
- The mechanisms of protection against overheating.
- Software monitoring of temperatures.

The control of the temperature of the components is critically important for the reliable operation of a computer. Overheating leads to the degradation and failure of components, to throttling and to shutdowns. The circuits of temperature monitoring and control track the heating and protect the equipment.

Temperature sensors are placed in the key nodes of a computer: inside the processor and the graphics processor (built-in digital sensors), on the motherboard near the VRM and the chipset, in the drives (through the S.M.A.R.T.), and sometimes in the case. They measure the temperature in real time.

The built-in sensors of the processor (the DTS — digital thermal sensors) measure the temperature of each core and transmit the data to the registers, from which the operating system and the utilities read them. These sensors are also used by the protection mechanisms of the processor itself.

The motherboard contains a hardware-monitoring controller (the Super I/O or a separate chip) that reads the analog and digital temperature sensors, the voltages and the speed of the fans. This controller provides the data for monitoring and controls the fans.

The reading of the sensors is performed over special buses (for example, I2C/SMBus). The monitoring utilities turn to the hardware-monitoring controller and the registers of the processor and the video card to obtain the values of the temperatures, the voltages and the frequencies.

The mechanisms of protection against overheating operate at several levels. The first level is the regulation of the speed of the fans depending on the temperature. The second level is throttling — the reduction of the frequency and voltage when the threshold temperature is reached. The third level is the emergency shutdown at a critical temperature.

Thermal throttling protects the components from damage but reduces the performance. The constant occurrence of throttling indicates an insufficient efficiency of cooling and requires maintenance — the cleaning of the radiators and the replacement of the thermal paste.

The emergency shutdown (thermal shutdown) occurs when the protection mechanisms cannot keep the temperature within the limits. This is the last line of defence, which prevents the physical destruction of the components. Frequent emergency shutdowns are a serious signal of a problem of cooling.

Software monitoring of temperatures is performed by utilities (HWMonitor, HWiNFO, AIDA64, Core Temp), which display the current, minimum and maximum values of the temperatures of all the nodes. The analysis of these data allows one to assess the state of cooling and to detect the overheating nodes.

The setting of the monitoring includes the choice of the displayed sensors, the setting of the alarm thresholds and the logging of the temperatures over time. The logging is useful for detecting overheating under specific loads and for diagnosing 'floating' problems.

The correct interpretation of the readings of the sensors requires the knowledge of the permissible values for each component. The temperatures at rest and under load are compared with the recommendations of the manufacturers, which allows one to assess whether the cooling system copes with its task.

Thus, the analysis of the circuits of temperature monitoring and control includes the understanding of the placement and operation of the sensors, the circuits of their reading, the multi-level mechanisms of protection against overheating and the software monitoring, which together ensure the thermal safety of a computer.

Control Questions

1. Where are the temperature sensors placed in a computer?
2. What controller of the motherboard reads the temperatures, the voltages and the speed of the fans?
3. What levels of protection against overheating exist?
4. What does the constant occurrence of thermal throttling indicate?
5. What utilities are used for the software monitoring of temperatures?

4. PRACTICAL CLASS MATERIALS

LO 1.2 Determine the operability of computer systems at the circuit-engineering level

Practical work 1.2.3 Analysis of the types of capacitors and their application in computer devices

Objective: To form practical skills in identifying the types of capacitors, reading their markings and measuring their parameters, and in determining their role in the circuits of computer devices.

Theoretical background

In the circuits of a computer, capacitors of various types are used: ceramic capacitors for high-frequency decoupling, electrolytic and polymer capacitors for power filters, and film capacitors for signal circuits. The ability to identify the type of capacitor and to read its markings is a basic skill of diagnostics.

The marking of capacitors may be alphanumeric (on the case directly) or in code (three digits for ceramic capacitors). The parameters can be checked with a multimeter in the capacitance-measurement mode and with an ESR meter without unsoldering, with the power off.

Equipment and materials

- A set of capacitors of various types (ceramic, electrolytic, film, tantalum).
- A motherboard and a power supply for studying the placement of capacitors.
- A multimeter with a capacitance-measurement function and an ESR meter.
- A magnifying glass and a report form.

Procedure

1. Examine the proposed capacitors, determine their type by appearance and design, and enter the data in a table.
2. Read the markings of the capacitors and determine the capacitance and the rated voltage of each.
3. Measure the actual capacitance of the capacitors with a multimeter and compare it with the markings.
4. Examine the motherboard and the power supply, find the capacitors and determine in which circuits they are used (power filters, decoupling).
5. Find capacitors with signs of faults (swelling, leakage of electrolyte) and measure their ESR.

6. Draw conclusions about the conformity of the parameters of the capacitors with the markings and about the presence of faulty elements.

Tasks for independent completion

1. Compile a table of the examined capacitors indicating their type, capacitance, rated voltage and place of application.
2. Determine which capacitors on the board belong to the power filters and which to the decoupling circuits.
3. Prepare a conclusion on the technical condition of the examined capacitors.

Conclusion: As a result of the work, the student has mastered the skills of identifying the types of capacitors, reading their markings and measuring their parameters, and of determining their role in the circuits of computer devices.

Control Questions

1. What types of capacitors are used in computer devices, and where?
2. How is the markings of ceramic and electrolytic capacitors deciphered?
3. How is the capacitance and the ESR of a capacitor measured?
4. What external signs indicate a faulty electrolytic capacitor?
5. What role do capacitors play in the power filters of a computer?

Practical work 1.2.4 Detecting faults in the power supplies of computer systems

Objective: To form practical skills in diagnosing the power supplies of computer systems, measuring the output voltages and detecting typical faults.

Theoretical background

The power supply unit (PSU) converts the alternating mains voltage into the stabilized direct voltages +3.3 V, +5 V and +12 V that are required by the components of a computer. A faulty power supply is one of the most frequent causes of the failure of a computer to start and of unstable operation.

The diagnostics of a power supply includes a visual inspection, checking the start signal (Power Good), measuring the output voltages under load and without load, and checking the ripple. The measurements are performed with a multimeter, and the deviation of the voltages from the nominal value must not exceed the permissible tolerances (usually $\pm 5\%$).

Equipment and materials

- An ATX power supply for diagnostics.
- A multimeter and a power-supply tester.
- A load (a load resistor or a serviceable motherboard).
- An anti-static strap and a report form.

Procedure

1. Perform a visual inspection of the power supply: check the integrity of the case, the cables, the connectors and the fan, and look for swollen capacitors.
2. Connect the power-supply tester and check the presence of all the output voltages and the Power Good signal.
3. Start the power supply (by closing the PS_ON contact to ground) and measure the output voltages with a multimeter without a load.
4. Connect a load and again measure the output voltages, comparing them with the nominal values and the tolerances.
5. Assess the operation of the fan and the degree of heating of the power supply during operation.
6. Make a conclusion about the serviceability of the power supply and about the detected faults.

Tasks for independent completion

1. Prepare a table of the measured output voltages with an indication of the conformity with the tolerances.
2. Describe the typical faults of power supplies and their external manifestations.
3. Formulate a conclusion about the serviceability of the diagnosed power supply.

Conclusion: As a result of the work, the student has formed the skills of diagnosing power supplies, measuring the output voltages and detecting the typical faults of the power circuits of a computer.

Control Questions

1. What voltages does a computer power supply produce, and what are they intended for?
2. What does the diagnostics of a power supply include?
3. What is the Power Good signal and what is its role?
4. What deviation of the output voltages from the nominal value is considered permissible?
5. What are the typical faults of computer power supplies?

Practical work 1.2.5 Investigation of the causes of faults of the power supplies of computer systems

Objective: To form practical skills in analysing the causes of faults of power supplies, localizing the faulty nodes and determining the methods of their elimination.

Theoretical background

The faults of a power supply can be located in various nodes: the input filter and the rectifier, the high-voltage part with the switching transistors, the pulse transformer, the secondary rectifiers and filters, and the stabilization and protection circuits. Each node has its own characteristic signs of a fault.

The most frequent causes of faults are dried-out and swollen electrolytic capacitors, failed switching transistors, blown fuses as a result of a short circuit, and faults of the standby-voltage circuit. The analysis of the causes allows not only the elimination of the fault but also the prevention of its recurrence.

Equipment and materials

- Faulty power supplies for analysis.
- A multimeter and an ESR meter.
- A schematic diagram of a typical ATX power supply.
- A tool for disassembly and a report form.

Procedure

1. Study the block diagram of a pulse power supply and determine the function of each node.
2. Disassemble the faulty power supply, observing safety measures (discharge the high-voltage capacitors).
3. Perform a visual inspection and detect the components with external signs of a fault (swelling, darkening, traces of overheating).
4. Check the fuse, the switching transistors and the diodes of the rectifiers with a multimeter.
5. Measure the ESR of the electrolytic capacitors of the secondary circuits and detect the degraded ones.
6. Determine the cause of the fault, the node in which it is located, and the method of its elimination.

Tasks for independent completion

1. Compile a list of the typical causes of faults of power supplies with an indication of their external manifestations.
2. For the analysed power supply, determine the faulty node and the cause of the fault.
3. Propose a method of eliminating the detected fault and measures to prevent its recurrence.

Conclusion: As a result of the work, the student has mastered the methodology of analysing the causes of faults of power supplies, localizing the faulty nodes and determining the methods of their elimination.

Control Questions

1. What nodes does a pulse power supply consist of?
2. What are the most frequent causes of faults of power supplies?
3. How are the switching transistors and the rectifier diodes checked?
4. Why is it important to discharge the high-voltage capacitors before servicing?
5. How does the analysis of the cause of a fault help to prevent its recurrence?

Practical work 1.2.13 Investigation of the methods of diagnosing graphics adapters and monitors

Objective: To form practical skills in diagnosing graphics adapters and monitors, detecting typical faults and determining the methods of their elimination.

Theoretical background

A graphics adapter (video card) forms the image and outputs it to the monitor. Its main nodes are the graphics processor (GPU), the video memory, the power circuits (VRM), the cooling system and the output connectors. The faults of a video card are connected with overheating, the power circuits and the GPU.

A monitor displays the image. Its faults may be located in the power supply, the backlight, the matrix or the control board. The diagnostics of the 'video card – monitor' link begins with determining where the problem is located — in the source of the image or in the device that displays it.

Equipment and materials

- A graphics adapter and a monitor for diagnostics.
- A serviceable computer for substitution checks.
- A multimeter and diagnostic software (FurMark, GPU-Z).
- A report form.

Procedure

1. Perform a visual inspection of the graphics adapter: check the condition of the cooling system, the capacitors and the connectors.
2. Install the video card in a serviceable computer and check the passage of the POST, the appearance of the image and the recognition of the card by the system.
3. Run a stress test of the video card (FurMark) and monitor the temperature, the frequency and the absence of artifacts on the image.
4. Check the monitor by connecting it to a serviceable source through different interfaces (HDMI, DisplayPort) and assess the quality of the image.
5. Localize the fault by the method of substitution: determine whether the problem is in the video card, the cable or the monitor.
6. Make a conclusion about the technical condition of the graphics adapter and the monitor and about the methods of eliminating the detected faults.

Tasks for independent completion

1. Compile a list of typical faults of graphics adapters and their external manifestations.
2. Determine, by the method of substitution, where the fault of the image is located in a given case.
3. Prepare a conclusion on the technical condition of the diagnosed video card and monitor.

Conclusion: As a result of the work, the student has mastered the methods of diagnosing graphics adapters and monitors, of detecting typical faults and of localizing them by the method of substitution.

Control Questions

1. What are the main nodes of a graphics adapter?
2. What faults are typical of video cards?
3. How is the location of a fault of the image determined — in the card or the monitor?
4. What does a stress test of a video card check?
5. What faults can be located in a monitor?

Practical work 1.2.15 Analysis of conductors, dielectrics and electromagnetic phenomena in electronic circuits

Objective: To form practical skills in analysing the behaviour of conductors and dielectrics and in detecting the influence of electromagnetic phenomena in the electronic circuits of a computer.

Theoretical background

In the electronic circuits of a computer, the properties of conductors (the tracks of boards, wires) and dielectrics (the base of the board, insulation) work together. Faults are often connected with the violation of these properties: short circuits, breaks, leakage currents.

Electromagnetic phenomena manifest themselves in the operation of chokes, transformers and motors, and also in the form of interference. The analysis of these phenomena helps to understand the operation of the power circuits and to detect the sources of interference.

Equipment and materials

- A motherboard and a power supply for analysis.
- A multimeter and an oscilloscope.
- A set of conductors and dielectrics for experiments.
- A report form.

Procedure

1. Check the continuity of the conductive tracks of a board with a multimeter and detect possible breaks.
2. Measure the resistance between the power circuits and ground and detect possible short circuits or leakages.
3. Examine the inductive components (chokes, transformers) of the power supply and the VRM of the motherboard and describe their purpose.
4. Observe the signal on the choke of the VRM with an oscilloscope and analyse its shape.
5. Investigate the influence of the screening on the level of interference by comparing the signals with and without a screen.
6. Formulate conclusions about the role of conductors, dielectrics and electromagnetic phenomena in the operation of the analysed circuits.

Tasks for independent completion

1. Compile a map of the power circuits of the board indicating the conductors and the inductive components.
2. Describe the faults connected with the violation of the properties of conductors and dielectrics.
3. Explain how screening reduces the level of electromagnetic interference.

Conclusion: As a result of the work, the student has mastered the methods of analysing the behaviour of conductors and dielectrics and of detecting electromagnetic phenomena in the electronic circuits of a computer.

Control Questions

1. How is the continuity of conductors and the absence of short circuits checked?
2. What faults are connected with the violation of the properties of conductors and dielectrics?

3. What inductive components are present in the power circuits of a computer?
4. How does screening affect the level of electromagnetic interference?
5. How does the oscilloscope help to analyse the operation of the power circuits?

Practical work 1.2.17 The role of the BIOS and the methods of its diagnostics in computers

Objective: To form practical skills in working with the BIOS/UEFI, in diagnosing the problems associated with it and in restoring its operability.

Theoretical background

The BIOS (and its modern form, UEFI) is the basic firmware of a motherboard, stored in a special chip. It performs the initialization of the equipment at start-up, the power-on self-test (POST), and the transfer of control to the boot loader of the operating system.

The problems associated with the BIOS/UEFI include the loss of settings (a discharged CMOS battery), incorrect settings that prevent the start, the corruption of the firmware, and incompatibility with new equipment, requiring an update. The diagnostics determines the nature of the problem and the method of its elimination.

Equipment and materials

- A computer with access to the BIOS/UEFI.
- A CMOS battery and a multimeter for checking it.
- A flash drive with a firmware file for the update.
- A report form.

Procedure

1. Enter the BIOS/UEFI and study its main sections: information about the system, the boot settings, the parameters of the equipment.
2. Check and set the correct date and time and verify the voltage of the CMOS battery with a multimeter.
3. Study the boot settings (Boot), the boot order of the devices and the operating modes of the disks (AHCI, the security settings).
4. Perform a reset of the BIOS/UEFI settings to the default values and describe the situations in which this is necessary.
5. Study the procedure for updating the BIOS/UEFI from a flash drive (without actually updating, if it is not required) and describe its stages and risks.
6. Make a conclusion about the role of the BIOS/UEFI and about the methods of diagnosing and eliminating the associated problems.

Tasks for independent completion

1. Make a list of the main sections and settings of the BIOS/UEFI.
2. Describe the signs of a discharged CMOS battery and the procedure for replacing it.
3. Compile an algorithm for safely updating the BIOS/UEFI with an indication of the precautions.

Conclusion: As a result of the work, the student has formed the skills of working with the BIOS/UEFI, of diagnosing the associated problems and of restoring the operability of the system by resetting the settings and replacing the CMOS battery.

Control Questions

1. What functions does the BIOS/UEFI perform at the start of a computer?
2. What problems are connected with the BIOS/UEFI?
3. What are the signs of a discharged CMOS battery?
4. In which cases is it necessary to reset the BIOS/UEFI settings?
5. What precautions must be observed when updating the BIOS/UEFI?

Practical work 1.2.18 Analysis of the methods of diagnosing the BIOS and the network interfaces of a computer

Objective: To form practical skills in diagnosing problems of the BIOS/UEFI and the network interfaces of a computer and in restoring their operability.

Theoretical background

The network interface (the network card) provides the connection of a computer to a local network and to the Internet. It may be integrated into the motherboard or made as a separate card. The faults of the network interface manifest themselves as the absence of a connection, low speed or instability of the link.

The diagnostics of network interfaces combines a hardware check (the integrity of the connector, the indicators of the link, the cable) and a software check (the presence and operation of the driver, the network settings, the test of the connection with utilities).

Equipment and materials

- A computer with an integrated and (or) a separate network card.
- A network cable and a cable tester.
- Diagnostic utilities (ping, ipconfig, the device manager).
- A report form.

Procedure

1. Check the settings of the BIOS/UEFI that relate to the network interface (the enabling of the integrated network card, the Wake-on-LAN modes).
2. Examine the network connector and the link indicators and check the network cable with a cable tester.
3. Check the presence and the operation of the driver of the network card in the device manager and update it if necessary.
4. View the network settings (the IP address, the gateway) with the ipconfig utility and check the connection with the ping utility.
5. Localize the problem: determine whether it is located in the BIOS settings, the hardware, the driver or the network settings.
6. Make a conclusion about the state of the network interface and about the methods of eliminating the detected problems.

Tasks for independent completion

1. Compile a checklist for diagnosing a network interface.
2. Describe the typical faults of network cards and the methods of their detection.
3. Show how the ping and ipconfig utilities help in the diagnostics of a network connection.

Conclusion: As a result of the work, the student has mastered the methods of diagnosing the BIOS/UEFI and the network interfaces of a computer and has learned to localize and eliminate the associated problems.

Control Questions

1. What settings of the network interface are located in the BIOS/UEFI?
2. What hardware and software checks does the diagnostics of a network card include?
3. How are the ping and ipconfig utilities used?
4. What are the typical faults of network interfaces?
5. How is the location of a network problem determined?

Practical work 1.2.19 Testing and repair of network interfaces and network cards

Objective: To form practical skills in testing and repairing network interfaces and network cards, including checking the hardware and restoring the operability of the network.

Theoretical background

A network card contains a network controller (PHY/MAC), a connector (RJ-45) with built-in transformers and indicators, and power circuits. The faults may be located in the controller, the connector, the magnetic components or the power circuits.

Testing includes checking the physical link (the indicators, the cable test), the operation at different speeds (10/100/1000 Mbit/s) and the stability of the data transmission. The repair may consist in replacing the connector, restoring the soldering or replacing a separate card.

Equipment and materials

- Network cards (integrated and separate) for testing.
- A network cable, a switch and a cable tester.
- A multimeter and a soldering tool.
- A report form.

Procedure

1. Examine the network card, check the integrity of the RJ-45 connector and the condition of the link indicators.
2. Connect the card to a network through a serviceable cable and a switch and check the establishment of the link.
3. Test the operation of the card at different speeds and assess the stability of the data transmission with the ping utility in a continuous mode.
4. If there is no link, check the cable with a tester, the driver and the settings, and localize the cause of the fault.
5. Describe the procedure for repairing the typical faults (replacement of the connector, restoration of the soldering, replacement of the card).
6. Make a conclusion about the operability of the tested network cards and about the necessary repair measures.

Tasks for independent completion

1. Prepare a protocol for testing a network card with the results of checking the link and the speed.
2. Describe the procedure for replacing a faulty RJ-45 connector.
3. Formulate recommendations for choosing between repairing and replacing a network card.

Conclusion: As a result of the work, the student has formed the skills of testing and repairing network interfaces and network cards and of restoring the operability of a network connection.

Control Questions

1. What nodes does a network card consist of?
2. What does the testing of a network card include?
3. How is the cause of the absence of a link determined?
4. What typical faults of network cards can be repaired?
5. In which cases is it more advisable to replace a network card than to repair it?

Practical work 1.2.21 Investigation of faults of data-storage devices and cooling systems of computers

Objective: To form practical skills in investigating the faults of data-storage devices and cooling systems of computers and in determining the methods of their elimination.

Theoretical background

The faults of drives are detected by the analysis of the S.M.A.R.T. and by surface testing, and the faults of cooling systems by checking the rotation of the fans, the temperature of the components and the condition of the radiators and the thermal paste.

There is a close connection between cooling and the operability of the components: overheating accelerates the degradation of the processor, the video card and the drives, and causes throttling and shutdowns. Therefore the diagnostics of cooling is an important part of maintenance.

Equipment and materials

- A computer with drives and a cooling system.
- Diagnostic utilities (CrystalDiskInfo, Victoria, HWMonitor).
- A tool for cleaning and replacing the thermal paste.
- A report form.

Procedure

1. Read and analyse the S.M.A.R.T. attributes of the drives and assess their state.
2. Perform a surface test of the drive (a reading test) and detect bad sectors or degraded blocks.
3. Check the operation of the fans of the cooling system: the rotation, the speed, the absence of extraneous sounds.
4. Measure the temperature of the processor and the video card at rest and under load with monitoring utilities.
5. Assess the condition of the radiators and the thermal paste and, if necessary, clean the radiator and replace the thermal paste.
6. Make a conclusion about the state of the drives and the cooling system and about the necessary measures.

Tasks for independent completion

1. Prepare a report on the analysis of the S.M.A.R.T. of the drive with an assessment of its state.
2. Compile a temperature map of the components at rest and under load.
3. Formulate recommendations for the maintenance of the cooling system.

Conclusion: As a result of the work, the student has mastered the methods of investigating the faults of data-storage devices and cooling systems and of determining the measures for their maintenance and restoration.

Control Questions

1. How are the faults of a drive detected by the analysis of the S.M.A.R.T.?
2. What does a surface test of a drive reveal?
3. How is the operability of a cooling system checked?
4. What is the connection between overheating and the degradation of the components?
5. When is it necessary to replace the thermal paste and clean the radiator?

Practical work 1.2.22 The principles of operation and the analysis of the circuits of cooling systems of computers

Objective: To form practical skills in analysing the circuits of cooling systems of computers, in controlling the speed of the fans and in assessing the efficiency of cooling.

Theoretical background

The cooling system removes the heat from the components of a computer. It includes radiators, fans, heat pipes and (in liquid systems) a pump, a radiator and a coolant. The control of the fans is performed by the motherboard on the basis of the temperature sensors.

The fans are controlled in two ways: by voltage and by the PWM (pulse-width modulation) signal. The motherboard changes the speed of the fans depending on the temperature, which ensures a balance between cooling and noise.

Equipment and materials

- A computer with an air cooling system.
- Fans with voltage and PWM control.
- Monitoring utilities and the settings of the BIOS/UEFI.
- A multimeter and a report form.

Procedure

1. Examine the cooling system of the computer and describe its elements: the radiators, the fans, the heat pipes.
2. Determine the type of control of the fans (3-pin by voltage or 4-pin by PWM) by the number of contacts of the connector.
3. Configure the fan-control curve in the BIOS/UEFI or in a utility depending on the temperature.
4. Measure the speed of the fans and the temperature of the components at different load levels and build a dependence.
5. Assess the efficiency of cooling by comparing the temperatures with the permissible values and detect the bottlenecks.
6. Formulate conclusions and recommendations for improving the cooling and reducing the noise.

Tasks for independent completion

1. Compile a description of the cooling system of the computer with an indication of the type of control of the fans.
2. Build a graph of the dependence of the temperature of the components on the speed of the fans.
3. Propose measures to improve the efficiency of the cooling system.

Conclusion: As a result of the work, the student has formed the skills of analysing the circuits of cooling systems, of controlling the speed of the fans and of assessing the efficiency of cooling.

Control Questions

1. What elements does the air cooling system of a computer consist of?
2. What methods of controlling the speed of the fans exist?
3. How is the fan-control curve configured?
4. How is the efficiency of a cooling system assessed?
5. What measures help to improve cooling and reduce noise?

Practical work 1.2.23 Methods of diagnosing and measuring the parameters of the electrical circuits of a computer

Objective: To form practical skills in the comprehensive diagnostics and measurement of the parameters of the electrical circuits of a computer with the use of measuring instruments.

Theoretical background

The comprehensive diagnostics of the electrical circuits of a computer combines the measurement of the supply voltages, the analysis of the signals, the detection of short circuits and breaks, and the assessment of the consumed current. For this, a multimeter, an oscilloscope and a POST card are used.

The measurements are performed at the control points: the connectors of the power supply, the chokes of the VRM, the contacts of the components. The results are compared with the nominal values, which allows the faulty circuits to be detected.

Equipment and materials

- A motherboard and a power supply for diagnostics.

- A multimeter, an oscilloscope and a POST card.
- A reference table of the nominal voltages.
- A report form.

Procedure

1. Measure the output voltages of the power supply at the connectors and compare them with the nominal values.
2. Measure the supply voltages of the processor, the memory and the chipset at the control points of the board.
3. Check the absence of short circuits in the power circuits by measuring the resistance to ground with the power off.
4. Observe the clock and control signals with the oscilloscope and assess their shape and amplitude.
5. Insert a POST card and analyse the codes of the passage of the POST.
6. Compile a protocol of the measurements and make a conclusion about the state of the electrical circuits of the computer.

Tasks for independent completion

1. Prepare a protocol of the measurements of the voltages with a comparison with the nominal values.
2. Describe the procedure for detecting a short circuit in the power circuits.
3. Make a conclusion about the operability of the electrical circuits of the diagnosed board.

Conclusion: As a result of the work, the student has mastered the methods of the comprehensive diagnostics and measurement of the parameters of the electrical circuits of a computer with the use of a multimeter, an oscilloscope and a POST card.

Control Questions

1. At which control points are the supply voltages of a computer measured?
2. How is the absence of short circuits in the power circuits checked?
3. What does the oscilloscope allow one to assess?
4. How does a POST card help in the diagnostics?
5. How are the results of the measurements compared with the nominal values?

Practical work 1.2.25 Testing and analysis of the circuits of USB and other input/output ports

Objective: To form practical skills in testing and analysing the circuits of USB and other input/output ports and in detecting their faults.

Theoretical background

USB ports provide the connection of peripheral devices and their power supply. Each USB port has data lines and power lines (+5 V and ground). The faults of the ports are connected with the connectors, the power circuits and the controllers.

In addition to USB, a computer has other input/output ports: PS/2, the audio jacks, the video outputs, and the network connector. The testing of the ports includes checking the supply voltage, the data exchange and the integrity of the connectors.

Equipment and materials

- A computer with a set of USB and other ports.
- Test USB devices (a flash drive, a mouse).
- A multimeter and a USB tester.
- A report form.

Procedure

1. Examine the USB ports of the computer and check the integrity of the connectors and the absence of mechanical damage.

2. Measure the supply voltage (+5 V) at the USB ports with a multimeter or a USB tester.
3. Check the operation of the ports by connecting test devices and assessing their recognition by the system.
4. Test the data-transmission speed of the USB ports of different versions (2.0, 3.0) using a flash drive.
5. Check the operation of the other ports (audio, network, video) by connecting the corresponding devices.
6. Make a conclusion about the operability of the ports and about the detected faults.

Tasks for independent completion

1. Compile a table of the ports of the computer with the results of testing each of them.
2. Describe the typical faults of USB ports and the methods of their detection.
3. Measure and compare the data-transmission speed of USB 2.0 and USB 3.0 ports.

Conclusion: As a result of the work, the student has formed the skills of testing and analysing the circuits of USB and other input/output ports and of detecting their faults.

Control Questions

1. What lines does a USB port contain, and what voltage is supplied to it?
2. How is the supply voltage of a USB port checked?
3. What are the typical faults of USB ports?
4. How is the data-transmission speed of USB ports tested?
5. What other input/output ports does a computer have?

Practical work 1.2.27 Analysis of the typical faults of the central processor, the random-access memory and the motherboard

Objective: To form practical skills in analysing and localizing the typical faults of the central processor, the random-access memory and the motherboard.

Theoretical background

The faults of the processor, the memory and the motherboard often have similar manifestations (the failure to start, freezes, restarts), which complicates the diagnostics. The task of the technician is to localize the fault to a specific node by the method of exclusion and substitution.

The diagnostics is built on the principle of the minimal configuration and the gradual substitution of components: the board, the processor, one memory module and the power supply. The substitution of components for obviously serviceable ones allows one to determine the faulty node.

Equipment and materials

- A set of components (a board, a processor, memory) for diagnostics.
- Serviceable components for substitution.
- A POST card and a multimeter.
- A report form.

Procedure

1. Assemble a minimal configuration and check the start and the passage of the POST.
2. Use a POST card to determine the stage at which the start stops.
3. Check the memory by substituting the modules and reseating them in different slots.
4. Check the processor by inspecting the socket, the contacts and (if possible) by substitution.
5. Check the power circuits of the board by measuring the supply voltages of the processor and the memory.
6. Localize the faulty node and make a conclusion about the cause of the fault.

Tasks for independent completion

1. Compile a table of the typical faults of the processor, the memory and the board with their manifestations.
2. Describe the procedure for localizing a fault by the method of the minimal configuration.
3. Make a conclusion about the faulty node in a given case and propose a method of elimination.

Conclusion: As a result of the work, the student has mastered the methods of analysing and localizing the typical faults of the processor, the memory and the motherboard by the methods of the minimal configuration and substitution.

Control Questions

1. Why do the faults of the processor, the memory and the board have similar manifestations?
2. What is the method of the minimal configuration?
3. How is a memory fault localized?
4. How is the serviceability of the power circuits of the board checked?
5. How does a POST card help to localize a fault?

Practical work 1.2.28 The role of circuit engineering in protecting computer systems from overheating and overvoltage

Objective: To form practical skills in analysing the circuit-engineering means of protecting computer systems from overheating and overvoltage and in assessing their operability.

Theoretical background

The circuit-engineering protection of computer systems includes the means of protection against overheating (temperature monitoring, throttling, the control of the fans) and against overvoltage (filters, stabilizers, protective diodes, fuses).

The protection against overvoltage in the power supply and on the motherboard prevents the damage of the components in the event of voltage surges and faults. The understanding of these means allows one to assess the reliability of the system and to diagnose the protection circuits.

Equipment and materials

- A power supply and a motherboard for analysis.
- A multimeter and an oscilloscope.
- A schematic diagram of the protection circuits.
- A report form.

Procedure

1. Study the means of protection against overheating in the system: the sensors, the throttling, the control of the fans.
2. Find the elements of protection against overvoltage in the power supply (the input filter, the fuse, the protective circuits).
3. Examine the protective elements on the motherboard (the filters, the protective diodes at the power inputs and the ports).
4. Check the operability of the temperature protection by observing the reaction of the system to a load and heating.
5. Describe how the protection circuits prevent the damage of the components in the event of faults.
6. Make a conclusion about the operability of the protection means of the analysed system.

Tasks for independent completion

1. Compile a list of the circuit-engineering means of protection against overheating and overvoltage.
2. Describe the operation of the protection against overvoltage in a power supply.
3. Assess the operability of the temperature protection of the analysed system.

Conclusion: As a result of the work, the student has formed the skills of analysing the circuit-engineering means of protecting computer systems from overheating and overvoltage and of assessing their operability.

Control Questions

1. What circuit-engineering means protect a computer from overheating?
2. What elements protect a computer from overvoltage?
3. How does the protection against overvoltage in a power supply work?
4. How is the operability of the temperature protection checked?
5. Why is the protection of the power inputs and the ports important?

Practical work 1.2.29 Modern methods of diagnosing integrated circuits, microcontrollers and monitoring systems of computers

Objective: To form practical skills in applying modern methods of diagnosing integrated circuits, microcontrollers and monitoring systems of computers.

Theoretical background

Modern computer systems are built on integrated circuits and microcontrollers that perform the functions of control and monitoring. Their diagnostics combines hardware methods (the measurement of the power supply, the signals, the use of an oscilloscope) and software methods (reading the data of the sensors, analysing the logs).

Monitoring systems collect data on the state of the equipment (temperatures, voltages, frequencies, the speed of the fans) and provide it to the operating system and the utilities. The analysis of these data is a modern non-destructive method of diagnostics.

Equipment and materials

- A computer with monitoring systems.
- Diagnostic utilities (HWiNFO, AIDA64).
- An oscilloscope and a multimeter.
- A report form.

Procedure

1. Read the data of all the available sensors of the computer with a monitoring utility and analyse them.
2. Identify the microcontrollers and the monitoring circuits on the motherboard (the Super I/O, the hardware-monitoring controller).
3. Check the power supply and the clock signals of the microcontrollers with a multimeter and an oscilloscope.
4. Compare the readings of the software monitoring with the results of the hardware measurements.
5. Detect deviations of the parameters and localize the possible causes.
6. Make a conclusion about the state of the integrated circuits and the monitoring systems.

Tasks for independent completion

1. Prepare a report on the analysis of the data of the monitoring of a computer.
2. Compare the readings of the software and hardware diagnostics.
3. Describe the modern non-destructive methods of diagnosing integrated circuits.

Conclusion: As a result of the work, the student has mastered the modern methods of diagnosing integrated circuits, microcontrollers and monitoring systems, combining hardware and software approaches.

Control Questions

1. On what basis are modern computer systems built?

2. What hardware and software methods of diagnostics are combined?
3. What monitoring circuits are located on a motherboard?
4. How are the readings of the software and hardware diagnostics compared?
5. What are the advantages of the non-destructive methods of diagnostics?

Practical work 1.2.30 Diagnostics and repair of integrated circuits and microcontrollers in computers

Objective: To consolidate and check, in a comprehensive form, the skills of diagnosing and repairing integrated circuits and microcontrollers in computers, summing up the learning outcome on circuit engineering.

Theoretical background

This lesson generalizes the material of the learning outcome on determining the operability of computer systems at the circuit-engineering level. The student must demonstrate the ability to diagnose comprehensively, from the power supply to the integrated circuits, and to determine the methods of repair.

The repair of integrated circuits and microcontrollers requires special equipment (a soldering and an infrared station for BGA), measuring instruments and skills of working with miniature components. In many cases, the repair is reduced to the replacement of a board or a module.

Equipment and materials

- A motherboard and components for comprehensive diagnostics.
- A multimeter, an oscilloscope, a POST card and a soldering station.
- A set of situational problems on diagnostics and repair.
- A final-assessment form.

Procedure

1. Perform a comprehensive diagnostics of the proposed board: a visual inspection, the checking of the power supply, the clock signals and the POST.
2. Localize the fault to a specific node or integrated circuit using all the studied methods.
3. Determine whether the detected fault can be repaired and what equipment is required for this.
4. Describe the technology of replacing a faulty microcircuit (the preparation, the dismantling, the installation, the verification).
5. Solve the proposed situational problems on the diagnostics and repair of computer systems.
6. Make a final conclusion about the state of the board and about the method of restoring its operability.

Tasks for independent completion

1. Prepare a comprehensive diagnostic report for the proposed board.
2. Describe the technology of replacing a faulty integrated circuit.
3. Solve a situational problem on determining the cause of a fault and the method of its elimination.

Conclusion: As a result of the work, the student has demonstrated comprehensive skills of diagnosing and repairing integrated circuits and microcontrollers and the ability to determine the operability of computer systems at the circuit-engineering level.

Control Questions

1. What does the comprehensive diagnostics of a motherboard include?
2. How is a fault localized to a specific integrated circuit?
3. What equipment is required for the repair of microcircuits?
4. What is the technology of replacing a faulty microcircuit?
5. In which cases is the repair of an integrated circuit not advisable?

5. LIST OF SOFTWARE AND MULTIMEDIA SUPPORT FOR TRAINING SESSIONS

- Multimedia projector and interactive (smart) board; digital podium; white marker board.
- PowerPoint presentations for every topic of the module.
- Personal computers with Internet access and access to local networks.
- Disassembled training PC stands; sets of motherboards, power supplies, RAM modules, storage drives and graphics adapters.
- Peripheral and office equipment: printers, scanners, multifunction devices (MFPs), plotters.
- Measuring instruments: multimeters, POST-cards, power-supply and cable testers.
- Anti-static wrist straps and mats; PC assembly and repair tool kits; soldering stations.
- Hardware-diagnostic software: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo, HWMonitor, OCCT.
- Programming and scripting tools: Python 3 (with psutil, WMI, pywin32), PowerShell, Windows Subsystem for Linux.
- Consumables of office equipment (cartridges, toner, ink, ribbons) for industrial training.
- Learning Management System (LMS) for hosting materials and assignments; video lectures on PC maintenance and repair.

6. LIST OF LITERATURE

Core (fundamental) literature:

1. Phil Hughes MBE, Ed Ferrett. Introduction to Health and Safety at Work. ISBN: 978-0-415-72308-4.
2. Mark A. Friend, James P. Kohn. Fundamentals of Occupational Safety and Health. ISBN-13: 978-0-86587-171-7.
3. Charles K. Alexander, Matthew N. O. Sadiku. Fundamentals of Electric Circuits. ISBN: 978-0-07-352955-4.
4. John L. Hennessy, David A. Patterson. Computer Architecture: A Quantitative Approach (5th Edition). ISBN-13: 978-8178672663.
5. Jim Ledin. Modern Computer Architecture and Organization (2nd Edition). ISBN-13: 978-1803234519.

Additional literature:

1. Saurabh Shrivastava, Neelanjali Srivastav. Solutions Architect's Handbook (2nd Edition). ISBN-13: 978-1801816618.
2. Joseph Ingeno. Software Architect's Handbook. ISBN-13: 978-1788624060.
3. Robert C. Martin. Clean Architecture: A Craftsman's Guide to Software Structure and Design. ISBN-13: 978-0134494166.

Internet resources:

1. Computer Architecture Topics — https://en.wikipedia.org/wiki/Computer_architecture
2. Fire Protection and Prevention (OSHA) — <https://www.osha.gov/sites/default/files/2019-03/fireprotection.pdf>
3. <https://learn.microsoft.com> — official Microsoft documentation on Windows and administration.
4. <https://support.microsoft.com> — technical support and instructions for configuring hardware and software.
5. <https://docs.python.org> — documentation on automation and system administration in Python.

IV. LO 1.3 Operate and control the work of computers and peripheral devices

1. ACADEMIC CALENDAR

LO 1.3 Operate and control the work of computers and peripheral devices

Instructor: D.M. Yermekov

Specialty: 06130100 – «Software Engineering (by types)»

Total hours: 36 hours (18 lessons of 2 academic hours each)

№	Name of topics	Date of lesson	Type of activity
1	Modern peripheral devices: overview and classification		Study of new material
2	Types and purpose of the peripheral devices of a PC		Skills-improvement lesson
3	Analysis of modern peripheral devices and the areas of their application		Application and consolidation of knowledge
4	Operation of peripheral devices		Combined lesson
5	Storage devices: SSD, NVMe, external drives, NAS		Study of new material
6	Investigation of modern data-storage technologies (SSD, NVMe, NAS)		Application and consolidation of knowledge
7	Input/output. Functions and characteristics		Combined lesson
8	Comparative analysis of the input and output interfaces of computer systems		Application and consolidation of knowledge
9	The fundamentals of operating computers: power-on, power-off, initialization of the operating system		Skills-improvement lesson
10	Features of the installation, updating and removal of software in Windows and Linux		Application and consolidation of knowledge
11	Management of software: installation, updating, removal of applications in Windows/Linux		Skills-improvement lesson
12	Monitoring and control of the work of hard disk drives and the storage of data		Study of new material
13	Management of network connections and security in networks		Combined lesson
14	Methods of ensuring the security of network connections and the protection of data		Application and consolidation of knowledge
15	Effective management of the energy consumption of computer systems		Skills-improvement

			lesson
16	Technical maintenance of peripherals: diagnostics, cleaning, replacement of consumables		Combined lesson
17	Preventive maintenance and diagnostics of faults of peripheral devices		Application and consolidation of knowledge
18	Elimination of typical faults of peripheral devices and the PC		Comprehensive assessment of knowledge

2. GLOSSARY

The glossary contains the main definitions, terms and concepts that reflect the core content of this learning outcome. The terms are arranged in alphabetical order.

Term	Definition
Driver	A program that allows the operating system to control a specific device and to interact with it.
Input/output (I/O)	The exchange of data between a computer and external devices or the environment through the corresponding interfaces and ports.
NAS (Network Attached Storage)	A network data store connected to a local network and providing shared access to files.
NVMe	A high-speed protocol for accessing solid-state drives connected over the PCI Express bus.
Peripheral device	A device connected to a computer to extend its capabilities of input, output or storage of information (printer, scanner, monitor, keyboard, etc.).
Power management	A set of hardware and software means for controlling the energy consumption of a computer and its components.
Preventive maintenance	Planned maintenance carried out to prevent faults: cleaning, inspection, replacement of consumables, updating of software.
SSD (Solid-State Drive)	A data-storage device based on flash memory without moving parts.
USB (Universal Serial Bus)	A universal serial interface for connecting peripheral devices and supplying them with power.
Software management	The process of installing, updating, configuring and removing programs in an operating system.

3. LECTURE MATERIALS

LO 1.3 Operate and control the work of computers and peripheral devices

Lecture 1.3.1 Modern peripheral devices: overview and classification

Content

The concept and purpose of peripheral devices.

Classification of peripheral devices.
Input, output and storage devices.
Interfaces for connecting peripherals.

Peripheral devices are devices connected to a computer to extend its capabilities of input, output, processing and storage of information. They are not part of the central core of the computer (the processor and the memory) but are necessary for the interaction of the user and the computer with the external environment.

Peripheral devices are classified by their purpose into input devices, output devices, input/output devices and storage devices. This classification helps to understand the role of each device in the operation of a computer system.

Input devices serve to enter information and commands into a computer. These are the keyboard, the mouse, scanners, graphics tablets, web cameras, microphones, and various sensors. They convert the actions of the user or the signals of the environment into a digital form understandable to the computer.

Output devices serve to display the results of the work of a computer. These are monitors, printers, plotters, speakers, headphones and projectors. They convert the digital data into a form perceptible to the human being — an image, a printout, sound.

Input/output devices combine both functions. These are touch screens, multifunction devices (which print, scan and copy), network devices, and modems. They both receive and transmit information.

Storage devices, although sometimes considered separately, are also referred to peripherals in a broad sense. These are external hard disk drives and SSDs, flash drives, memory cards, optical drives and network stores (NAS). They serve for the long-term storage and transfer of data.

By the method of connection, peripheral devices are divided into internal (installed inside the case) and external (connected from the outside). By the type of interface — into devices with a USB, HDMI, DisplayPort, Ethernet, Bluetooth, Wi-Fi connection and others.

The interface determines the method and the speed of the exchange of data between the computer and the peripheral device. USB is the most universal interface for connecting most external devices; HDMI and DisplayPort serve for transmitting video and audio; Ethernet and Wi-Fi serve for the network connection.

Modern peripheral devices increasingly support wireless connection (Bluetooth, Wi-Fi), which increases the convenience of use but requires attention to the security and the stability of the connection. Wired connections, in turn, provide greater reliability and speed.

An important characteristic of peripheral devices is their compatibility with the operating system and the availability of drivers. A driver is a program that allows the operating system to control the device. Without a suitable driver, the device may not work or work with limited functionality.

The development of peripheral devices goes in the direction of increasing the speed of the interfaces, the universality of the connection, the wireless technologies and the integration of functions. The technician must keep track of these trends in order to competently select, connect and service the devices.

Thus, peripheral devices are a diverse class of devices that extend the capabilities of a computer. Their classification by purpose, method of connection and interface, and the understanding of their characteristics, are the basis for their correct operation and control.

Control Questions

1. What are peripheral devices, and what are they for?
2. How are peripheral devices classified by purpose?
3. What is the difference between input, output and input/output devices?

4. What interfaces are used to connect peripheral devices?
5. Why is the availability of a suitable driver important for the operation of a peripheral device?

Lecture 1.3.4 Operation of peripheral devices

Content

- The rules for operating peripheral devices.
- Connection and configuration of devices.
- Maintenance and care of peripherals.
- Typical operation problems and their elimination.

The operation of peripheral devices is the process of their correct use and maintenance to ensure reliable and long-term operation. The competent operation prolongs the service life of the devices, prevents faults and ensures the quality of their work.

The operation begins with the correct connection of the device. The type of interface, the polarity and the orientation of the connectors, and the need to install drivers are taken into account. Some devices require connection with the power off, others support a 'hot' connection (USB).

After the connection, the device is configured: the installation of drivers, the setting of the operating parameters (the resolution, the print quality, the speed), and the verification of the operability. The correct configuration ensures the optimal operation of the device.

Each type of peripheral device has its own operation features. Printers require the correct loading of paper and consumables; monitors require the setting of brightness and resolution; scanners require the correct placement of the originals; keyboards and mice require maintenance of cleanliness.

The maintenance and care of peripheral devices includes the regular cleaning of dust and dirt, the timely replacement of consumables, the updating of drivers and firmware, and the verification of the condition of the cables and the connectors. The regular maintenance prevents many faults.

The cleaning of devices is performed with means that are suitable for the specific type of device. Monitors are cleaned with soft cloths and special liquids; keyboards are cleaned of dust and crumbs; printers require the cleaning of the print heads and the rollers. The use of unsuitable means may damage the device.

The replacement of consumables (cartridges, toner, paper, batteries) is performed in a timely manner and with the use of suitable materials. The use of low-quality consumables may worsen the quality of the work and damage the device.

The updating of drivers and firmware improves the operation of devices, eliminates errors and adds new functions. The updating is performed from the official sources of the manufacturers, observing the precautions to avoid damaging the device.

The typical operation problems include the absence of a connection, the incorrect operation of the device, the deterioration of the quality of the work, and error messages. Most of these problems are solved by checking the connection, the drivers and the settings.

The diagnostics of operation problems is built from the simple to the complex: the verification of the connection and the power, the verification of the drivers and the settings, and only then the search for a hardware fault. A significant part of the problems is solved at the level of the connection and the configuration.

Observing the rules of operation, the temperature and humidity conditions, and protecting the devices from overloads and mechanical damage are the basis for their reliable operation. The technician must know and apply these rules and teach them to the users.

Thus, the operation of peripheral devices includes their correct connection and configuration, the regular maintenance and care, and the timely elimination of the typical problems, which ensures the reliable and long-term operation of the devices.

Control Questions

1. What does the operation of peripheral devices include?
2. What is taken into account when connecting and configuring a device?
3. What does the maintenance and care of peripheral devices include?
4. How is the replacement of consumables performed correctly?
5. What are the typical operation problems, and how are they eliminated?

Lecture 1.3.5 Storage devices: SSD, NVMe, external drives, NAS

Content

- Modern types of storage devices.
- SSD and the NVMe protocol.
- External drives and their interfaces.
- Network stores (NAS).

Storage devices provide the long-term storage of the data of a computer. Modern technologies offer a variety of solutions, from high-speed SSDs to network stores, each of which has its own characteristics, advantages and areas of application.

A solid-state drive (SSD) stores data in flash memory and has no moving parts. Compared with a hard disk drive, the SSD provides a much higher speed of access and data transfer, lower energy consumption, noiselessness and resistance to shocks. The SSD has become the main drive for the operating system and programs.

SSDs are made in various form factors and with various interfaces. The SATA SSD uses the SATA interface and the form factor of a 2.5-inch disk. The M.2 SSD has a compact form factor and may use either the SATA interface or the high-speed PCI Express bus.

The NVMe (Non-Volatile Memory Express) protocol is designed specifically for solid-state drives connected over the PCI Express bus. It provides a much higher speed and a lower latency than SATA, since it is optimized for the parallel operation of flash memory. NVMe drives are the fastest mass-market drives.

Hard disk drives (HDD) retain their relevance for storing large volumes of data at a low cost per gigabyte. They are slower than SSDs but offer a large capacity at an affordable price, which makes them suitable for archives, backups and media libraries.

External drives are storage devices connected from the outside, usually over the USB interface. They serve for the transfer of data, the backing-up and the extension of the storage volume. External drives may be based on either an HDD or an SSD.

The choice of the interface of an external drive affects the speed of the data transfer. USB 3.0 and higher provide a high speed sufficient for an external SSD. Modern interfaces (USB-C, Thunderbolt) provide an even higher speed.

Network stores (NAS — Network Attached Storage) are devices connected to a local network and providing shared access to files for several users and devices. A NAS contains one or several drives, often united into a RAID array for reliability.

A NAS is used for the centralized storage of data, the organization of backups, the media library and the shared access in a home or office. Modern NAS devices have their own operating system and provide many functions, including remote access.

The combination of the RAID technology in a NAS increases the reliability of the storage: the data is duplicated or distributed among several drives so that the failure of one of them does not lead to the loss of information. The choice of the RAID level determines the balance between the reliability, the speed and the usable volume.

When choosing a storage device, one takes into account the required volume, the speed, the reliability, the budget and the scenario of use. For the system and programs, an NVMe SSD is optimal; for the archive, an HDD; for the shared access, a NAS; for the transfer of data, an external drive.

Thus, modern storage devices — SSD, NVMe, external drives and NAS — offer a wide range of solutions for various tasks. The understanding of their characteristics and areas of application allows the technician to competently select and operate them.

Control Questions

1. What advantages does an SSD have over a hard disk drive?
2. What is the NVMe protocol, and why is it faster than SATA?
3. In which cases do hard disk drives retain their relevance?
4. What are external drives for, and what interfaces do they use?
5. What is a NAS, and what tasks does it solve?

Lecture 1.3.7 Input/output. Functions and characteristics

Content

The concept of input/output and its role in a computer system.

Input/output interfaces and ports.

Methods of organizing input/output.

Characteristics of input/output systems.

Input/output (I/O) is the process of the exchange of data between the central part of a computer and the external devices and the environment. The input/output system provides the connection of the computer with the outside world and is one of the most important parts of the computer architecture.

Without an input/output system, a computer would be useless, since it could not receive data and commands or display the results. The input/output combines the work of the processor, the memory, the buses, the controllers and the peripheral devices into a single system.

Input/output is performed through interfaces and ports. An interface is a set of rules and means for connecting and exchanging data. A port is a specific connector or a logical access point. The interfaces determine the speed, the protocol and the physical method of the exchange.

Modern computers have a variety of input/output interfaces: USB for universal connection, SATA and NVMe for drives, PCI Express for expansion cards, HDMI and DisplayPort for video, Ethernet and Wi-Fi for networks, and Bluetooth for wireless peripherals.

There are several methods of organizing input/output. Programmed input/output is controlled directly by the processor, which constantly polls the device. This method is simple but loads the processor and is used for slow devices.

Input/output by interruptions allows the processor to perform other work and to be distracted to the device only when it is ready, having received a signal — an interruption. This method increases the efficiency of the use of the processor and is widely applied.

Direct memory access (DMA) allows the device to exchange data with the memory directly, bypassing the processor. This is the most efficient method for fast devices (drives, network cards), since it frees the processor from the transfer of large volumes of data.

A controller is an intermediary between the processor and a peripheral device. It takes upon itself the management of the device, the buffering of the data and the coordination of the speeds. Modern controllers are integrated into the chipset or made as separate microcircuits.

The characteristics of input/output systems include the bandwidth (the speed of the data transfer), the latency (the time of the response), the number of connected devices, and the support of various protocols. These characteristics determine the performance of the input/output.

The bandwidth of the interfaces is constantly growing: from the first versions of USB with a speed of a few megabits per second to the modern versions with a speed of tens of gigabits per second. The growth of the bandwidth is necessary for working with high-speed drives and devices.

The balance of the input/output system is important for the performance of the entire computer. A fast processor and memory will not show their potential if the input/output is a bottleneck. Therefore the development of input/output interfaces goes in step with the development of the other components.

Thus, the input/output system provides the connection of a computer with the external devices and the environment through interfaces and ports, using various methods of organization (programmed, by interruptions, DMA), and its characteristics determine the efficiency of the exchange of data.

Control Questions

1. What is input/output, and what is its role in a computer system?
2. What is the difference between an interface and a port?
3. What methods of organizing input/output exist?
4. What is direct memory access (DMA), and what is its advantage?
5. What characteristics determine the performance of an input/output system?

Lecture 1.3.12 Monitoring and control of the work of hard disk drives and the storage of data

Content

The parameters of the state of drives and their monitoring.

S.M.A.R.T. technology and the analysis of its attributes.

Control of the free space and the file system.

Prevention of the loss of data.

The monitoring and control of the work of drives is an important part of the operation of a computer system. The timely detection of the deterioration of the state of a drive allows one to prevent the loss of data and to plan the replacement of the equipment.

The state of a drive is characterized by a number of parameters: the temperature, the number of bad sectors, the wear of the resource (for SSDs), the time of operation, the number of switch-ons, and the speed of reading and writing. The monitoring of these parameters allows one to assess the 'health' of the drive.

The S.M.A.R.T. (Self-Monitoring, Analysis and Reporting Technology) technology is built into modern drives and continuously collects the data on their state in the form of attributes. Each attribute has a current value, a threshold value and the worst recorded value.

The analysis of the S.M.A.R.T. attributes allows one to detect the deterioration of the state of a drive before its failure. The key attributes are the number of reallocated sectors, the number of pending sectors for reallocation, the number of read errors, the temperature, and (for SSDs) the wear of the resource.

The deterioration of the values of the critical attributes is a signal of an approaching failure of the drive. In such cases, it is necessary to urgently back up the data and to plan the replacement of the drive. The monitoring of the S.M.A.R.T. is the main method of predicting the failures of drives.

The utilities for monitoring drives (CrystalDiskInfo, the manufacturers' utilities) read the S.M.A.R.T., display the attributes in a convenient form and give a general assessment of the state of the drive ('good', 'caution', 'bad'). The regular check of the state of the drives is a part of the maintenance.

The control of the free space on the drives is also important for the stable operation of the system. The filling of the system disk to a critical level slows down the operation of the computer and may lead to errors. It is recommended to leave a sufficient amount of free space.

The control of the state of the file system includes the checking and the correction of errors with the help of the system tools (chkdsk in Windows, fsck in Linux). The errors of the file system may arise as a result of incorrect shutdowns and faults of the drive.

The defragmentation of hard disk drives (HDD) improves the speed of access to the data by ordering the fragments of the files. For SSDs, the defragmentation is not required and is even harmful; instead, the TRIM mechanism is used, which maintains the performance of the SSD.

The prevention of the loss of data is built on a combination of the monitoring of the state of the drives, the regular backing-up and the observance of the rules of operation. The most important rule is that the data must be duplicated, since any drive may fail.

The planning of the replacement of drives on the basis of the data of the monitoring allows one to avoid an unexpected loss of data. The drives with a deteriorating S.M.A.R.T. are replaced in a planned manner, transferring the data to new drives in advance.

Thus, the monitoring and control of the work of drives includes the analysis of the S.M.A.R.T., the control of the free space and the file system, the prevention of the loss of data and the planning of the replacement of the equipment, which ensures the reliability of the storage of information.

Control Questions

1. What parameters characterize the state of a drive?
2. What is the S.M.A.R.T. technology, and what attributes does it contain?
3. How does the analysis of the S.M.A.R.T. help to predict a failure?
4. Why is the control of the free space on a drive important?
5. What is the difference between the maintenance of an HDD and an SSD (defragmentation, TRIM)?

Lecture 1.3.13 Management of network connections and security in networks

Content

The basics of network connections of a computer.

The setting of network parameters.

Network security and protection against threats.

Diagnostics of network connections.

The management of network connections is an important part of the operation of a computer system, since most computers operate in networks and have access to the Internet. The competent management ensures the stable, fast and safe operation of the network.

A network connection of a computer is provided by a network adapter (wired or wireless) and the corresponding settings. The wired connection (Ethernet) provides a high speed and reliability; the wireless one (Wi-Fi) provides mobility and convenience.

The setting of the network parameters includes the IP address, the subnet mask, the gateway and the DNS servers. These parameters may be assigned automatically by the DHCP server or set manually. The correct setting ensures the connection of the computer to the network and the Internet.

The IP address uniquely identifies a device in a network. The DNS server converts the symbolic names of sites into IP addresses. The gateway provides the access of the computer to other networks

and the Internet. The understanding of these parameters is necessary for the management of the network.

Network security is the protection of a computer and the data from network threats: unauthorized access, malware, the interception of data, and network attacks. The security is ensured by a complex of hardware and software means.

A firewall controls the network traffic and blocks unwanted connections, protecting the computer from network attacks. It may be software (built into the operating system) or hardware (in a router). The correct setting of the firewall is the basis of network security.

Antivirus protection protects the computer from malware that may penetrate from the network. The regular updating of the antivirus databases and the scanning of the system are necessary for effective protection.

The encryption of data and connections protects the information from interception. The protocols HTTPS, VPN and the encryption of Wi-Fi (WPA2/WPA3) ensure the confidentiality of the data transmitted over the network. The use of unprotected connections is dangerous.

The protection of the wireless network includes the setting of a strong password, the use of modern encryption (WPA3), the hiding of the network name and the limitation of the access by the addresses of the devices. An unprotected Wi-Fi network is vulnerable to unauthorized access.

The diagnostics of network connections is performed by the system utilities: ping (the check of the availability of a node), ipconfig/ifconfig (the viewing of the settings), tracert (the tracing of the route), nslookup (the check of the DNS). These utilities help to detect and eliminate network problems.

The typical network problems are the absence of a connection, the low speed, the instability and the inaccessibility of certain resources. The diagnostics is built from the check of the physical connection and the settings to the analysis of the route and the operation of the services.

Thus, the management of network connections includes the setting of the network parameters, the provision of network security and the diagnostics of connections, which together ensure the stable and safe operation of a computer in a network.

Control Questions

1. What provides the network connection of a computer?
2. What network parameters must be set, and what is each of them for?
3. What is a firewall, and how does it protect a computer?
4. How is the security of a wireless network ensured?
5. What utilities are used for the diagnostics of network connections?

Lecture 1.3.16 Technical maintenance of peripherals: diagnostics, cleaning, replacement of consumables

Content

The concept and goals of the technical maintenance of peripherals.

Diagnostics of the state of peripheral devices.

Cleaning and care of peripherals.

Replacement of consumables.

The technical maintenance of peripheral devices is a complex of measures aimed at maintaining them in working order, preventing faults and prolonging their service life. The regular maintenance is more economical than the repair of faults that have already occurred.

The goals of the technical maintenance are the prevention of faults, the maintenance of the quality of the work of the devices, the prolongation of their service life and the provision of safety. The maintenance may be planned (according to a schedule) or carried out as needed.

The diagnostics of the state of peripheral devices is the first stage of the maintenance. It includes the visual inspection, the check of the operability, the assessment of the quality of the work and the detection of signs of wear or faults. The diagnostics determines the necessary maintenance measures.

For different types of devices, different diagnostic methods are applied. For printers, the quality of the printing and the state of the print heads are checked; for monitors, the quality of the image; for the keyboard and the mouse, the operability of all the keys and buttons; for the cooling, the rotation of the fans.

The cleaning of peripheral devices is an important part of the maintenance. Dust, dirt and the products of wear accumulate in the devices and worsen their work. Monitors are cleaned of dust and prints; keyboards are cleaned of crumbs and dust; printers require the cleaning of the print mechanism.

The cleaning is performed with means suitable for the specific type of device: soft cloths, brushes, compressed air, special liquids. The use of unsuitable means (aggressive solvents, abrasives) may damage the device.

The replacement of consumables is a regular operation of the maintenance of peripherals. Consumables are materials that are used up in the process of the work of the devices: cartridges, toner, ink, paper, ribbons, batteries. Their timely replacement ensures the quality of the work.

The replacement of consumables is performed with the use of suitable materials and observing the technology. The use of low-quality or unsuitable consumables may worsen the quality of the work, damage the device or void the warranty.

The cartridges and the toner of printers require careful handling: the observance of the storage rules, the correct installation and the timely replacement. The spilling of the toner is dangerous to health, so the replacement is performed carefully.

The maintenance of the cables and the connectors includes the check of their integrity, the cleaning of the contacts and the replacement of the damaged ones. The poor contact in the connections is a frequent cause of the unstable operation of the devices.

The keeping of the records of the maintenance (the schedule, the journal of the carried-out works, the records of the replacement of consumables) allows one to plan the maintenance, to track the state of the devices and to forecast the need for consumables and repair.

Thus, the technical maintenance of peripheral devices includes the diagnostics of their state, the cleaning and the care, and the replacement of consumables, which ensures the reliable operation and the prolongation of the service life of the devices.

Control Questions

1. What is the technical maintenance of peripheral devices, and what are its goals?
2. What does the diagnostics of the state of a peripheral device include?
3. How is the cleaning of peripheral devices performed correctly?
4. What are consumables, and why is their timely replacement important?
5. Why is it necessary to keep records of the maintenance of peripherals?

4. PRACTICAL CLASS MATERIALS

LO 1.3 Operate and control the work of computers and peripheral devices

Practical work 1.3.2 Types and purpose of the peripheral devices of a PC

Objective: To form practical skills in identifying the types of peripheral devices, determining their purpose and characteristics, and connecting them to a computer.

Theoretical background

The practical work with peripheral devices begins with the ability to identify their type, to determine the purpose and the characteristics, and to choose the correct interface for connection. Each type of device has its own connection and configuration features.

When connecting a peripheral device, the type of interface, the availability of a driver and the settings of the operating system are taken into account. After the connection, the operability of the device and the correctness of its operation are checked.

Equipment and materials

- A set of peripheral devices (keyboard, mouse, printer, scanner, web camera, headphones).
- A computer with various ports.
- Cables and connectors for connection.
- A report form.

Procedure

1. Examine the proposed peripheral devices and determine the type and the purpose of each.
2. Determine the interface for connecting each device and find the corresponding port on the computer.
3. Connect the devices to the computer and check their recognition by the operating system.
4. Install or update the drivers of the devices if necessary and check their operability.
5. Determine the main characteristics of each device (resolution, speed, type of connection).
6. Compile a table of the connected devices with their purpose and characteristics.

Tasks for independent completion

1. Compile a table of peripheral devices indicating their type, purpose, interface and characteristics.
2. Connect and configure at least three peripheral devices of different types.
3. Describe the procedure for installing a driver for a peripheral device.

Conclusion: As a result of the work, the student has formed the skills of identifying the types of peripheral devices, of determining their purpose and characteristics, and of connecting and configuring them.

Control Questions

1. What types of peripheral devices were examined in the work?
2. How is the interface for connecting a device determined?
3. What is necessary for the correct operation of a peripheral device after connection?
4. What characteristics of peripheral devices are the most important?
5. How is the operability of a connected device checked?

Practical work 1.3.3 Analysis of modern peripheral devices and the areas of their application

Objective: To form practical skills in analysing the characteristics of modern peripheral devices and in determining the optimal devices for various areas of application.

Theoretical background

The choice of a peripheral device depends on the area of its application. For office work, reliable and economical devices are needed; for design, devices with high accuracy and resolution; for gaming, devices with a high response speed. The analysis of the characteristics allows one to make the optimal choice.

The main parameters for comparison are the performance, the resolution, the speed of the interface, the energy consumption, the reliability and the cost. The comparison of devices by these parameters is the basis for a reasoned choice.

Equipment and materials

- Technical specifications of various peripheral devices.

- Access to the catalogues and reviews of manufacturers.
- A computer for working with information.
- A report form.

Procedure

1. Choose a category of peripheral devices for analysis (printers, monitors, scanners or others).
2. Select several models of devices of this category with different characteristics.
3. Compile a comparative table of the characteristics of the chosen models.
4. Determine the areas of application for which each model is optimal.
5. Formulate recommendations for choosing a device for a given task (office, design, home use).
6. Make a conclusion about the criteria for choosing a peripheral device for a specific area.

Tasks for independent completion

1. Prepare a comparative table of several models of peripheral devices of one category.
2. Determine the optimal device for an office, for design and for home use.
3. Justify the choice of a device for a given task by its characteristics.

Conclusion: As a result of the work, the student has mastered the methods of analysing the characteristics of peripheral devices and of choosing the optimal devices for various areas of application.

Control Questions

1. On what does the choice of a peripheral device depend?
2. What parameters are used to compare devices?
3. What devices are optimal for office work and which for design?
4. How is the choice of a device justified by its characteristics?
5. Why is it important to take into account the area of application when choosing a device?

Practical work 1.3.6 Investigation of modern data-storage technologies (SSD, NVMe, NAS)

Objective: To form practical skills in investigating and comparing modern data-storage technologies and in measuring their performance.

Theoretical background

The investigation of data-storage technologies includes the comparison of their characteristics, the measurement of the speed of reading and writing, and the assessment of the suitability for various tasks. For the measurement of the performance, the utilities CrystalDiskMark and the manufacturers' utilities are used.

The key parameters of the performance are the speed of sequential reading and writing, the speed of random access (IOPS), and the latency. These parameters are compared for SSD, NVMe and HDD.

Equipment and materials

- Drives of various types (HDD, SATA SSD, NVMe SSD).
- A computer for connecting and testing the drives.
- Utilities for measuring the performance (CrystalDiskMark, CrystalDiskInfo).
- A report form.

Procedure

1. Connect the drives of various types to the computer and ensure their recognition by the system.
2. Read the characteristics and the S.M.A.R.T. of each drive with the CrystalDiskInfo utility.
3. Measure the speed of sequential and random reading and writing of each drive with CrystalDiskMark.
4. Compile a comparative table of the performance of the drives.

5. Investigate the operating principle of a NAS (or describe it) and determine the areas of its application.
6. Make a conclusion about the suitability of each technology for various tasks.

Tasks for independent completion

1. Prepare a comparative table of the performance of HDD, SATA SSD and NVMe SSD.
2. Determine which drive is optimal for the system, for the archive and for the transfer of data.
3. Describe the advantages of a NAS for the organization of shared storage.

Conclusion: As a result of the work, the student has mastered the methods of investigating and comparing modern data-storage technologies and of measuring their performance.

Control Questions

1. What parameters characterize the performance of a drive?
2. How is the speed of reading and writing of a drive measured?
3. Why is an NVMe SSD faster than a SATA SSD?
4. What utility shows the S.M.A.R.T. and the characteristics of a drive?
5. For what tasks is a NAS optimal?

Practical work 1.3.8 Comparative analysis of the input and output interfaces of computer systems

Objective: To form practical skills in the comparative analysis of the input and output interfaces of computer systems by their characteristics and areas of application.

Theoretical background

Modern computers use many input/output interfaces, each of which is optimized for certain tasks. The comparative analysis by the bandwidth, the latency, the universality and the areas of application allows one to choose the optimal interface for a specific task.

The key parameters for the comparison are the bandwidth (the maximum speed), the support of power supply, the maximum length of the cable, the number of connected devices, and the type of transmitted data (universal, video, network, storage).

Equipment and materials

- A computer with various interfaces.
- Reference data on the characteristics of the interfaces.
- Cables and devices with different interfaces.
- A report form.

Procedure

1. Examine the interfaces available on the computer and determine their type and version.
2. Compile a table of the characteristics of the main interfaces (USB of various versions, SATA, NVMe, HDMI, DisplayPort, Ethernet).
3. Compare the interfaces by the bandwidth, the universality and the areas of application.
4. Determine which interface is optimal for connecting a specific device (a drive, a monitor, a network, a peripheral).
5. Measure the actual data-transmission speed over different interfaces (for example, USB 2.0 and USB 3.0).
6. Make a conclusion about the criteria for choosing an interface for various tasks.

Tasks for independent completion

1. Prepare a comparative table of the input/output interfaces by their characteristics.
2. Determine the optimal interface for connecting a drive, a monitor and a network.
3. Justify the choice of an interface for a given task.

Conclusion: As a result of the work, the student has mastered the methods of the comparative analysis of the input and output interfaces of computer systems and of choosing the optimal interface for a specific task.

Control Questions

1. What input/output interfaces are available on a modern computer?
2. What parameters are used to compare interfaces?
3. Which interface is optimal for connecting a high-speed drive?
4. What is the difference between HDMI and DisplayPort?
5. How is the choice of an interface for a specific device justified?

Practical work 1.3.9 The fundamentals of operating computers: power-on, power-off, initialization of the operating system

Objective: To form practical skills in the correct power-on and power-off of a computer and in understanding the process of the initialization of the operating system.

Theoretical background

The correct operation of a computer begins with the proper power-on and power-off. The process of the power-on includes the initialization of the equipment by the BIOS/UEFI, the power-on self-test (POST), the search for and the launch of the boot loader, and the loading of the operating system.

The incorrect power-off (the abrupt disconnection of the power) may lead to the loss of data and to errors of the file system. The correct power-off includes the completion of the work of the programs and the shutdown of the operating system, which correctly completes all the processes.

Equipment and materials

- A computer with an installed operating system.
- Access to the BIOS/UEFI.
- Tools for monitoring the loading.
- A report form.

Procedure

1. Power on the computer and observe the process of the initialization of the equipment and the passage of the POST.
2. Enter the BIOS/UEFI and study the boot settings and the order of the boot devices.
3. Observe the process of the loading of the operating system and determine its stages.
4. Study the modes of the power-off and the energy saving (shutdown, sleep, hibernation) and the difference between them.
5. Perform the correct power-off of the computer and explain why the abrupt disconnection of the power is undesirable.
6. Make a conclusion about the rules of the correct operation of a computer.

Tasks for independent completion

1. Describe the stages of the loading of a computer from the power-on to the loading of the operating system.
2. Compile a table of the modes of the power-off and the energy saving with their differences.
3. Formulate the rules of the correct power-on and power-off of a computer.

Conclusion: As a result of the work, the student has formed the skills of the correct power-on and power-off of a computer and the understanding of the process of the initialization of the operating system.

Control Questions

1. What stages does the process of the power-on of a computer include?
2. What does the BIOS/UEFI do at the start of a computer?
3. What modes of the power-off and the energy saving exist?
4. Why is the abrupt disconnection of the power of a computer undesirable?
5. What does the correct power-off of a computer include?

Practical work 1.3.10 Features of the installation, updating and removal of software in Windows and Linux

Objective: To form practical skills in installing, updating and removing software in the Windows and Linux operating systems.

Theoretical background

The management of software differs in Windows and Linux. In Windows, programs are installed from installation files (installers) or from the store; in Linux, package managers and repositories are used, which automatically resolve the dependencies.

The correct installation, updating and removal of software maintains the system in working order, frees up resources and ensures security. The incorrect removal may leave 'tails' and disrupt the operation of the system.

Equipment and materials

- Computers with the Windows and Linux operating systems (or virtual machines).
- Installation files and access to the repositories.
- Administrator rights.
- A report form.

Procedure

1. Install a program in Windows from an installation file and configure its parameters.
2. Install a program in Linux through the package manager from the repository and observe the resolution of the dependencies.
3. Update an installed program and the system in both operating systems.
4. Correctly remove a program in Windows (through the control panel) and in Linux (through the package manager).
5. Compare the processes of the management of software in Windows and Linux.
6. Make a conclusion about the features of the management of software in different operating systems.

Tasks for independent completion

1. Describe the procedure for installing, updating and removing a program in Windows.
2. Describe the procedure for installing, updating and removing a program in Linux through the package manager.
3. Compile a comparative table of the management of software in Windows and Linux.

Conclusion: As a result of the work, the student has formed the skills of installing, updating and removing software in the Windows and Linux operating systems and understands the differences between them.

Control Questions

1. How does the installation of software in Windows and in Linux differ?
2. What is a package manager, and how does it resolve dependencies?
3. Why is the correct removal of a program important?
4. How is the updating of programs and the system performed?
5. What are the advantages of repositories in Linux?

Practical work 1.3.11 Management of software: installation, updating, removal of applications in Windows/Linux

Objective: To consolidate the practical skills of the comprehensive management of software in Windows and Linux, including the management of versions and the resolution of conflicts.

Theoretical background

The comprehensive management of software includes not only the installation and removal, but also the management of the versions, the resolution of conflicts and dependencies, the maintenance of the relevance of the programs and the cleaning of the system from unnecessary software.

The maintenance of the system in an up-to-date and clean state increases its security, performance and reliability. The technician must master the tools of the management of software of both operating systems.

Equipment and materials

- Computers with Windows and Linux.
- Package managers and installation tools.
- Administrator rights.
- A report form.

Procedure

1. Make a list of the installed programs in Windows and in Linux and assess their relevance.
2. Update all the installed programs and the system to the current versions.
3. Remove unnecessary and outdated programs, cleaning the system of residual files.
4. Resolve a conflict of versions or dependencies (model the situation and describe the solution).
5. Configure the automatic updating of the programs and the system.
6. Make a conclusion about the methods of maintaining the system in an up-to-date and clean state.

Tasks for independent completion

1. Prepare a list of the installed programs with an assessment of their relevance.
2. Describe the procedure for resolving a conflict of versions or dependencies.
3. Formulate recommendations for maintaining the system in a clean and up-to-date state.

Conclusion: As a result of the work, the student has consolidated the skills of the comprehensive management of software in Windows and Linux, including the management of versions and the maintenance of the system in an up-to-date state.

Control Questions

1. What does the comprehensive management of software include?
2. How is the relevance of the installed programs assessed?
3. How is a conflict of versions or dependencies resolved?
4. Why is it necessary to clean the system of unnecessary programs?
5. How is the automatic updating of programs configured?

Practical work 1.3.14 Methods of ensuring the security of network connections and the protection of data

Objective: To form practical skills in configuring the means of ensuring the security of network connections and the protection of data.

Theoretical background

The security of network connections is ensured by a complex of measures: the setting of the firewall, the use of encryption, the protection of the wireless network, and the differentiation of access. The practical mastery of these means is necessary for the technician.

The protection of data includes encryption, access control, backing-up and antivirus protection. The combination of these measures ensures the confidentiality, integrity and availability of the data.

Equipment and materials

- A computer with access to a network.
- A router for configuring the wireless network.
- Software for protection (firewall, antivirus).
- A report form.

Procedure

1. Configure the firewall of the operating system: create rules for allowing and blocking connections.
2. Configure the protection of the wireless network: set the WPA3/WPA2 encryption and a strong password.
3. Check the operation of the antivirus protection and update its databases.
4. Configure a secure connection (HTTPS, VPN) and verify the encryption of the data.
5. Configure the differentiation of access to network resources.
6. Make a conclusion about the complex of measures for ensuring the security of network connections.

Tasks for independent completion

1. Describe the procedure for configuring the firewall of the operating system.
2. Configure the protection of a wireless network and describe the chosen settings.
3. Compile a list of measures for ensuring the security of network connections and the protection of data.

Conclusion: As a result of the work, the student has formed the skills of configuring the means of ensuring the security of network connections and the protection of data.

Control Questions

1. What measures ensure the security of network connections?
2. How is the firewall of the operating system configured?
3. What encryption is used to protect a wireless network?
4. How does antivirus protection protect against network threats?
5. What measures ensure the protection of data?

Practical work 1.3.15 Effective management of the energy consumption of computer systems

Objective: To form practical skills in configuring the energy-consumption management of computer systems and in optimizing the energy efficiency.

Theoretical background

The management of the energy consumption allows one to reduce the energy costs, to prolong the operating time of mobile devices from the battery and to reduce the heat release. The operating systems provide flexible means of the management of the power supply.

The power-supply schemes (plans) determine the behaviour of the computer: the brightness of the screen, the switching-off of the disks, the sleep modes, and the performance of the processor. The choice and the configuration of the scheme allow one to find a balance between the performance and the energy saving.

Equipment and materials

- A computer (preferably a laptop) with an operating system.
- Tools for the management of the power supply.

- Utilities for monitoring the energy consumption.
- A report form.

Procedure

1. Study the power-supply schemes available in the operating system and their parameters.
2. Configure the parameters of the power supply: the time of switching-off the screen and the disks, the sleep mode.
3. Configure the performance modes of the processor for various scenarios (maximum performance, balanced, energy saving).
4. Measure or assess the energy consumption in different modes.
5. Configure the energy-saving modes for a mobile device to prolong the battery life.
6. Make a conclusion about the methods of optimizing the energy efficiency of a computer system.

Tasks for independent completion

1. Describe the power-supply schemes and their parameters.
2. Configure an optimal power-supply scheme for a given scenario of use.
3. Formulate recommendations for prolonging the battery life of a mobile device.

Conclusion: As a result of the work, the student has formed the skills of configuring the energy-consumption management of computer systems and of optimizing the energy efficiency.

Control Questions

1. What does the management of the energy consumption of a computer allow one to achieve?
2. What is a power-supply scheme, and what does it determine?
3. What parameters of the power supply can be configured?
4. How are the performance modes of the processor configured?
5. What measures help to prolong the battery life of a mobile device?

Practical work 1.3.17 Preventive maintenance and diagnostics of faults of peripheral devices

Objective: To form practical skills in carrying out the preventive maintenance and the diagnostics of faults of peripheral devices.

Theoretical background

Preventive maintenance is the planned maintenance carried out to prevent faults. It includes the inspection, the cleaning, the check of the operability and the replacement of the worn elements before they cause a failure.

The diagnostics of faults of peripheral devices is built from the simple to the complex: the check of the connection and the power, the check of the drivers and the settings, and the search for a hardware fault. Most problems are solved at the early stages.

Equipment and materials

- A set of peripheral devices for maintenance.
- Tools and means for cleaning.
- Consumables for replacement.
- A report form.

Procedure

1. Compile a plan of the preventive maintenance for a given set of peripheral devices.
2. Carry out the visual inspection and the diagnostics of the state of the devices.
3. Perform the cleaning of the devices from dust and dirt.
4. Check the operability of the devices and detect possible faults.
5. Eliminate the detected problems (the connection, the drivers, the settings) or describe the procedure for their elimination.

6. Make a conclusion about the state of the devices and the carried-out maintenance.

Tasks for independent completion

1. Compile a plan of the preventive maintenance for a set of peripheral devices.
2. Carry out the diagnostics and the cleaning of at least two peripheral devices.
3. Describe the procedure for the diagnostics of a fault of a peripheral device.

Conclusion: As a result of the work, the student has formed the skills of carrying out the preventive maintenance and the diagnostics of faults of peripheral devices.

Control Questions

1. What is preventive maintenance, and what does it include?
2. How is the plan of the preventive maintenance compiled?
3. In what sequence is the diagnostics of a fault performed?
4. What problems are solved at the early stages of the diagnostics?
5. Why is preventive maintenance more economical than repair?

Practical work 1.3.18 Elimination of typical faults of peripheral devices and the PC

Objective: To check and consolidate, in a comprehensive form, the skills of operating, monitoring and eliminating the typical faults of computers and peripheral devices, summing up the learning outcome.

Theoretical background

This lesson generalizes the material of the learning outcome on the operation and control of the work of computers and peripheral devices. The student must demonstrate the ability to comprehensively operate the equipment, to monitor its state and to eliminate the typical faults.

The elimination of faults is built on a systematic approach: the determination of the symptom, the formulation of the hypotheses about the cause, the check of the hypotheses from the most probable, the elimination of the cause and the verification of the result.

Equipment and materials

- Computers and peripheral devices with model faults.
- Diagnostic tools and utilities.
- A set of situational problems.
- A final-assessment form.

Procedure

1. Determine the symptoms of the faults of the proposed devices.
2. Formulate the hypotheses about the causes of the faults and arrange them by the probability.
3. Check the hypotheses, beginning with the most probable, and localize the causes.
4. Eliminate the detected faults (the connection, the drivers, the settings, the replacement of elements) or describe the procedure for their elimination.
5. Solve the proposed situational problems on the diagnostics and the elimination of faults.
6. Make a final conclusion about the state of the devices and the carried-out works.

Tasks for independent completion

1. Prepare a report on the diagnostics and the elimination of the faults of the proposed devices.
2. Solve a situational problem on the elimination of a typical fault of a peripheral device.
3. Compile a memo on the typical faults of peripheral devices and the methods of their elimination.

Conclusion: As a result of the work, the student has demonstrated comprehensive skills of operating, monitoring and eliminating the typical faults of computers and peripheral devices.

Control Questions

1. What systematic approach is applied to the elimination of faults?

2. How are the hypotheses about the causes of a fault formulated and checked?
3. What are the typical faults of peripheral devices?
4. How is the result of the elimination of a fault verified?
5. Why is it important to begin the check with the most probable causes?

5. LIST OF SOFTWARE AND MULTIMEDIA SUPPORT FOR TRAINING SESSIONS

- Multimedia projector and interactive (smart) board; digital podium; white marker board.
- PowerPoint presentations for every topic of the module.
- Personal computers with Internet access and access to local networks.
- Disassembled training PC stands; sets of motherboards, power supplies, RAM modules, storage drives and graphics adapters.
- Peripheral and office equipment: printers, scanners, multifunction devices (MFPs), plotters.
- Measuring instruments: multimeters, POST-cards, power-supply and cable testers.
- Anti-static wrist straps and mats; PC assembly and repair tool kits; soldering stations.
- Hardware-diagnostic software: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo, HWMonitor, OCCT.
- Programming and scripting tools: Python 3 (with psutil, WMI, pywin32), PowerShell, Windows Subsystem for Linux.
- Consumables of office equipment (cartridges, toner, ink, ribbons) for industrial training.
- Learning Management System (LMS) for hosting materials and assignments; video lectures on PC maintenance and repair.

6. LIST OF LITERATURE

Core (fundamental) literature:

1. Phil Hughes MBE, Ed Ferrett. Introduction to Health and Safety at Work. ISBN: 978-0-415-72308-4.
2. Mark A. Friend, James P. Kohn. Fundamentals of Occupational Safety and Health. ISBN-13: 978-0-86587-171-7.
3. Charles K. Alexander, Matthew N. O. Sadiku. Fundamentals of Electric Circuits. ISBN: 978-0-07-352955-4.
4. John L. Hennessy, David A. Patterson. Computer Architecture: A Quantitative Approach (5th Edition). ISBN-13: 978-8178672663.
5. Jim Ledin. Modern Computer Architecture and Organization (2nd Edition). ISBN-13: 978-1803234519.

Additional literature:

1. Saurabh Shrivastava, Neelanjali Srivastav. Solutions Architect's Handbook (2nd Edition). ISBN-13: 978-1801816618.
2. Joseph Ingeno. Software Architect's Handbook. ISBN-13: 978-1788624060.
3. Robert C. Martin. Clean Architecture: A Craftsman's Guide to Software Structure and Design. ISBN-13: 978-0134494166.

Internet resources:

1. Computer Architecture Topics — https://en.wikipedia.org/wiki/Computer_architecture
2. Fire Protection and Prevention (OSHA) — <https://www.osha.gov/sites/default/files/2019-03/fireprotection.pdf>

3. <https://learn.microsoft.com> — official Microsoft documentation on Windows and administration.
4. <https://support.microsoft.com> — technical support and instructions for configuring hardware and software.
5. <https://docs.python.org> — documentation on automation and system administration in Python.

V. LO 1.4 Perform diagnostics and eliminate malfunctions and failures of the hardware of computing equipment

1. ACADEMIC CALENDAR

LO 1.4 Perform diagnostics and eliminate malfunctions and failures of the hardware of computing equipment

Instructor: D.M. Yermekov

Specialty: 06130100 – «Software Engineering (by types)»

Total hours: 36 hours (18 lessons of 2 academic hours each)

№	Name of topics	Date of lesson	Type of activity
1	Composition and types of the hardware of computer systems		Study of new material
2	The structure of the system unit and the purpose of its components		Combined lesson
3	Comparative analysis of the components of a modern system unit		Application and consolidation of knowledge
4	Capacitors and their application in computer equipment		Combined lesson
5	Investigation of the application of capacitors in computer equipment		Application and consolidation of knowledge
6	The fundamentals of the diagnostics and the check of the operability of computer equipment		Skills-improvement lesson
7	The use of diagnostic programs and utilities for the analysis of the hardware		Combined lesson
8	Overview of modern software tools for the diagnostics of the hardware		Application and consolidation of knowledge
9	Performing the diagnostics and the configuration of the BIOS/UEFI parameters of a computer system		Skills-improvement lesson
10	Diagnostics and configuration of the central processor of a computer system		Study of new material
11	Analysis and repair of the memory (RAM) in computer systems		Skills-improvement lesson
12	Diagnostics and repair of motherboards and the system logic		Combined lesson
13	Analysis of the typical faults of processors, the random-access memory and motherboards		Application and consolidation of knowledge
14	Assessment and repair of video cards, monitors and graphics devices		Skills-improvement

			lesson
15	Investigation of the methods of the diagnostics of data-storage devices and power supplies		Application and consolidation of knowledge
16	Diagnostics and restoration of the operability of power supplies and power supply		Combined lesson
17	Preventive maintenance of computer systems and the prolongation of the service life of equipment		Study of new material
18	Performing the check of the state of data-storage devices with the use of diagnostic utilities		Skills-improvement lesson

2. GLOSSARY

The glossary contains the main definitions, terms and concepts that reflect the core content of this learning outcome. The terms are arranged in alphabetical order.

Term	Definition
BIOS/UEFI	The basic firmware of a motherboard that initializes the equipment at start-up and transfers control to the operating-system loader.
Diagnostics	The process of determining the technical state of equipment and detecting the causes of faults.
Diagnostic utility	A program for checking, testing and obtaining information about the state of the hardware components of a computer.
Hardware	The physical components of a computer system: the system unit, the boards, the drives, the peripheral devices.
POST (Power-On Self-Test)	The self-test of the equipment performed by the motherboard at power-on; its result is signalled by codes and sound signals.
Preventive maintenance	Planned maintenance aimed at preventing faults and prolonging the service life of equipment.
System unit	The main part of a stationary computer that contains the motherboard, the processor, the memory, the drives and the power supply.
Thermal paste	A heat-conducting compound applied between a component and a radiator to improve the heat removal.
Troubleshooting	The systematic search for and elimination of the causes of the faults of equipment.
Upgrade (modernization)	The replacement or addition of the components of a computer to increase its performance or capabilities.

3. LECTURE MATERIALS

LO 1.4 Perform diagnostics and eliminate malfunctions and failures of the hardware of computing equipment

Lecture 1.4.1 Composition and types of the hardware of computer systems

Content

- The concept of hardware and its role.
- The main components of a computer system.
- The classification of hardware.
- The interaction of the components.

The hardware of a computer system is the totality of its physical components: the electronic, electrical and mechanical devices that ensure the operation of the computer. The understanding of the composition and the types of the hardware is the basis for the diagnostics and the elimination of its faults.

The central core of a computer is formed by the central processor and the random-access memory. The processor performs the calculations and controls the operation of the system; the memory stores the data and the code of the running programs. These components determine the performance of the computer.

The motherboard unites all the components of the computer, providing their interaction. It contains the processor socket, the memory slots, the chipset, the expansion slots, the controllers and the connectors. The motherboard is the basis of the architecture of the computer.

The power supply unit converts the alternating mains voltage into the direct stabilized voltages required by the components. The reliability of the power supply directly affects the stability of the operation of the entire computer.

The data-storage devices (HDD, SSD) provide the long-term storage of the information. The graphics adapter forms the image and outputs it to the monitor. These components extend and complete the functionality of the computer.

The cooling system removes the heat from the components, preventing their overheating. It includes radiators, fans and (in some systems) liquid cooling. The efficient cooling is necessary for the reliable operation of the computer.

The case unites and protects the internal components, provides their placement and the air circulation. The peripheral devices (the monitor, the keyboard, the mouse, the printer) ensure the interaction of the user with the computer.

The hardware is classified into internal (located inside the case) and external (connected from the outside); into the main (without which the computer will not work) and the additional; and into the components by their functional purpose (computing, storage, input/output).

The components of a computer interact over buses — the sets of conductors for the transfer of data, addresses and control signals. The modern buses (PCI Express, DMI) provide a high speed of the exchange between the components.

The compatibility of the components is an important factor when assembling and modernizing a computer. The processor must correspond to the socket of the motherboard; the memory must correspond to the supported standard; the power supply must provide a sufficient power. The incompatibility of the components leads to the inoperability of the system.

The knowledge of the composition and the types of the hardware allows the technician to understand how the computer works, to determine the purpose of each component and to localize the faults to a specific component during the diagnostics.

Thus, the hardware of a computer system is a totality of interacting components, each of which performs its function. The understanding of their composition, types and interaction is the foundation for the diagnostics and the elimination of the faults of the hardware.

Control Questions

1. What is the hardware of a computer system?
2. What components form the central core of a computer?

3. What function does the motherboard perform?
4. How is the hardware of a computer classified?
5. Why is the compatibility of the components important?

Lecture 1.4.2 The structure of the system unit and the purpose of its components

Content

- The structure of the system unit.
- The placement of the components in the case.
- The purpose of each component.
- The organization of the air flows.

The system unit is the main part of a stationary computer, which contains its key components. The understanding of the structure of the system unit and the placement of the components in it is necessary for the assembly, the maintenance and the diagnostics of the computer.

The case of the system unit serves for the placement and the protection of the components, the organization of the cooling and the provision of access for the maintenance. The cases differ in their dimensions (form factor), the number of bays for the drives and the capabilities of the cooling.

The motherboard is fixed to the rear or the side wall of the case and is the basis on which the other components are installed. Its form factor (ATX, microATX, mini-ITX) must correspond to the case.

The central processor is installed in the socket of the motherboard and is covered with a cooling system — a radiator with a fan or a liquid cooling system. The quality of the thermal contact between the processor and the radiator is critically important for the heat removal.

The random-access-memory modules are installed in the slots of the motherboard near the processor. The number and the volume of the modules determine the total volume of the random-access memory of the computer.

The data-storage devices (HDD, SSD) are placed in the special bays of the case or are installed directly on the motherboard (M.2 SSD). They are connected to the motherboard over the SATA or NVMe interface and to the power supply over the power cables.

The power supply unit is installed in the special compartment of the case (usually at the top or the bottom) and supplies all the components with the necessary voltages over the cables. Its power must be sufficient for all the installed components.

The graphics adapter (if it is not integrated into the processor) is installed in the PCI Express slot of the motherboard and is connected to the monitor over the video connectors. Powerful video cards require additional power from the power supply.

The fans of the case provide the through ventilation: the cool air enters from the front and the bottom, and the warm air is removed from the rear and the top. The correct organization of the air flows ensures the efficient cooling of all the components.

The placement of the components in the case takes into account the convenience of the assembly and the maintenance, the organization of the cooling, and the management of the cables. The neat laying of the cables improves the air circulation and the appearance and simplifies the maintenance.

The understanding of the placement and the connection of the components allows the technician to assemble and disassemble the computer correctly, to connect all the cables, to organize the cooling and to perform the maintenance and the diagnostics.

Thus, the system unit is an organized structure in which each component has its place and purpose, and the correct placement and connection of the components ensures the reliable operation, the cooling and the convenience of the maintenance of the computer.

Control Questions

1. What is the system unit, and what does it contain?
2. What function does the case of the system unit perform?
3. Where are the processor, the memory and the drives placed?
4. How are the air flows organized in the system unit?
5. Why is the management of the cables important?

Lecture 1.4.4 Capacitors and their application in computer equipment

Content

- The role of capacitors in computer equipment.
- Capacitors in the power circuits.
- Capacitors in the signal circuits.
- Faults of capacitors and their diagnostics.

Capacitors are one of the most widespread components of computer equipment. They are used in the power circuits, the signal circuits and the timing circuits, performing the functions of filtering, decoupling, accumulating the charge and forming delays. The understanding of their role is necessary for the diagnostics.

In the power circuits, capacitors perform the function of filtering — smoothing the ripples of the voltage after the rectification and the conversion. The electrolytic and polymer capacitors of the power filters accumulate the charge and supply it at the moments of the increase of the consumption, maintaining the stability of the voltage.

The decoupling capacitors are installed near the microcircuits and supply them with energy at the moments of the fast switching, smoothing the local drops of the voltage. Without the decoupling capacitors, the operation of the high-speed microcircuits would be unstable.

In the voltage-regulator modules (VRM) of the processor and the memory, the capacitors smooth the output voltage and ensure its stability under the changing load. The quality and the state of these capacitors directly affect the stability of the operation of the processor.

In the signal circuits, capacitors are used for the transmission of the alternating component of the signal with the blocking of the constant one (the decoupling capacitors of the signal), and also in the filters that separate the signals by the frequency. In the timing circuits, the capacitors together with the resistors form the time delays.

The electrolytic capacitors of the power circuits are the most frequent point of the faults. Over time and under the action of the heat, the electrolyte dries out, the equivalent series resistance (ESR) increases, and the capacitor ceases to perform its function of filtering, which leads to the instability of the operation.

The external signs of a faulty electrolytic capacitor are the swelling of the case (the bulging of the upper cover), the leakage of the electrolyte, and the change of the colour. The swollen capacitors of the motherboards and the power supplies are a classic cause of the faults.

The diagnostics of capacitors includes the visual inspection (the detection of the swelling and the leakage), the measurement of the capacitance with a multimeter, and the measurement of the ESR with a special meter. The increased ESR indicates the degradation of the capacitor even in the absence of the external signs.

The measurement of the ESR can be performed without unsoldering the capacitor, which speeds up the diagnostics. However, for the exact measurement of the capacitance, the capacitor often has to be unsoldered, since the parallel elements of the circuit affect the readings.

The replacement of faulty capacitors is one of the most frequent repair operations. The replacement is performed with capacitors of the same capacitance and the same or a higher rated voltage, observing the polarity for the electrolytic ones and a low ESR for the power circuits.

The prevention of the faults of capacitors consists in the provision of a good cooling (since the heat accelerates their degradation), the avoidance of the operation in the overload modes, and the timely replacement of the aging capacitors during the planned maintenance.

Thus, capacitors play a key role in the power and signal circuits of computer equipment, and their faults are a frequent cause of the instability of the operation, which makes the ability to diagnose and replace them an important skill of the technician.

Control Questions

1. What functions do capacitors perform in the power circuits?
2. What are the decoupling capacitors for?
3. Why are the electrolytic capacitors a frequent point of the faults?
4. What are the external signs of a faulty electrolytic capacitor?
5. How is the diagnostics of capacitors performed?

Lecture 1.4.7 The use of diagnostic programs and utilities for the analysis of the hardware

Content

The purpose and the types of diagnostic programs.

Programs for obtaining information about the system.

Programs for testing the components.

The interpretation of the results of the diagnostics.

The diagnostic programs and utilities are an important tool of the modern technician. They allow one to obtain detailed information about the components of a computer, to test their operability and to detect the faults without disassembling the equipment.

The diagnostic programs are divided into informational (which provide the data about the components and their state), testing (which check the operability and the stability), and monitoring (which track the parameters in real time). Each type solves its own tasks.

The informational programs (AIDA64, HWiNFO, CPU-Z, GPU-Z) read the data about the processor, the memory, the motherboard, the video card and the drives — the model, the characteristics, the current parameters. They give a complete picture of the configuration and the state of the computer.

The programs for testing the memory (MemTest86, TestMem5) write the test patterns into the memory and check the correctness of the reading, detecting the errors of the cells. The testing of the memory is performed for a long time and in several passes to detect the unstable errors.

The programs for testing the processor (Prime95, OCCT, LinX) load the processor with the calculations and check the stability and the absence of the errors under the maximum load, while monitoring the temperature. They are used to check the stability and after the modernization.

The programs for testing the drives (Victoria, CrystalDiskMark, the manufacturers' utilities) check the surface, measure the speed, read the S.M.A.R.T. and detect the bad sectors. They allow one to assess the state and the performance of the drives.

The programs for testing the video card (FurMark, 3DMark) load the graphics processor with the calculations and check the stability, the temperature and the absence of the artifacts. They are used to check the video card after the assembly or the repair.

The monitoring programs (HWMonitor, HWiNFO) track the temperatures, the voltages, the frequencies and the speed of the fans in real time. They are indispensable for the diagnostics of the problems of cooling and the power supply.

The interpretation of the results of the diagnostics requires the knowledge of the norms and the typical values. The technician compares the readings of the programs with the permissible values and detects the deviations indicating the problems. The correct interpretation determines the accuracy of the diagnosis.

The advantage of the software diagnostics is the speed, the safety and the depth: the programs obtain the information directly from the sensors and the registers of the components without the need to disassemble the computer. However, the software diagnostics complements but does not replace the hardware methods.

The combination of the software and the hardware diagnostics gives the most complete picture of the state of the equipment. The programs detect and localize the problems, and the hardware methods (the measurements, the inspection) confirm and specify the causes.

Thus, the diagnostic programs and utilities provide the technician with the powerful means of obtaining the information, the testing and the monitoring of the components, and the skill of their use and the interpretation of the results is an important part of the modern diagnostics.

Control Questions

1. Into what types are the diagnostic programs divided?
2. What programs provide the information about the components of a computer?
3. What programs are used to test the memory and the processor?
4. What is necessary for the correct interpretation of the results of the diagnostics?
5. Why does the software diagnostics complement but not replace the hardware methods?

Lecture 1.4.10 Diagnostics and configuration of the central processor of a computer system

Content

The parameters and the operating modes of the processor.

The diagnostics of the state of the processor.

The configuration of the parameters of the processor.

The cooling and the thermal mode.

The central processor is the main computing component of a computer. The diagnostics and the configuration of the processor require the understanding of its parameters, the operating modes, the methods of the check of the state and the provision of the thermal mode.

The main parameters of the processor are the number of cores and threads, the clock frequency, the volume of the cache memory, and the heat-release power (TDP). The current parameters (the frequency, the voltage, the temperature, the load) are read by the monitoring utilities.

The processor operates in various modes depending on the load: at a low load, the frequency and the voltage are reduced for the energy saving; at a high load, the frequency is increased (the Turbo modes) within the limits of the thermal package. The understanding of these modes is necessary for the diagnostics.

The diagnostics of the state of the processor includes the check of the start (the passage of the POST), the testing of the stability under the load and the monitoring of the temperature and the parameters. The instability or the errors under the load indicate a problem of the processor, its power supply or the cooling.

Before concluding that the processor is faulty, one must exclude the problems of its strapping: check the power supply (the VRM), the contacts of the socket, the correctness of the installation and the quality of the thermal contact. Often the problem is located not in the processor itself.

The configuration of the parameters of the processor is performed through the BIOS/UEFI: the frequency, the voltage, the operating modes and the power-supply limits. The default settings ensure the stable operation; the manual configuration is used for the optimization or the overclocking.

The overclocking of the processor is the increase of its frequency above the nominal one for the increase of the performance. It requires the increase of the voltage and the improved cooling and is accompanied by the risk of the instability and the reduction of the service life. The overclocking is performed consciously and with the testing of the stability.

The temperature of the processor is a critical parameter. Under the load, it must not exceed the limit value (usually about 90-100 degrees, with the start of the throttling earlier). The overheating leads to the throttling and the shutdowns.

The cooling system of the processor must correspond to its heat release. A poor thermal contact, the dried-out thermal paste or an insufficiently powerful cooler lead to the overheating and the throttling. The replacement of the thermal paste and the cleaning of the radiator restore the efficiency of the cooling.

The thermal paste between the processor and the radiator ensures the efficient heat removal. Over time, it dries out and loses its properties, which leads to the overheating. The periodic replacement of the thermal paste is a part of the maintenance of the computer.

The monitoring of the parameters of the processor during the operation (the temperature, the frequency, the voltage, the load) allows one to assess its state and the efficiency of the cooling and to detect the problems in a timely manner.

Thus, the diagnostics and the configuration of the central processor combine the understanding of its parameters and modes, the check of the state and the stability, the configuration through the BIOS/UEFI and the provision of the proper thermal mode.

Control Questions

1. What are the main parameters of the central processor?
2. In what modes does the processor operate depending on the load?
3. What does the diagnostics of the state of the processor include?
4. What must be excluded before concluding that the processor is faulty?
5. Why is the timely replacement of the thermal paste important?

Lecture 1.4.12 Diagnostics and repair of motherboards and the system logic

Content

The structure of a motherboard and the system logic.

The diagnostics of the start of a motherboard.

The typical faults of motherboards.

The methods of repairing motherboards.

The motherboard is the basic component of a computer that unites all the others. Its diagnostics and repair are among the most complex tasks, requiring the understanding of the structure, the system logic and the start sequence of the board.

The system logic (the chipset) organizes the interaction of the processor with the memory, the buses and the peripheral devices. In the modern platforms, the part of the functions is integrated into the processor, and the remaining part is performed by a single chip (the PCH). The chipset is the centre of the connections of the board.

The start of a motherboard is a strictly defined sequence: the appearance of the standby voltage, the formation of the power-on signal, the start of the power supply, the appearance of all the voltages, the clock signals, the removal of the reset, and the start of the processor with the execution of the BIOS/UEFI code.

The diagnostics of a motherboard is built on the check of each step of this sequence. The technician checks the standby voltage, the power-on, the supply voltages, the clock and the reset, gradually localizing the place where the start sequence breaks down.

The POST card is an effective tool for the diagnostics of a motherboard. It displays the codes of the passage of the POST, and the last displayed code indicates the stage at which the start stopped, which helps to localize the faulty node.

The visual inspection reveals the swollen capacitors, the traces of the overheating of the VRM, the mechanical damage, the cracks of the board, the faults of the connectors and the traces of the moisture and the oxidation. Many faults of motherboards are detected precisely visually.

The measurement of the voltages at the control points and the comparison with the nominal values reveal the faults of the power circuits. The measurement of the resistance to ground at the power circuits reveals the short circuits, which are a frequent cause of the failure of the board to start.

The typical faults of motherboards are the faults of the VRM, the BIOS/UEFI chip, the clock generator, the chipset, the connectors and the slots, the breaks of the tracks and the short circuits. Some faults are eliminated by the replacement of the components, others require complex repair.

The faults of the power circuits (the VRM) lead to the inability to start or to the instability. The diagnostics of the VRM includes the check of the supply voltages, the chokes, the transistors and the capacitors. The replacement of the faulty elements restores the operability.

The faults of the BIOS/UEFI chip (the corruption of the firmware) lead to the inability to start. They are eliminated by the reflashing of the chip with a programmer or by the replacement of the chip. The modern boards have the means of restoring the BIOS.

The repair of motherboards requires the soldering tool, the soldering and the infrared stations for the work with the BGA chips, the measuring instruments and the replacement components. The repair of the complex faults (the chipset, the multilayer tracks) is not always advisable.

Thus, the diagnostics and the repair of motherboards are based on the knowledge of their structure and start sequence, on the use of a POST card and the measurements, on the visual inspection and the localization of the faults, and on the mastery of the methods of repair.

Control Questions

1. What function does the system logic (the chipset) perform?
2. What is the start sequence of a motherboard?
3. How does a POST card help in the diagnostics?
4. What are the typical faults of motherboards?
5. How is the diagnostics of the power circuits (the VRM) performed?

Lecture 1.4.16 Diagnostics and restoration of the operability of power supplies and power supply

Content

The structure and the operating principle of a pulse power supply.

The diagnostics of the power supply.

The typical faults and the causes of their occurrence.

The methods of restoring the operability.

The power supply unit (PSU) provides the conversion of the alternating mains voltage into the direct stabilized voltages required by the components of a computer. The diagnostics and the

restoration of the operability of a power supply require the understanding of its structure and the typical faults.

A pulse power supply consists of several nodes: the input filter and the rectifier, the high-voltage part with the switching transistors, the pulse transformer, the secondary rectifiers and filters, and the stabilization and protection circuits. Each node has its own function and characteristic faults.

The input part rectifies and filters the mains voltage and charges the high-voltage capacitor. The switching transistors convert the constant voltage into a high-frequency one, which is fed to the pulse transformer. The transformer converts and isolates the voltage.

The secondary rectifiers and filters form the output voltages +3.3 V, +5 V and +12 V. The stabilization circuits maintain these voltages constant under the changing load. The protection circuits disconnect the power supply in the event of an overload or a short circuit.

The diagnostics of a power supply begins with the visual inspection: the integrity of the case, the cables, the connectors and the fan, and the detection of the swollen capacitors and the traces of the overheating. Many faults are detected at this stage.

The check of the output voltages is performed with a power-supply tester and a multimeter under the load and without the load. The deviation of the voltages from the nominal values beyond the tolerances (usually 5 percent), their absence or instability indicate a fault.

The most frequent causes of the faults of power supplies are the dried-out and swollen electrolytic capacitors, the failed switching transistors, the blown fuses as a result of a short circuit, and the faults of the standby-voltage circuit.

The dried-out capacitors of the secondary circuits increase the ripple of the output voltages, which leads to the instability of the operation of the computer. Their replacement is one of the most frequent operations of the repair of power supplies.

The faults of the switching transistors and the input rectifier most often occur as a result of the voltage surges in the mains and lead to the complete inoperability of the power supply, often with the blowing of the fuse. The repair requires the replacement of the failed elements.

The restoration of the operability of a power supply requires the observance of the safety measures: the disconnection from the mains, the discharge of the high-voltage capacitors, and the caution when working with the high voltage. The high-voltage part of a power supply is dangerous even after the disconnection.

After the repair, the power supply is necessarily checked under the load before the connection to the computer, in order to be sure of the correctness of all the output voltages and the absence of the ripple. The connection of a faulty power supply may damage the components of the computer.

Thus, the diagnostics and the restoration of the operability of power supplies are based on the understanding of their structure and the typical faults, on the diagnostics by the inspection and the measurements, and on the observance of the safety measures during the repair.

Control Questions

1. What nodes does a pulse power supply consist of?
2. What output voltages does a power supply form?
3. What does the diagnostics of a power supply include?
4. What are the most frequent causes of the faults of power supplies?
5. What safety measures must be observed when repairing a power supply?

Lecture 1.4.17 Preventive maintenance of computer systems and the prolongation of the service life of equipment

Content

The concept and the goals of the preventive maintenance.
The types and the frequency of the preventive maintenance.
The operations of the preventive maintenance.
The prolongation of the service life of equipment.

The preventive maintenance of computer systems is a complex of the planned measures aimed at preventing the faults, maintaining the operability and prolonging the service life of the equipment. The regular preventive maintenance is more economical than the repair of the faults that have already occurred.

The goals of the preventive maintenance are the prevention of the faults, the maintenance of the performance and the reliability, the prolongation of the service life of the equipment and the reduction of the costs of the repair and the downtime. The preventive maintenance is the basis of the proactive approach to the maintenance.

The preventive maintenance is divided into the planned (carried out according to a schedule at certain intervals) and the unplanned (carried out as needed). The planned preventive maintenance is the most effective, since it prevents the faults.

The frequency of the preventive maintenance depends on the operating conditions of the equipment: in the dusty and hot conditions, the maintenance is carried out more often. Usually the cleaning of the computers is carried out once every few months, and the full preventive maintenance once or twice a year.

The main operation of the preventive maintenance is the cleaning of the equipment from the dust. The dust accumulates on the radiators, the fans and the boards, worsening the cooling and causing the overheating. The cleaning is performed with compressed air, brushes and a vacuum cleaner.

The replacement of the thermal paste between the processor and the radiator is an important operation of the preventive maintenance. Over time, the thermal paste dries out and loses its properties, which leads to the overheating. The periodic replacement of the thermal paste restores the efficiency of the cooling.

The check of the state of the components includes the inspection of the capacitors, the check of the operation of the fans, the analysis of the S.M.A.R.T. of the drives and the monitoring of the temperatures. The detection of the signs of wear allows one to plan the replacement of the components before their failure.

The maintenance of the software is also a part of the preventive maintenance: the updating of the operating system and the drivers, the cleaning of the disk from the unnecessary files, the check for malware, and the defragmentation (for HDD). This maintains the performance of the system.

The check and the maintenance of the cooling system include the cleaning of the radiators and the fans, the check of their operation, the replacement of the thermal paste and the thermal pads. The efficient cooling is the key to the long service life of the components.

The keeping of the records of the preventive maintenance (the schedule, the journal of the carried-out works) allows one to plan the maintenance, to track the state of the equipment and to forecast the need for the replacement of the components and consumables.

The prolongation of the service life of the equipment is achieved by the regular preventive maintenance, the provision of the proper operating conditions (the temperature, the humidity, the cleanliness, the stable power supply), and the timely replacement of the wearing components.

Thus, the preventive maintenance of computer systems includes the cleaning, the replacement of the thermal paste, the check of the state of the components and the maintenance of the software, which prevents the faults and prolongs the service life of the equipment.

Control Questions

1. What is the preventive maintenance, and what are its goals?

2. What is the difference between the planned and the unplanned preventive maintenance?
3. What determines the frequency of the preventive maintenance?
4. What are the main operations of the preventive maintenance?
5. How is the prolongation of the service life of the equipment achieved?

4. PRACTICAL CLASS MATERIALS

LO 1.4 Perform diagnostics and eliminate malfunctions and failures of the hardware of computing equipment

Practical work 1.4.3 Comparative analysis of the components of a modern system unit

Objective: To form practical skills in the comparative analysis of the components of a modern system unit and in selecting the components for various tasks.

Theoretical background

The choice of the components of a system unit depends on the tasks: an office computer, a gaming one, a workstation. The comparative analysis of the components by their characteristics, the compatibility and the cost allows one to assemble an optimal configuration.

The compatibility of the components is a key factor: the processor must correspond to the socket of the motherboard, the memory to the supported standard, the power supply to the total power consumption, and the case to the form factor of the board.

Equipment and materials

- Technical specifications of the components (processors, motherboards, memory, drives, video cards, power supplies).
- A computer for working with information.
- Tools for assessing the compatibility.
- A report form.

Procedure

1. Choose a task for which a system unit is to be assembled (office, gaming, workstation).
2. Select the components corresponding to the task and compile their list.
3. Compile a comparative table of the characteristics of the chosen components.
4. Check the compatibility of the components (the socket, the memory standard, the form factor, the power).
5. Assess the total power consumption and select a suitable power supply.
6. Make a conclusion about the optimality of the assembled configuration for the chosen task.

Tasks for independent completion

1. Compile the configuration of a system unit for a given task with a justification of the choice of the components.
2. Check and confirm the compatibility of the chosen components.
3. Calculate the total power consumption and select a power supply with a reserve.

Conclusion: As a result of the work, the student has mastered the methods of the comparative analysis of the components of a system unit and of selecting a compatible configuration for various tasks.

Control Questions

1. On what does the choice of the components of a system unit depend?
2. What factors of the compatibility of the components must be taken into account?

3. How is the total power consumption calculated?
4. What components are critical for a gaming computer?
5. How is the optimality of the assembled configuration assessed?

Practical work 1.4.5 Investigation of the application of capacitors in computer equipment

Objective: To form practical skills in detecting and analysing the application of capacitors in the circuits of computer equipment and in diagnosing their faults.

Theoretical background

On a motherboard and in a power supply, dozens of capacitors of various types are used in the power, decoupling and signal circuits. The ability to find the capacitors, to determine their role and to check their state is an important skill of the diagnostics.

The diagnostics of capacitors is performed by the visual inspection (the swelling, the leakage) and by the measurement of the capacitance and the ESR. The faulty capacitors of the power circuits cause the instability of the operation of the equipment.

Equipment and materials

- A motherboard and a power supply for analysis.
- A multimeter and an ESR meter.
- A magnifying glass for the inspection.
- A report form.

Procedure

1. Examine the motherboard and find the capacitors in the power circuits of the processor (the VRM) and the memory.
2. Determine the role of the capacitors in various circuits (filtering, decoupling, timing).
3. Perform the visual inspection of the capacitors and detect the elements with the signs of the faults.
4. Measure the ESR of the electrolytic capacitors and compare it with the typical values.
5. Examine the capacitors of the power supply and assess their state.
6. Make a conclusion about the state of the capacitors and the necessity of their replacement.

Tasks for independent completion

1. Compile a map of the placement of the capacitors on the board with an indication of their role.
2. Detect the capacitors with the signs of the faults and measure their ESR.
3. Formulate a conclusion about the state of the capacitors of the analysed equipment.

Conclusion: As a result of the work, the student has mastered the methods of detecting and analysing the application of capacitors in computer equipment and of diagnosing their faults.

Control Questions

1. In what circuits of a motherboard are capacitors used?
2. How is the role of a capacitor in a circuit determined?
3. What signs indicate a faulty capacitor?
4. How is the ESR of a capacitor measured?
5. When is it necessary to replace a capacitor?

Practical work 1.4.6 The fundamentals of the diagnostics and the check of the operability of computer equipment

Objective: To form practical skills in carrying out the basic diagnostics and the check of the operability of computer equipment.

Theoretical background

The diagnostics of computer equipment is the process of determining its technical state and the detection of the causes of the faults. It is built on a systematic approach: from the check of the obvious causes to the in-depth analysis of the components.

The basic diagnostics includes the visual inspection, the check of the connections and the power, the analysis of the symptoms, the passage of the POST and the use of the diagnostic utilities. Most problems are detected at these stages.

Equipment and materials

- A computer for the diagnostics.
- Diagnostic utilities (AIDA64, HWiNFO).
- A multimeter and a POST card.
- A report form.

Procedure

1. Perform the visual inspection of the computer: the integrity of the components, the connections, the signs of overheating and the faults of the capacitors.
2. Check the connection and the power of the components and the operability of the power supply.
3. Power on the computer and observe the passage of the POST and the loading of the system.
4. Run the diagnostic utilities and obtain the information about the components and their state.
5. Analyse the symptoms of the problems and formulate the hypotheses about the causes.
6. Make a conclusion about the technical state of the computer and the detected problems.

Tasks for independent completion

1. Compile a checklist of the basic diagnostics of a computer.
2. Carry out the diagnostics of a computer and prepare a report on its state.
3. Describe the sequence of the actions of the diagnostics from the simple to the complex.

Conclusion: As a result of the work, the student has formed the skills of carrying out the basic diagnostics and the check of the operability of computer equipment.

Control Questions

1. What is the diagnostics of computer equipment?
2. What does the basic diagnostics include?
3. Why is the diagnostics built from the simple to the complex?
4. What utilities are used for the diagnostics?
5. What does the visual inspection of a computer reveal?

Practical work 1.4.8 Overview of modern software tools for the diagnostics of the hardware

Objective: To form practical skills in working with modern software tools for the diagnostics of the hardware and in analysing the obtained data.

Theoretical background

Modern software tools for the diagnostics provide a wide range of capabilities: from obtaining the information about the configuration to the deep testing of the stability. The technician must master the main tools and be able to choose the right one for the task.

The analysis of the data of the diagnostics includes the comparison of the obtained parameters with the norms, the detection of the deviations and the formulation of the conclusions about the state of the equipment.

Equipment and materials

- A computer with a set of diagnostic programs.
- Internet access for downloading the utilities.

- Reference data on the norms of the parameters.
- A report form.

Procedure

1. Install and run the informational utility (AIDA64 or HWiNFO) and obtain the data about all the components.
2. Run the monitoring of the temperatures, the voltages and the frequencies and analyse the values.
3. Perform the testing of the memory with the MemTest86 utility (or the built-in tool) and assess the result.
4. Perform the check of the drive with the CrystalDiskInfo utility and analyse the S.M.A.R.T.
5. Compile a summary report on the state of the computer on the basis of the data of all the utilities.
6. Make a conclusion about the technical state of the equipment and the detected problems.

Tasks for independent completion

1. Prepare a report on the configuration and the state of the computer on the basis of the diagnostic utilities.
2. Analyse the data of the monitoring and detect the possible deviations.
3. Compile a list of the recommended diagnostic utilities with an indication of their purpose.

Conclusion: As a result of the work, the student has mastered the modern software tools for the diagnostics of the hardware and the methods of analysing the obtained data.

Control Questions

1. What software tools are used for the diagnostics of the hardware?
2. How is the monitoring of the temperatures and the voltages performed?
3. How are the results of the testing of the memory analysed?
4. What does the analysis of the S.M.A.R.T. of a drive give?
5. How is a summary report on the state of a computer compiled?

Practical work 1.4.9 Performing the diagnostics and the configuration of the BIOS/UEFI parameters of a computer system

Objective: To form practical skills in the diagnostics and the configuration of the BIOS/UEFI parameters of a computer system.

Theoretical background

The BIOS/UEFI is the basic firmware of a motherboard that initializes the equipment and provides its configuration. Through the BIOS/UEFI, one can obtain the information about the equipment, configure its parameters and diagnose the problems of the start.

The main sections of the BIOS/UEFI: the information about the system, the boot settings (Boot), the operating modes of the disks, the parameters of the memory and the processor, the monitoring of the temperatures and the voltages, and the security settings.

Equipment and materials

- A computer with access to the BIOS/UEFI.
- Documentation of the motherboard.
- A flash drive for the boot settings.
- A report form.

Procedure

1. Enter the BIOS/UEFI and study its main sections and their purpose.
2. Obtain the information about the equipment (the processor, the memory, the drives) from the BIOS/UEFI.
3. Configure the order of the boot devices and the operating mode of the disks (AHCI).

4. Study the monitoring of the temperatures, the voltages and the speed of the fans in the BIOS/UEFI.
5. Configure the fan-control curve and the energy-saving parameters.
6. Make a conclusion about the capabilities of the diagnostics and the configuration through the BIOS/UEFI.

Tasks for independent completion

1. Describe the main sections of the BIOS/UEFI and their purpose.
2. Configure the order of the boot and the operating mode of the disks.
3. Configure the monitoring and the management of the cooling in the BIOS/UEFI.

Conclusion: As a result of the work, the student has formed the skills of the diagnostics and the configuration of the BIOS/UEFI parameters of a computer system.

Control Questions

1. What functions does the BIOS/UEFI perform?
2. What are the main sections of the BIOS/UEFI?
3. How is the order of the boot devices configured?
4. What monitoring is available in the BIOS/UEFI?
5. How is the management of the cooling configured in the BIOS/UEFI?

Practical work 1.4.11 Analysis and repair of the memory (RAM) in computer systems

Objective: To form practical skills in analysing the state of the random-access memory, in testing it and in eliminating the typical problems.

Theoretical background

The random-access memory (RAM) stores the data and the code of the running programs. Its faults manifest themselves as the failure to start, the 'blue screens', the freezes and the errors. The diagnostics of the memory is performed by software (MemTest86) and by the method of exclusion.

The problems of the memory may be caused by the modules themselves, the poor contact in the slots, the incompatibility, or the incorrect settings in the BIOS/UEFI. The localization is performed by the check of the modules one by one in the different slots.

Equipment and materials

- A computer with the random-access memory.
- Memory modules for the substitution.
- The MemTest86 utility on a flash drive.
- A report form.

Procedure

1. Check the correct installation of the memory modules in the slots and the cleanliness of the contacts.
2. Run the testing of the memory with the MemTest86 utility in several passes.
3. If errors are detected, localize the faulty module by the method of exclusion (check the modules one by one).
4. Check the operation of the modules in the different slots and detect the possible problems of the slots.
5. Study the settings of the memory in the BIOS/UEFI and assess their influence on the stability.
6. Make a conclusion about the state of the memory and the methods of eliminating the problems.

Tasks for independent completion

1. Carry out the testing of the memory and prepare a report on its result.
2. Describe the procedure for localizing a faulty memory module.
3. Formulate the recommendations for eliminating the problems of the memory.

Conclusion: As a result of the work, the student has formed the skills of analysing the state of the random-access memory, of testing it and of eliminating the typical problems.

Control Questions

1. How do the faults of the random-access memory manifest themselves?
2. What utility is used to test the memory, and why in several passes?
3. How is a faulty memory module localized?
4. What may cause the problems of the memory besides the modules themselves?
5. How do the settings of the memory in the BIOS/UEFI affect the stability?

Practical work 1.4.13 Analysis of the typical faults of processors, the random-access memory and motherboards

Objective: To form practical skills in analysing and localizing the typical faults of processors, the random-access memory and motherboards by the methods of the minimal configuration and substitution.

Theoretical background

The faults of the processor, the memory and the motherboard often have similar manifestations (the failure to start, the freezes, the restarts). The task of the technician is to localize the fault to a specific component by the methods of exclusion and substitution.

The method of the minimal configuration consists in the assembly of the system from the minimum set of the components (the board, the processor, one memory module, the power supply) and the gradual connection of the others, which allows the faulty component to be detected.

Equipment and materials

- A set of the components for the diagnostics.
- Serviceable components for the substitution.
- A POST card and a multimeter.
- A report form.

Procedure

1. Assemble the minimal configuration and check the start and the passage of the POST.
2. Use a POST card to determine the stage at which the start stops.
3. Check the memory by the substitution of the modules and the reseating in the different slots.
4. Check the processor by the inspection of the socket and the contacts and (if possible) by the substitution.
5. Check the power circuits of the board by the measurement of the supply voltages.
6. Localize the faulty component and make a conclusion about the cause of the fault.

Tasks for independent completion

1. Compile a table of the typical faults of the processor, the memory and the board with their manifestations.
2. Localize the faulty component in the given case by the method of the minimal configuration.
3. Propose a method of eliminating the detected fault.

Conclusion: As a result of the work, the student has mastered the methods of analysing and localizing the typical faults of processors, the random-access memory and motherboards.

Control Questions

1. Why do the faults of the processor, the memory and the board have similar manifestations?
2. What is the method of the minimal configuration?
3. How is a memory fault localized?

4. How does a POST card help in the localization of a fault?
5. How is the serviceability of the power circuits of a board checked?

Practical work 1.4.14 Assessment and repair of video cards, monitors and graphics devices

Objective: To form practical skills in assessing the state and repairing video cards, monitors and graphics devices.

Theoretical background

The graphics subsystem includes the video card and the monitor. The faults of the video card are connected with the overheating, the power circuits and the GPU; the faults of the monitor are connected with the power supply, the backlight and the matrix. The diagnostics determines where the problem is located.

The assessment of the state includes the visual inspection, the testing under the load, the monitoring of the temperature and the detection of the artifacts. The repair may consist in the maintenance of the cooling, the replacement of the capacitors or more complex operations.

Equipment and materials

- A video card and a monitor for the diagnostics.
- A serviceable computer for the substitution checks.
- Tools for the maintenance and the testing utilities.
- A report form.

Procedure

1. Perform the visual inspection of the video card and assess the state of the cooling system and the capacitors.
2. Test the video card under the load (FurMark) and monitor the temperature and the artifacts.
3. Perform the maintenance of the cooling of the video card: the cleaning of the radiator and the replacement of the thermal paste.
4. Check the monitor by the connection to a serviceable source through the different interfaces.
5. Localize the fault by the method of substitution and determine the faulty node.
6. Make a conclusion about the state of the graphics devices and the methods of eliminating the faults.

Tasks for independent completion

1. Carry out the diagnostics and the maintenance of a video card.
2. Determine the location of a fault of the image by the method of substitution.
3. Describe the procedure for repairing the typical faults of a monitor.

Conclusion: As a result of the work, the student has formed the skills of assessing the state and repairing video cards, monitors and graphics devices.

Control Questions

1. What faults are typical of video cards?
2. How is the maintenance of the cooling of a video card performed?
3. How is the location of a fault of the image determined?
4. What faults are typical of monitors?
5. What does a stress test of a video card check?

Practical work 1.4.15 Investigation of the methods of the diagnostics of data-storage devices and power supplies

Objective: To form practical skills in the diagnostics of data-storage devices and power supplies and in detecting their faults.

Theoretical background

The diagnostics of drives is performed by the analysis of the S.M.A.R.T. and by the surface testing; the diagnostics of power supplies is performed by the measurement of the output voltages under the load and the visual inspection. Both components are critical for the operability of a computer.

The faults of a power supply lead to the failure to start, the instability and the restarts; the faults of a drive lead to the loss of data and the slow operation. The timely diagnostics allows one to prevent these problems.

Equipment and materials

- Drives and power supplies for the diagnostics.
- A multimeter and a power-supply tester.
- Utilities for the diagnostics of the drives.
- A report form.

Procedure

1. Read and analyse the S.M.A.R.T. of the drive and assess its state.
2. Perform a surface test of the drive and detect the bad sectors.
3. Perform the visual inspection of the power supply and detect the swollen capacitors.
4. Measure the output voltages of the power supply under the load and compare them with the nominal values.
5. Assess the operation of the fan of the power supply and the degree of its heating.
6. Make a conclusion about the state of the drive and the power supply.

Tasks for independent completion

1. Prepare a report on the analysis of the S.M.A.R.T. of the drive.
2. Prepare a protocol of the measurement of the output voltages of the power supply.
3. Formulate a conclusion about the state of the diagnosed components.

Conclusion: As a result of the work, the student has mastered the methods of the diagnostics of data-storage devices and power supplies and the detection of their faults.

Control Questions

1. How is the state of a drive assessed by the S.M.A.R.T.?
2. What does a surface test of a drive reveal?
3. How is the diagnostics of a power supply performed?
4. What deviation of the voltages of a power supply is permissible?
5. What faults of a power supply are detected by the visual inspection?

Practical work 1.4.18 Performing the check of the state of data-storage devices with the use of diagnostic utilities

Objective: To consolidate, in a comprehensive form, the skills of the diagnostics of the hardware on the example of the check of the state of data-storage devices, summing up the learning outcome.

Theoretical background

This lesson generalizes the material of the learning outcome on the diagnostics and the elimination of the faults of the hardware. The check of the state of the drives demonstrates the comprehensive application of the diagnostic utilities and the analysis of the obtained data.

The diagnostics of the drives includes the reading and the analysis of the S.M.A.R.T., the surface testing, the measurement of the speed and the check of the file system. The comprehensive assessment allows one to forecast the failures and to plan the maintenance.

Equipment and materials

- Drives of various types for the check.

- Diagnostic utilities (CrystalDiskInfo, Victoria, CrystalDiskMark).
- A computer for the connection of the drives.
- A final-assessment form.

Procedure

1. Connect the drives and read the S.M.A.R.T. of each of them with the CrystalDiskInfo utility.
2. Analyse the critical attributes of the S.M.A.R.T. and assess the state of the drives.
3. Perform a surface test of the drive with the Victoria utility and detect the bad sectors.
4. Measure the speed of reading and writing of the drives with the CrystalDiskMark utility.
5. Check the file system of the drive for errors with the system tool.
6. Make a comprehensive conclusion about the state of the drives and the recommendations for their maintenance and replacement.

Tasks for independent completion

1. Prepare a comprehensive report on the state of the checked drives.
2. Forecast the failure of a drive on the basis of the analysis of the S.M.A.R.T.
3. Formulate the recommendations for the maintenance and the timely replacement of the drives.

Conclusion: As a result of the work, the student has consolidated the comprehensive skills of the diagnostics of the hardware on the example of the check of the state of data-storage devices and the forecasting of their failures.

Control Questions

1. What does the comprehensive diagnostics of a drive include?
2. What critical attributes of the S.M.A.R.T. are analysed?
3. How is the surface test of a drive performed?
4. How is the failure of a drive forecast?
5. What recommendations are formulated by the results of the diagnostics?

5. LIST OF SOFTWARE AND MULTIMEDIA SUPPORT FOR TRAINING SESSIONS

- Multimedia projector and interactive (smart) board; digital podium; white marker board.
- PowerPoint presentations for every topic of the module.
- Personal computers with Internet access and access to local networks.
- Disassembled training PC stands; sets of motherboards, power supplies, RAM modules, storage drives and graphics adapters.
- Peripheral and office equipment: printers, scanners, multifunction devices (MFPs), plotters.
- Measuring instruments: multimeters, POST-cards, power-supply and cable testers.
- Anti-static wrist straps and mats; PC assembly and repair tool kits; soldering stations.
- Hardware-diagnostic software: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo, HWMonitor, OCCT.
- Programming and scripting tools: Python 3 (with psutil, WMI, pywin32), PowerShell, Windows Subsystem for Linux.
- Consumables of office equipment (cartridges, toner, ink, ribbons) for industrial training.
- Learning Management System (LMS) for hosting materials and assignments; video lectures on PC maintenance and repair.

6. LIST OF LITERATURE

Core (fundamental) literature:

1. Phil Hughes MBE, Ed Ferrett. Introduction to Health and Safety at Work. ISBN: 978-0-415-72308-4.
2. Mark A. Friend, James P. Kohn. Fundamentals of Occupational Safety and Health. ISBN-13: 978-0-86587-171-7.
3. Charles K. Alexander, Matthew N. O. Sadiku. Fundamentals of Electric Circuits. ISBN: 978-0-07-352955-4.
4. John L. Hennessy, David A. Patterson. Computer Architecture: A Quantitative Approach (5th Edition). ISBN-13: 978-8178672663.
5. Jim Ledin. Modern Computer Architecture and Organization (2nd Edition). ISBN-13: 978-1803234519.

Additional literature:

1. Saurabh Shrivastava, Neelanjali Srivastav. Solutions Architect's Handbook (2nd Edition). ISBN-13: 978-1801816618.
2. Joseph Ingeno. Software Architect's Handbook. ISBN-13: 978-1788624060.
3. Robert C. Martin. Clean Architecture: A Craftsman's Guide to Software Structure and Design. ISBN-13: 978-0134494166.

Internet resources:

1. Computer Architecture Topics — https://en.wikipedia.org/wiki/Computer_architecture
2. Fire Protection and Prevention (OSHA) — <https://www.osha.gov/sites/default/files/2019-03/fireprotection.pdf>
3. <https://learn.microsoft.com> — official Microsoft documentation on Windows and administration.
4. <https://support.microsoft.com> — technical support and instructions for configuring hardware and software.
5. <https://docs.python.org> — documentation on automation and system administration in Python.

VI. LO 1.5 Restore equipment after hardware and software failures and prevent them

1. ACADEMIC CALENDAR

LO 1.5 Restore equipment after hardware and software failures and prevent them

Instructor: D.M. Yermekov

Specialty: 06130100 – «Software Engineering (by types)»

Total hours: 90 hours (45 lessons of 2 academic hours each)

№	Name of topics	Date of lesson	Type of activity
1	The personal computer and its hardware part		Study of new material
2	Elimination of hardware failures of a personal computer		Combined lesson
3	External devices of a personal computer		Skills-improvement lesson
4	Investigation of the types of external devices and the connection interfaces of a personal computer		Application and consolidation of knowledge
5	Cables and their types		Skills-improvement lesson
6	Utilities for work		Study of new material
7	Analysis of the utilities for the diagnostics and the monitoring of computer systems		Application and consolidation of knowledge
8	The fundamentals of the restoration and the prevention of failures in computer equipment		Study of new material
9	Preparation of a brief report on the application of WMI for obtaining information about the hardware part of a computer		Application and consolidation of knowledge
10	Backing-up of data and the restoration of information after failures		Study of new material
11	Investigation of the methods of backing-up and the restoration of information after failures		Application and consolidation of knowledge
12	Proactive maintenance of the hardware for the prevention of failures		Study of new material
13	Comprehensive analysis of the state of the operating system and the hardware components of a PC		Application and consolidation of knowledge
14	Restoration of operating systems after failures and critical errors		Skills-improvement lesson
15	Protection against malware and viruses: methods of		Combined lesson

	prevention and removal		
16	Analysis of modern methods of protection against malware and viruses		Application and consolidation of knowledge
17	Restoration of the operation of network systems and connections after failures		Skills-improvement lesson
18	Work with backup systems and recovery plans in emergency situations (Disaster Recovery)		Study of new material
19	Development of scenarios for the restoration of the operability of a computer system		Application and consolidation of knowledge
20	Application of WMI for obtaining information about the hardware part of a computer		Study of new material
21	Comprehensive analysis of the state of the operating system and the hardware components of a PC		Application and consolidation of knowledge
22	Development of scenarios for the restoration of the operability of a computer system		Combined lesson
23	Creation of a script for the automatic diagnostics of a computer in Python		Comprehensive application of knowledge
24	Investigation of the capabilities of PowerShell for the diagnostics and the restoration of network connections		Application and consolidation of knowledge
25	Performing the diagnostics and the restoration of network connections with the use of PowerShell		Skills-improvement lesson
26	Modern means of the automation of the diagnostics and the monitoring of computer systems		Combined lesson
27	Development of a script for the automated check of the state of a computer system in Python		Comprehensive assessment of knowledge
28	Introduction to the diagnostics of computers and the software methods of restoration		Comprehensive application of knowledge
29	Analysis of the system logs: principles, formats, main files		Comprehensive application of knowledge
30	Use of Python for reading and processing logs		Comprehensive application of knowledge
31	The fundamentals of the system monitoring: parameters, methods, tools		Comprehensive application of knowledge
32	PowerShell for the diagnostics of the equipment: basic commands		Comprehensive application of knowledge

33	Creation of the first monitoring scripts in Python		Comprehensive application of knowledge
34	Work with the Windows API by means of Python (WMI)		Comprehensive application of knowledge
35	Reading and changing the parameters of the system through winreg		Comprehensive application of knowledge
36	Management of the processes and the services through subprocess		Comprehensive application of knowledge
37	Scripts for the cleaning and the optimization of the state of the system		Comprehensive application of knowledge
38	Automatic diagnostics of network problems		Comprehensive application of knowledge
39	Methods of the restoration of network connections through PowerShell		Comprehensive application of knowledge
40	Utilities for the check of the file system (Python + system tools)		Comprehensive application of knowledge
41	Diagnostics and fixing of the problems of the drivers		Comprehensive application of knowledge
42	Creation of a Python utility for the comprehensive check of the system		Comprehensive application of knowledge
43	Integration of Python and PowerShell in one maintenance process		Comprehensive application of knowledge
44	Creation of an automated recovery scenario (a mini-project)		Comprehensive application of knowledge
45	Testing and demonstration of the operability of the developed utility		Generalization and assessment of knowledge

2. GLOSSARY

The glossary contains the main definitions, terms and concepts that reflect the core content of this learning outcome. The terms are arranged in alphabetical order.

Term	Definition
Backup	A copy of data created for its restoration in the event of loss or damage of

	the originals.
Disaster Recovery	A plan and a set of measures for restoring the operability of IT systems after serious failures and emergency situations.
Event log	A system file in which the operating system records the events of the operation of the system, the applications and the security.
Hardware failure	A malfunction of the physical components of a computer that disrupts its operation.
psutil	A cross-platform Python library for obtaining information about the processes and the use of the system resources (CPU, memory, disks, network).
PowerShell	A task-automation and configuration-management framework from Microsoft, consisting of a command-line shell and a scripting language.
Restore point	A saved state of the system parameters and the files to which the system can be returned after a failure.
Script	A program in an interpreted language that automates a sequence of actions.
WMI (Windows Management Instrumentation)	An infrastructure for managing data and operations in Windows that provides access to the information about the equipment and the system.
winreg	A Python module for working with the Windows registry — reading and changing the system parameters.

3. LECTURE MATERIALS

LO 1.5 Restore equipment after hardware and software failures and prevent them

Lecture 1.5.1 The personal computer and its hardware part

Content

- The structure of a personal computer.
- The main hardware components.
- The interaction of the components.
- The reliability of the hardware and the causes of failures.

A personal computer is a universal computing device intended for the work of one user. The understanding of its hardware part is the basis for the diagnostics, the restoration of the operability and the prevention of failures.

The hardware part of a personal computer consists of the central processor, the random-access memory, the motherboard, the data-storage devices, the graphics adapter, the power supply, the cooling system and the case. Each of these components performs its function and affects the reliability of the whole.

The central processor performs the calculations and controls the operation of the system. The random-access memory stores the data and the code of the running programs. The motherboard unites all the components and provides their interaction over the buses.

The data-storage devices store the information permanently. The graphics adapter forms the image. The power supply provides all the components with the necessary voltages. The cooling system removes the heat, preventing the overheating.

The components of a computer interact over the buses and the interfaces, forming a single system. The reliability of the operation of the computer depends on the reliability of each component and the correctness of their interaction.

The hardware failures may be caused by various reasons: the natural wear of the components, the overheating, the voltage surges, the mechanical damage, the manufacturing defects and the violation of the operating conditions. The understanding of these causes helps to prevent the failures.

The natural wear concerns first of all the components with the moving parts (the hard disk drives, the fans) and the components subject to the thermal degradation (the capacitors, the thermal paste). The planned replacement of these components prevents the failures.

The overheating is one of the main causes of the failures. It accelerates the degradation of the components, causes the throttling and the shutdowns and may lead to the irreversible damage. The provision of the efficient cooling is the key factor of the reliability.

The voltage surges in the mains may damage the power supply and the components. The protection against the surges is provided by the surge protectors and the sources of uninterruptible power supply, which it is recommended to use to connect the computers.

The prevention of the hardware failures is based on the provision of the proper operating conditions (the temperature, the humidity, the cleanliness, the stable power supply), the regular preventive maintenance and the monitoring of the state of the components. The proactive approach prevents most of the failures.

The understanding of the hardware part of a personal computer, the interaction of its components and the causes of the failures is the foundation for the work on the restoration of the operability and the prevention of the failures, to which this learning outcome is devoted.

Thus, the personal computer is a system of interacting hardware components, the reliability of which depends on the operating conditions and the maintenance, and the understanding of its structure and the causes of the failures is the basis for their prevention and elimination.

Control Questions

1. What components form the hardware part of a personal computer?
2. How do the components of a computer interact?
3. What causes can lead to hardware failures?
4. Why is the overheating one of the main causes of failures?
5. On what is the prevention of hardware failures based?

Lecture 1.5.2 Elimination of hardware failures of a personal computer

Content

The classification of hardware failures.

The diagnostics of hardware failures.

The methods of eliminating failures.

The restoration of the operability after a failure.

The elimination of hardware failures is the process of detecting, localizing and eliminating the malfunctions of the physical components of a computer. It requires a systematic approach and the understanding of the operation of the components and their interaction.

The hardware failures are classified by their nature into the complete (the component does not work at all) and the partial (the component works incorrectly or unstably); by the location into the failures of the processor, the memory, the board, the drives and the other components; and by the manifestation into the obvious and the latent.

The diagnostics of a hardware failure begins with the analysis of the symptoms: the failure to start, the freezes, the restarts, the errors, the extraneous sounds, the absence of the image. The symptoms allow one to formulate the hypotheses about the possible causes of the failure.

The localization of the failure is performed by the methods of the minimal configuration and the substitution: the assembly of the system from the minimum set of the components and the gradual connection of the others, and the substitution of the suspicious components for the obviously serviceable ones.

The systematic approach to the localization consists in the consistent check of the components, beginning with the most probable causes. This allows one to gradually narrow the search area and to determine the faulty component.

After the localization of the failure, the method of its elimination is determined. The simple problems (the poor contact, the dust, the dried-out thermal paste) are eliminated by the maintenance. The faults of the components require their repair or replacement.

The replacement of a faulty component is the most frequent method of eliminating a hardware failure. The replacement is performed with a compatible serviceable component, observing the rules of the installation and the anti-static measures.

The repair of a component (for example, the replacement of the swollen capacitors of a board or a power supply) is possible for some types of faults and is economically justified for the expensive components. The complex repair requires special equipment and skills.

The restoration of the operability after the elimination of a failure includes the assembly of the computer, the check of all the connections, the start and the testing. It is important to make sure that the failure is completely eliminated and that no new problems have arisen.

After the restoration of the operability, the testing of the stability is performed: the check of the operation under the load, the monitoring of the temperatures and the parameters, the testing of the memory and the drives. This confirms the success of the elimination of the failure.

The documenting of the eliminated failures (the symptom, the cause, the method of elimination) accumulates the experience and helps in the diagnostics of the similar problems in the future. The keeping of the records is a part of the professional approach.

Thus, the elimination of hardware failures includes the diagnostics by the symptoms, the localization by the methods of the minimal configuration and the substitution, the elimination by the maintenance, the repair or the replacement, and the restoration and the testing of the operability.

Control Questions

1. How are hardware failures classified?
2. With what does the diagnostics of a hardware failure begin?
3. What methods are used to localize a failure?
4. What methods of eliminating failures exist?
5. What is performed after the restoration of the operability?

Lecture 1.5.6 Utilities for work

Content

The concept and the purpose of the system utilities.
The utilities for the diagnostics and the monitoring.
The utilities for the maintenance and the optimization.
The utilities for the restoration and the backing-up.

The system utilities are the auxiliary programs intended for the maintenance, the diagnostics, the optimization and the restoration of a computer system. They are an important tool of the technician for maintaining the operability and the reliability of the computers.

The utilities are classified by their purpose into the diagnostic, the monitoring, the maintenance, the optimization, the restoration and the backing-up ones. Each group solves its own tasks of maintaining the operability of the system.

The diagnostic utilities (AIDA64, HWiNFO, CrystalDiskInfo) provide the information about the components and their state and detect the problems. They are the basis for assessing the state of the computer and planning the maintenance.

The monitoring utilities (HWMonitor, HWiNFO) track the parameters of the system in real time: the temperatures, the voltages, the frequencies, the load and the speed of the fans. They allow one to detect the problems of the cooling and the power supply.

The maintenance and optimization utilities clean the system of the unnecessary files, optimize the autostart, defragment the disks (for HDD), and clean the registry. They maintain the performance and the order in the system.

The built-in utilities of the operating system (the disk cleanup, the disk checking, the system restore, the task manager) provide the basic capabilities of the maintenance without the installation of the additional software.

The restoration utilities (the system restore, the recovery tools, the boot-recovery means) allow one to restore the operability of the system after the failures. They return the system to a working state or restore its damaged components.

The backing-up utilities create the copies of the data and the system for their restoration in the event of the loss. The regular backing-up with the help of these utilities is the most reliable protection against the loss of data.

The antivirus utilities protect the system from malware. The regular scanning and the updating of the antivirus databases ensure the security of the system and prevent the failures caused by the malware.

The choice of the utilities depends on the task: for the diagnostics, the informational and the testing utilities are used; for the maintenance, the cleaning and the optimization utilities; for the protection, the antivirus and the backing-up ones. The technician must master the main utilities.

The correct use of the system utilities allows one to maintain the computer in working order, to diagnose and to eliminate the problems, to optimize the performance and to protect the data, which is the basis of the proactive maintenance.

Thus, the system utilities are a diverse set of tools for the diagnostics, the monitoring, the maintenance, the optimization, the restoration and the protection of a computer system, the mastery of which is necessary for the technician.

Control Questions

1. What are the system utilities, and what are they for?
2. How are the utilities classified by their purpose?
3. What utilities are used for the diagnostics and the monitoring?
4. What do the maintenance and optimization utilities do?
5. Why is the use of the backing-up utilities important?

Lecture 1.5.8 The fundamentals of the restoration and the prevention of failures in computer equipment

Content

The concept of the restoration of the operability.
The strategies of the prevention of failures.
The means of the restoration of the system.
The proactive maintenance.

The restoration of the operability and the prevention of failures are two interconnected directions of the work on the provision of the reliability of computer systems. The prevention reduces the probability of the failures, and the restoration returns the operability after the failures that have occurred.

The restoration of the operability is the process of returning a computer system to a working state after a hardware or software failure. It may include the repair or the replacement of the components, the restoration of the software, the data and the settings.

The strategies of the prevention of failures are based on the proactive approach: the regular maintenance, the monitoring of the state, the timely replacement of the wearing components and the backing-up of the data. The prevention is more economical than the elimination of the consequences of the failures.

The monitoring of the state of the system allows one to detect the signs of the approaching failures (the deterioration of the S.M.A.R.T., the overheating, the instability) and to take the measures before the failure occurs. This is the basis of the proactive maintenance.

The means of the restoration of the system include the restore points, the system images, the backups of the data, the recovery tools and the boot-recovery means. They allow one to restore the system or its components after the failures.

The restore points save the state of the system parameters and allow one to return the system to a working state after the failures caused by the changes of the settings, the installation of the programs or the drivers. They are created automatically and manually.

The system image is a complete copy of the system disk, which allows one to fully restore the system to the state at the moment of the creation of the image. The restoration from the image is used in the event of the serious failures and the corruption of the system.

The backups of the data are the copies of the important files created for their restoration in the event of the loss. The backing-up is the most reliable protection against the loss of data, since any drive may fail.

The proactive maintenance combines the monitoring of the state, the regular preventive maintenance, the backing-up and the planning of the replacement of the components. It allows one to prevent most of the failures and to minimize the consequences of those that have occurred.

The planning of the restoration includes the determination of the critical systems and data, the development of the procedures of the restoration, the provision of the backups and the means of the restoration, and the periodic check of their operability. The readiness for the restoration reduces the downtime.

The combination of the prevention and the readiness for the restoration ensures the high reliability of computer systems. The prevention reduces the number of the failures, and the readiness for the restoration minimizes the consequences of those that occur.

Thus, the fundamentals of the restoration and the prevention of failures include the strategies of the proactive maintenance, the monitoring of the state, the means of the restoration of the system and the backing-up of the data, which together ensure the reliability of computer systems.

Control Questions

1. What are the restoration of the operability and the prevention of failures?
2. On what are the strategies of the prevention of failures based?
3. What means of the restoration of the system exist?
4. What is the difference between a restore point and a system image?

5. What does the proactive maintenance combine?

Lecture 1.5.10 Backing-up of data and the restoration of information after failures

Content

- The concept and the importance of backing-up.
- The types of backups.
- The strategies and the schedules of backing-up.
- The restoration of data from a backup.

The backing-up of data is the creation of the copies of the information for its restoration in the event of the loss or the damage of the originals. The backing-up is the most reliable protection against the loss of data, since any drive may fail at any moment.

The importance of the backing-up is difficult to overestimate. The loss of data may occur as a result of the failure of a drive, the malware, the erroneous deletion, the damage of the file system or the theft of the equipment. The backing-up allows one to restore the data in all these cases.

There are several types of backups. A full backup copies all the data; it requires a lot of space and time but provides the simplest restoration. An incremental backup copies only the data changed since the previous copy; it is economical but the restoration is more complex.

A differential backup copies the data changed since the last full copy. It occupies an intermediate position between the full and the incremental copies in terms of the space and the speed of the restoration. The choice of the type depends on the requirements.

The strategy of the backing-up determines what data to copy, how often, where to store the copies and how long to keep them. A well-known rule is the rule '3-2-1': to keep three copies of the data on two different media, one of which is located off-site.

The schedule of the backing-up determines the frequency of the creation of the copies. The critical data is copied more often (daily or several times a day), the less important data less often. The frequency is chosen on the basis of the permissible volume of the loss of data.

The place of the storage of the backups affects their reliability. The copies are stored on the external drives, the network stores (NAS), in the cloud storages or on the specialized backup servers. The off-site storage protects against the local accidents.

The automation of the backing-up ensures the regularity and the reliability of the creation of the copies. The backing-up programs create the copies according to the schedule without the participation of the user, which excludes the human factor.

The restoration of the data from a backup is the reverse process — the recovery of the lost or the damaged data from the copy. The procedure of the restoration depends on the type of the copy and the backing-up program.

The check of the operability of the backups is a critically important but often neglected stage. A backup is useful only if it is possible to restore the data from it. The periodic check of the restoration confirms the operability of the copies.

The combination of the correct strategy, the automation, the reliable storage and the regular check of the backups ensures the reliable protection of the data against the loss. This is the basis of the data-protection policy of any organization.

Thus, the backing-up of data is the most reliable protection against the loss of information, and its effectiveness depends on the correct choice of the type and the strategy of the copying, the automation, the reliable storage and the regular check of the operability of the copies.

Control Questions

1. What is the backing-up of data, and why is it important?

2. What types of backups exist, and what is the difference between them?
3. What is the rule '3-2-1'?
4. How is the frequency of the backing-up chosen?
5. Why is it necessary to periodically check the operability of the backups?

Lecture 1.5.12 Proactive maintenance of the hardware for the prevention of failures

Content

The concept of the proactive maintenance.

The monitoring of the state of the equipment.

The planning of the maintenance and the replacement of the components.

The advantages of the proactive approach.

The proactive maintenance is an approach to the maintenance of the equipment in which the measures are taken before the failures occur, on the basis of the monitoring of the state and the forecasting. It is opposed to the reactive approach, in which the equipment is repaired after the failure.

The proactive maintenance is based on the monitoring of the state of the components, the analysis of the data, the forecasting of the failures and the planned execution of the maintenance and the replacement of the wearing components. This allows one to prevent most of the failures.

The monitoring of the state of the equipment includes the tracking of the temperatures, the voltages, the load, the state of the drives (the S.M.A.R.T.), the operation of the cooling and the other parameters. The deviation of the parameters from the norm is a signal of the possible problem.

The analysis of the data of the monitoring allows one to detect the trends of the deterioration of the state of the components and to forecast their failure. For example, the deterioration of the attributes of the S.M.A.R.T. of a drive indicates its approaching failure.

The forecasting of the failures on the basis of the monitoring allows one to plan the replacement of the components before their failure, which excludes the unexpected downtime and the loss of data. This is the key advantage of the proactive approach.

The planning of the maintenance includes the determination of the frequency of the preventive maintenance, the schedule of the cleaning, the replacement of the thermal paste and the wearing components, and the check of the state. The planned maintenance maintains the operability of the equipment.

The planning of the replacement of the components is based on the data of the monitoring and the service life of the components. The drives with a deteriorating S.M.A.R.T., the aging capacitors and the worn fans are replaced in a planned manner, before their failure.

The keeping of the records of the state and the maintenance of the equipment allows one to track the dynamics, to plan the maintenance and to forecast the need for the replacement of the components and the consumables. The documenting is the basis of the systematic maintenance.

The advantages of the proactive approach are the reduction of the number of the failures, the exclusion of the unexpected downtime, the prevention of the loss of data, the prolongation of the service life of the equipment and the reduction of the total costs of the maintenance.

The proactive maintenance requires the initial costs of the monitoring and the planned maintenance, but it pays off by the reduction of the costs of the elimination of the consequences of the failures, the downtime and the emergency repair.

The combination of the proactive maintenance, the monitoring, the backing-up and the readiness for the restoration ensures the high reliability and the availability of computer systems, which is especially important for the critical systems.

Thus, the proactive maintenance of the hardware is an approach based on the monitoring of the state, the forecasting of the failures and the planned maintenance, which allows one to prevent most of the failures and to ensure the reliability of the equipment.

Control Questions

1. What is the proactive maintenance, and how does it differ from the reactive approach?
2. What does the monitoring of the state of the equipment include?
3. How does the analysis of the data of the monitoring help to forecast the failures?
4. What does the planning of the maintenance and the replacement of the components include?
5. What are the advantages of the proactive approach?

Lecture 1.5.15 Protection against malware and viruses: methods of prevention and removal

Content

The types of malware.

The methods of the penetration of malware.

The means of protection against malware.

The detection and the removal of malware.

Malware (viruses, trojans, worms, ransomware, spyware) is one of the main threats to the security and the operability of computer systems. The protection against malware is an important part of the work on the prevention of the failures.

Malware is classified by the method of the propagation and the action. Viruses attach themselves to the files and propagate together with them. Worms propagate independently over the networks. Trojans disguise themselves as useful programs. Ransomware encrypts the data and demands a ransom.

Spyware collects the information about the user without his knowledge. Adware shows the unwanted advertising. Rootkits hide the presence of the malware in the system. Each type of the malware poses its own threat and requires the corresponding methods of protection.

Malware penetrates the system in various ways: through the infected files and the programs, the malicious sites and the links, the email attachments, the removable media, and the vulnerabilities of the software. The understanding of these ways helps to prevent the infection.

The means of protection against malware include the antivirus programs, the firewalls, the updating of the software, the differentiation of the access rights and the observance of the rules of the safe behaviour. The complex protection is the most effective.

The antivirus program is the main means of protection. It detects and blocks the malware by the signatures (the known samples) and by the behaviour (the heuristic analysis). The regular updating of the antivirus databases is necessary for the protection against the new threats.

The firewall controls the network traffic and blocks the unwanted connections, protecting the system from the network threats and the propagation of the malware over the network. The correct setting of the firewall is an important element of the protection.

The updating of the software eliminates the vulnerabilities through which the malware penetrates. The timely installation of the security updates of the operating system and the programs significantly reduces the risk of the infection.

The rules of the safe behaviour include the caution when opening the files and the links, the downloading of the programs only from the trusted sources, the non-use of the pirated software, and the attentiveness to the email attachments. The human factor is often the weak link.

The detection of the malware is performed by the scanning of the system with the antivirus, the analysis of the suspicious behaviour (the slowdown, the unusual network activity, the unexpected windows) and the check of the autostart and the processes. The timely detection limits the damage.

The removal of the malware is performed by the antivirus or manually. In the difficult cases, the scanning from bootable media or the specialized removal utilities is used. After the removal, the check of the system and the restoration of the damaged data are necessary.

Thus, the protection against malware combines the means of the prevention (the antivirus, the firewall, the updates, the safe behaviour) and the methods of the detection and the removal, which together ensure the security and the operability of computer systems.

Control Questions

1. What types of malware exist?
2. In what ways does malware penetrate a system?
3. What means of protection against malware are used?
4. How does an antivirus program detect malware?
5. How are malware detected and removed?

Lecture 1.5.18 Work with backup systems and recovery plans in emergency situations (Disaster Recovery)

Content

- The concept of Disaster Recovery.
- The components of a recovery plan.
- The metrics RTO and RPO.
- The testing of the recovery plan.

Disaster Recovery (DR) is a set of policies, tools and procedures for restoring the operability of the IT systems and the infrastructure after serious failures and emergency situations: the accidents, the natural disasters, the cyber-attacks and the large-scale failures of the equipment.

The need for Disaster Recovery is due to the fact that the modern organizations critically depend on their IT systems. The prolonged downtime or the loss of data may lead to the serious financial losses and even to the termination of the activity.

A recovery plan (DR plan) is a documented set of the procedures for restoring the operability of the systems. It includes the description of the critical systems, the procedures of the restoration, the responsible persons, the contacts and the resources necessary for the restoration.

The components of a DR plan include the inventory of the critical systems and data, the strategy of the backing-up, the procedures of the restoration, the alternative resources (the backup site, the equipment), and the plan of the communication and the coordination during the emergency situation.

The metric RTO (Recovery Time Objective) is the maximum permissible time of the restoration of the operability of a system after a failure. It determines how quickly the system must be restored and affects the choice of the means and the strategies of the restoration.

The metric RPO (Recovery Point Objective) is the maximum permissible volume of the loss of data, expressed in the time. It determines how often it is necessary to make the backups: the smaller the RPO, the more frequent the backing-up must be.

The RTO and the RPO are the key metrics that determine the requirements for the Disaster Recovery system. On their basis, the strategies of the backing-up and the restoration, the means and the resources necessary for achieving the required indicators are chosen.

The strategies of the Disaster Recovery differ in the cost and the speed of the restoration: from the simple backing-up with the restoration on the new equipment to the hot reserve sites that provide the restoration in the minutes. The choice depends on the criticality of the systems and the budget.

The backup site is an alternative infrastructure to which the operation is switched in the event of a failure of the main one. It may be 'cold' (the equipment without the data), 'warm' (the partially prepared) or 'hot' (the fully ready and the synchronized).

The testing of the recovery plan is a critically important stage. A plan that has not been tested may turn out to be inoperable at the decisive moment. The regular exercises of the restoration confirm the operability of the plan and reveal its weak points.

The maintenance of the relevance of the DR plan is necessary, since the IT systems change. The plan is regularly reviewed and updated taking into account the changes of the infrastructure, the systems and the requirements of the business.

Thus, Disaster Recovery is a comprehensive approach to the restoration of the IT systems after serious failures, based on the recovery plan, the metrics RTO and RPO, the choice of the strategies and the resources, and the regular testing of the readiness.

Control Questions

1. What is Disaster Recovery, and why is it necessary?
2. What components does a recovery plan include?
3. What is the metric RTO, and what does it determine?
4. What is the metric RPO, and how is it connected with the frequency of the backing-up?
5. Why is the testing of the recovery plan necessary?

Lecture 1.5.20 Application of WMI for obtaining information about the hardware part of a computer

Content

The architecture of WMI.

The WMI classes and the namespaces.

The methods of accessing WMI.

The application of WMI for the diagnostics and the monitoring.

WMI (Windows Management Instrumentation) is the implementation by Microsoft of the standard of the management of the IT infrastructure. It provides a unified interface for obtaining the information about the system and the equipment and for managing them, which makes it a powerful tool of the technician.

The architecture of WMI includes the providers (which supply the data about the various components), the repository (the storage of the descriptions of the classes), the WMI service (the intermediary), and the consumers (the programs and the scripts that request the data).

The data in WMI is organized in the form of the classes, which describe the manageable objects (the processor, the disk, the service, the process). Each class has the properties (the characteristics of the object) and the methods (the operations on the object). The instances of the classes represent the specific objects.

The classes of WMI are grouped into the namespaces. The main namespace for the management of the system is root\CIMV2, which contains the classes for the work with the equipment and the operating system, such as Win32_Processor, Win32_PhysicalMemory and Win32_DiskDrive.

The access to WMI is possible in several ways: through PowerShell (the cmdlets Get-CimInstance and Get-WmiObject), through the command-line utility wmic, through the scripts (VBScript, Python with the WMI library) and from the programs in various languages.

PowerShell is the most convenient and the modern way of working with WMI. The cmdlet Get-CimInstance allows one to request the instances of the classes and to obtain their properties. For example, the query of the class Win32_Processor returns the information about the processor.

The Win32_Processor class provides the information about the processor: the model, the number of cores, the frequency, the load. The Win32_PhysicalMemory class provides the information about the memory modules. The Win32_DiskDrive class provides the information about the drives.

The Win32_OperatingSystem class provides the information about the operating system: the version, the build, the volume of the memory, the time of the operation. The Win32_BIOS class provides the information about the BIOS. These classes are the basis for the inventory of the system.

The application of WMI for the diagnostics consists in obtaining the data about the state of the components: the temperatures, the load, the state of the drives. WMI provides the access to this information in a unified form, which is convenient for the automation.

The application of WMI for the monitoring consists in the periodic obtaining of the parameters of the system and the tracking of their changes. On the basis of WMI, the monitoring scripts that collect and analyse the data about the state of the equipment are built.

The advantage of WMI is the universality and the standardization: a unified interface for obtaining the information about the various components and for managing them. This makes WMI the basis for the automation of the administration and the diagnostics in Windows.

Thus, WMI is a powerful and universal tool for obtaining the information about the equipment and the system and for managing them, the mastery of which opens up wide possibilities for the diagnostics, the monitoring and the automation.

Control Questions

1. What is WMI, and what does it provide?
2. What does the architecture of WMI include?
3. How is the data organized in WMI?
4. In what ways is the access to WMI possible?
5. How is WMI applied for the diagnostics and the monitoring?

Lecture 1.5.22 Development of scenarios for the restoration of the operability of a computer system

Content

The principles of the development of the scenarios of the restoration.

The structure of a scenario of the restoration.

The automation of the scenarios.

The testing and the maintenance of the scenarios.

The development of the scenarios for the restoration of the operability is an important part of the work on the provision of the reliability of computer systems. A well-developed scenario allows one to quickly and correctly restore the operability after a failure, minimizing the downtime and the loss of data.

The principles of the development of the scenarios include the completeness (the coverage of all the typical failures), the detail (the description of every step), the clarity (the understandability for the executor), the realism (the correspondence to the real conditions) and the testability.

A scenario of the restoration is developed for a specific type of the failure: the failure of a drive, the corruption of the operating system, the malware, the loss of data, the network failure. Each scenario takes into account the features of the corresponding failure.

The structure of a scenario of the restoration includes the description of the failure and its symptoms, the conditions of the application of the scenario, the necessary resources, the step-by-step procedure of the restoration, the criteria of the successful restoration and the actions in the event of the impossibility of the restoration.

The step-by-step procedure of the restoration describes in detail the sequence of the actions: the diagnostics, the preparation of the resources, the execution of the restoration, the check of the result. Each step is formulated clearly and unambiguously.

The necessary resources include the backups, the bootable media, the recovery means, the replacement components and the time. The availability and the readiness of these resources are a condition for the successful restoration.

The automation of the scenarios of the restoration with the help of the scripts (PowerShell, Python) increases the speed and the reliability of the restoration and excludes the errors of the manual execution. The automated scenarios are especially useful for the typical and the frequent failures.

The scripts of the restoration can automate the diagnostics, the backing-up, the restoration of the data and the settings, the restart of the services and the check of the result. The combination of the automation and the manual control ensures the flexibility and the reliability.

The testing of the scenarios of the restoration is a critically important stage. A scenario that has not been tested may turn out to be inoperable at the decisive moment. The regular exercises of the restoration confirm the operability of the scenarios and reveal their weak points.

The maintenance of the relevance of the scenarios is necessary, since the computer systems change. The scenarios are regularly reviewed and updated taking into account the changes of the configuration, the software and the requirements.

The documenting of the scenarios and the keeping of the records of their application accumulate the experience and improve the scenarios. The analysis of the real cases of the restoration helps to improve the scenarios and to prevent the repeated failures.

Thus, the development of the scenarios for the restoration of the operability is based on the principles of the completeness, the detail and the testability, includes the clear structure and the procedure, the automation with the scripts and the regular testing and the maintenance of the relevance.

Control Questions

1. What principles underlie the development of the scenarios of the restoration?
2. What does the structure of a scenario of the restoration include?
3. What resources are necessary for the restoration?
4. How does the automation of the scenarios increase the reliability of the restoration?
5. Why is the testing and the maintenance of the relevance of the scenarios necessary?

Lecture 1.5.26 Modern means of the automation of the diagnostics and the monitoring of computer systems

Content

The need for the automation of the diagnostics and the monitoring.

The means and the languages of the automation.

The monitoring systems.

The integration of the means of the automation.

The automation of the diagnostics and the monitoring is a modern direction of the work of the technician, which allows one to increase the speed, the accuracy and the scale of the maintenance of computer systems. The manual diagnostics of a large number of computers is laborious and is subject to the errors.

The need for the automation grows with the increase of the number of the serviced computers. The automated means allow one to simultaneously diagnose and to monitor many systems, to collect and to analyse the data centrally and to react to the problems in a timely manner.

For the automation of the diagnostics and the monitoring, the scripting languages are used: PowerShell (for Windows), Python (cross-platform), Bash (for Linux). They allow one to create the scripts that automate the collection of the data, the analysis and the actions of the maintenance.

PowerShell is the standard means of the automation of the administration of Windows. It provides the access to WMI, the registry, the services, the network and the other components of the system, which makes it a universal tool for the automation of the diagnostics and the maintenance.

Python is a cross-platform language with the rich set of the libraries (psutil, WMI, paramiko) for the work with the system. It is convenient for the creation of the complex scripts and the utilities of the diagnostics and the monitoring with the analysis and the visualization of the data.

The monitoring systems (Zabbix, Nagios, Prometheus and others) are the specialized means for the centralized monitoring of the state of the computers and the network. They collect the data from many systems, analyse them and notify of the problems.

The monitoring systems track the parameters of the equipment and the software (the load, the temperatures, the availability, the errors) and notify the administrators of the deviations. This allows one to react to the problems before they lead to the failures.

The agents of the monitoring are installed on the serviced systems and collect the data, transmitting them to the central server. The agentless monitoring uses the standard protocols (WMI, SNMP) for the collection of the data without the installation of the additional software.

The integration of the means of the automation allows one to build the complex solutions: the monitoring detects the problem, the script performs the diagnostics and the restoration, and the system notifies the administrator. This is the basis of the modern automated maintenance.

The automation of the routine tasks frees the technician from the monotonous work and allows him to concentrate on the complex problems. At the same time, the technician must understand the operation of the automated means and be able to create and to maintain the scripts.

The development of the means of the automation goes in the direction of the integration with the cloud technologies, the use of the machine learning for the forecasting of the failures and the creation of the self-healing systems that restore the operability without the participation of the person.

Thus, the modern means of the automation of the diagnostics and the monitoring — the scripting languages and the monitoring systems — allow one to increase the efficiency and the scale of the maintenance of computer systems and are an important tool of the modern technician.

Control Questions

1. Why is the automation of the diagnostics and the monitoring necessary?
2. What languages are used for the automation?
3. What are the advantages of PowerShell and Python for the automation?
4. What are the monitoring systems for?
5. What does the integration of the means of the automation allow one to achieve?

4. PRACTICAL CLASS MATERIALS

LO 1.5 Restore equipment after hardware and software failures and prevent them

Practical work 1.5.3 External devices of a personal computer

Objective: To form practical skills in connecting, configuring and diagnosing the external devices of a personal computer.

Theoretical background

The external devices of a personal computer are connected from the outside and extend its capabilities: the monitors, the printers, the scanners, the keyboards, the mice, the external drives and others. They are connected over various interfaces and require the corresponding drivers.

The diagnostics of the external devices includes the check of the connection, the power, the drivers and the settings. Most problems of the external devices are solved at these levels.

Equipment and materials

- A set of external devices.
- A computer with various ports.
- Cables and drivers.
- A report form.

Procedure

1. Examine the external devices and determine the interface for connecting each of them.
2. Connect the devices to the computer and check their recognition by the system.
3. Install or update the drivers of the devices and configure their parameters.
4. Check the operability of each device and detect the possible problems.
5. Perform the diagnostics of a non-working device (the connection, the power, the driver, the settings).
6. Make a conclusion about the state of the external devices and the methods of eliminating the problems.

Tasks for independent completion

1. Connect and configure at least three external devices.
2. Carry out the diagnostics of a non-working external device.
3. Compile a memo on the connection and the diagnostics of external devices.

Conclusion: As a result of the work, the student has formed the skills of connecting, configuring and diagnosing the external devices of a personal computer.

Control Questions

1. What are the external devices of a personal computer?
2. Over what interfaces are the external devices connected?
3. What does the diagnostics of an external device include?
4. Why is a driver necessary for the operation of a device?
5. How is a non-working external device diagnosed?

Practical work 1.5.4 Investigation of the types of external devices and the connection interfaces of a personal computer

Objective: To form practical skills in investigating and classifying the external devices and the connection interfaces of a personal computer.

Theoretical background

The external devices are connected over various interfaces: USB, HDMI, DisplayPort, Ethernet, Bluetooth, Wi-Fi. Each interface has its own characteristics of the speed, the universality and the area of application. The choice of the interface affects the operation of the device.

The classification of the external devices by their purpose, the type of connection and the interface helps to systematize the knowledge and to choose the optimal method of connection.

Equipment and materials

- A set of external devices with various interfaces.
- A computer with various ports.
- Reference data on the interfaces.

- A report form.

Procedure

1. Examine the external devices and classify them by their purpose and the type of connection.
2. Determine the interface of each device and its characteristics (the speed, the universality).
3. Compile a comparative table of the interfaces used by the devices.
4. Connect the devices over the different interfaces and assess the convenience and the speed.
5. Determine the optimal interface for each type of device.
6. Make a conclusion about the classification of the external devices and the criteria for choosing the interface.

Tasks for independent completion

1. Compile a classification of the external devices by their purpose and the interface.
2. Prepare a comparative table of the connection interfaces.
3. Justify the choice of the interface for each type of device.

Conclusion: As a result of the work, the student has mastered the methods of investigating and classifying the external devices and the connection interfaces of a personal computer.

Control Questions

1. What interfaces are used to connect external devices?
2. How are external devices classified?
3. What are the characteristics of the interface that affect the operation of a device?
4. How is the optimal interface for a device chosen?
5. What is the difference between the wired and the wireless interfaces?

Practical work 1.5.5 Cables and their types

Objective: To form practical skills in identifying the types of cables, determining their purpose and checking their serviceability.

Theoretical background

Cables provide the connection of the components and the devices of a computer and the transmission of the data and the power. There are many types of cables: the power cables, the data cables (SATA, USB, HDMI, DisplayPort, Ethernet), and the internal cables of the computer.

The serviceability of the cables affects the reliability of the connections. A faulty cable is a frequent cause of the problems of the connection of the devices. The check of the cables is performed by the visual inspection and by the testing.

Equipment and materials

- A set of cables of various types.
- A cable tester and a multimeter.
- A computer for the check of the connections.
- A report form.

Procedure

1. Examine the cables and determine the type and the purpose of each of them.
2. Determine the connectors of the cables and the devices for the connection of which they are intended.
3. Perform the visual inspection of the cables and detect the damage.
4. Check the serviceability of the cables with a cable tester or a multimeter (the continuity).
5. Check the operation of the connection through the cables on the computer.
6. Make a conclusion about the types and the serviceability of the examined cables.

Tasks for independent completion

1. Compile a table of the types of cables with an indication of their purpose and the connectors.
2. Check the serviceability of at least five cables.
3. Describe the typical faults of the cables and the methods of their detection.

Conclusion: As a result of the work, the student has formed the skills of identifying the types of cables, of determining their purpose and of checking their serviceability.

Control Questions

1. What types of cables are used in a computer?
2. How is the type and the purpose of a cable determined?
3. How is the serviceability of a cable checked?
4. What are the typical faults of cables?
5. Why is the serviceability of the cables important for the reliability of the connections?

Practical work 1.5.7 Analysis of the utilities for the diagnostics and the monitoring of computer systems

Objective: To form practical skills in working with the utilities for the diagnostics and the monitoring of computer systems and in analysing the obtained data.

Theoretical background

The utilities for the diagnostics and the monitoring provide the data about the state of the components and the parameters of the operation of the system. The analysis of these data allows one to assess the state of the computer and to detect the problems.

The monitoring in real time allows one to track the dynamics of the parameters under the load and to detect the problems that appear only in certain modes of operation.

Equipment and materials

- A computer with the diagnostic and the monitoring utilities.
- Reference data on the norms of the parameters.
- A tool for creating the load.
- A report form.

Procedure

1. Run the informational utility and obtain the data about all the components of the computer.
2. Run the monitoring of the temperatures, the voltages and the frequencies in real time.
3. Create the load on the computer and observe the change of the parameters.
4. Analyse the obtained data and compare them with the norms.
5. Detect the possible deviations and the problems.
6. Make a conclusion about the state of the computer on the basis of the analysis of the data.

Tasks for independent completion

1. Prepare a report on the state of the computer on the basis of the diagnostic utilities.
2. Analyse the dynamics of the parameters under the load.
3. Detect and describe the possible deviations of the parameters.

Conclusion: As a result of the work, the student has mastered the methods of working with the utilities for the diagnostics and the monitoring and of analysing the obtained data.

Control Questions

1. What utilities are used for the diagnostics and the monitoring?
2. What does the monitoring in real time give?
3. How are the obtained data compared with the norms?
4. Why is it important to observe the parameters under the load?

5. How is a conclusion about the state of a computer made?

Practical work 1.5.9 Preparation of a brief report on the application of WMI for obtaining information about the hardware part of a computer

Objective: To form practical skills in using WMI for obtaining information about the hardware part of a computer and in preparing a report on the obtained data.

Theoretical background

WMI (Windows Management Instrumentation) is an infrastructure for managing data and operations in Windows that provides access to the information about the equipment, the system and the processes. WMI is accessed through PowerShell, the wmic utility or from programs.

Through WMI, one can obtain the information about the processor, the memory, the drives, the motherboard, the BIOS, the operating system and other components. This information is used for the inventory, the diagnostics and the monitoring.

Equipment and materials

- A computer with the Windows operating system.
- PowerShell or the wmic utility.
- Examples of the WMI queries.
- A report form.

Procedure

1. Study the structure of WMI and the main classes for obtaining the information about the equipment.
2. Obtain the information about the processor through WMI (the class Win32_Processor) with the help of PowerShell.
3. Obtain the information about the memory, the drives and the motherboard through the corresponding WMI classes.
4. Obtain the information about the BIOS and the operating system.
5. Systematize the obtained data and prepare a report on the hardware part of the computer.
6. Make a conclusion about the capabilities of WMI for obtaining the information about the equipment.

Tasks for independent completion

1. Prepare a report on the hardware part of a computer on the basis of the data of WMI.
2. Compile a list of the WMI classes for obtaining the information about the main components.
3. Describe the procedure for obtaining the information about a component through WMI.

Conclusion: As a result of the work, the student has formed the skills of using WMI for obtaining the information about the hardware part of a computer and of preparing a report on the obtained data.

Control Questions

1. What is WMI, and what does it provide access to?
2. Through what means is WMI accessed?
3. What WMI class is used to obtain the information about the processor?
4. What information about the equipment can be obtained through WMI?
5. How is WMI used for the inventory and the diagnostics?

Practical work 1.5.11 Investigation of the methods of backing-up and the restoration of information after failures

Objective: To form practical skills in performing the backing-up of data by various methods and the restoration of the information from the copies.

Theoretical background

The practical mastery of the backing-up includes the creation of the copies of various types (full, incremental, differential), the configuration of the schedule and the storage, and the restoration of the data from the copies. The built-in and the specialized tools are used for this.

The check of the restoration is a mandatory stage that confirms the operability of the backup. The backing-up without the check of the restoration does not guarantee the protection of the data.

Equipment and materials

- A computer with the data for the copying.
- An external drive or a network store for the storage of the copies.
- Backing-up software (the built-in tools or the specialized ones).
- A report form.

Procedure

1. Create a full backup of the selected data to an external drive.
2. Configure an incremental or a differential backup according to a schedule.
3. Model the loss of the data (delete the test files) and restore them from the copy.
4. Check the completeness and the correctness of the restoration of the data.
5. Configure the automatic backing-up according to a schedule.
6. Make a conclusion about the methods of the backing-up and the restoration of the data.

Tasks for independent completion

1. Create a full and an incremental backup of the data.
2. Restore the data from the copy and check the correctness of the restoration.
3. Configure the automatic backing-up according to a schedule and describe its parameters.

Conclusion: As a result of the work, the student has formed the skills of performing the backing-up of data by various methods and the restoration of the information from the copies.

Control Questions

1. What types of backups were created in the work?
2. How is the automatic backing-up configured?
3. How is the data restored from a copy?
4. Why is it necessary to check the correctness of the restoration?
5. Where is it recommended to store the backups?

Practical work 1.5.13 Comprehensive analysis of the state of the operating system and the hardware components of a PC

Objective: To form practical skills in the comprehensive analysis of the state of the operating system and the hardware components of a computer.

Theoretical background

The comprehensive analysis of the state of a computer combines the assessment of the hardware components (the processor, the memory, the drives, the cooling) and the operating system (the performance, the errors, the security). This gives a complete picture of the state of the computer.

The analysis is performed with the help of the diagnostic and the monitoring utilities, the system tools and the analysis of the event logs. The result is a conclusion about the state and the recommendations for the maintenance.

Equipment and materials

- A computer for the analysis.
- The diagnostic and the monitoring utilities.
- The system tools (the event viewer, the resource monitor).

- A report form.

Procedure

1. Analyse the state of the hardware components: the processor, the memory, the drives, the cooling.
2. Analyse the state of the operating system: the performance, the load of the resources, the errors.
3. View the event logs of the system and detect the warnings and the errors.
4. Check the security of the system: the antivirus, the updates, the firewall.
5. Systematize the results of the analysis and compile a comprehensive report.
6. Make a conclusion about the state of the computer and the recommendations for the maintenance.

Tasks for independent completion

1. Prepare a comprehensive report on the state of the hardware and the operating system.
2. Analyse the event logs and detect the problems.
3. Formulate the recommendations for the maintenance on the basis of the analysis.

Conclusion: As a result of the work, the student has mastered the methods of the comprehensive analysis of the state of the operating system and the hardware components of a computer.

Control Questions

1. What does the comprehensive analysis of the state of a computer combine?
2. What tools are used for the analysis?
3. What do the event logs of the system give?
4. How is the security of the system checked?
5. How are the recommendations for the maintenance formulated?

Practical work 1.5.14 Restoration of operating systems after failures and critical errors

Objective: To form practical skills in restoring operating systems after failures and critical errors with the use of the built-in and the additional means.

Theoretical background

The failures of the operating system may be caused by the corruption of the system files, the erroneous changes of the settings, the conflicts of the drivers, the malware or the hardware problems. For the restoration, the built-in and the additional means are used.

The means of the restoration of Windows include the restore points, the recovery environment (WinRE), the system-file checker (sfc), the recovery of the loading (bootrec) and the restoration from a system image. The choice of the means depends on the nature of the failure.

Equipment and materials

- A computer with an operating system (or a virtual machine).
- Bootable recovery media.
- Restore points and a system image.
- A report form.

Procedure

1. Study the means of the restoration of the operating system and the situations of their application.
2. Perform the restoration of the system from a restore point.
3. Use the system-file checker (sfc) to restore the corrupted system files.
4. Study the recovery environment (WinRE) and its capabilities.
5. Perform the restoration of the loading of the system (model the problem and describe the solution).
6. Make a conclusion about the means and the methods of the restoration of the operating system.

Tasks for independent completion

1. Describe the means of the restoration of the operating system and the situations of their application.
2. Perform the restoration of the system from a restore point.

3. Compile an algorithm for the restoration of the operating system after a critical error.

Conclusion: As a result of the work, the student has formed the skills of restoring operating systems after failures and critical errors with the use of the built-in and the additional means.

Control Questions

1. What may cause the failures of the operating system?
2. What means of the restoration of Windows exist?
3. How is the restoration of the system from a restore point performed?
4. What does the system-file checker (sfc) do?
5. What is the recovery environment (WinRE)?

Practical work 1.5.16 Analysis of modern methods of protection against malware and viruses

Objective: To form practical skills in applying modern methods of protection against malware and in detecting and removing it.

Theoretical background

The modern methods of protection against malware include the antivirus protection with the heuristic and the behavioural analysis, the protection in real time, the cloud technologies of the detection, the sandboxes and the protection against ransomware.

The practical mastery of these methods includes the configuration of the antivirus protection, the scanning of the system, the detection and the removal of the malware, and the restoration after the infection.

Equipment and materials

- A computer with an antivirus program.
- Tools for the analysis of the processes and the autostart.
- Test (safe) samples for the demonstration.
- A report form.

Procedure

1. Configure the antivirus protection: the protection in real time, the scheduled scanning, the updating of the databases.
2. Perform a full scan of the system and analyse the results.
3. Analyse the autostart and the running processes for the suspicious elements.
4. Study the methods of protection against ransomware and configure the corresponding protection.
5. Describe the procedure for detecting and removing the malware and restoring the system.
6. Make a conclusion about the modern methods of protection against malware.

Tasks for independent completion

1. Configure the antivirus protection and perform a scan of the system.
2. Analyse the autostart and the processes for the suspicious elements.
3. Compile an algorithm for the actions when malware is detected.

Conclusion: As a result of the work, the student has formed the skills of applying modern methods of protection against malware and of detecting and removing it.

Control Questions

1. What modern methods of protection against malware exist?
2. How is the antivirus protection configured?
3. How is the autostart analysed for the suspicious elements?
4. What is the protection against ransomware?
5. What is the algorithm of the actions when malware is detected?

Practical work 1.5.17 Restoration of the operation of network systems and connections after failures

Objective: To form practical skills in diagnosing and restoring the operation of network systems and connections after failures.

Theoretical background

The failures of the network connection may be caused by the hardware problems (the cable, the network card, the equipment), the incorrect settings, the faults of the drivers or the network services. The diagnostics determines the cause and the method of the restoration.

For the diagnostics and the restoration of the network, the system utilities are used: ping, ipconfig, tracert, nslookup, netsh. They allow one to detect and to eliminate the network problems.

Equipment and materials

- A computer with a network connection.
- Network cables and equipment.
- Network diagnostic utilities.
- A report form.

Procedure

1. Check the physical connection of the network: the cable, the connectors, the indicators of the link.
2. View the network settings with the ipconfig utility and check their correctness.
3. Check the availability of the network nodes with the ping utility.
4. Trace the route to a remote node with the tracert utility and detect the place of the problem.
5. Restore the network connection: reset the settings, restart the adapter, reinstall the driver.
6. Make a conclusion about the cause of the failure and the method of the restoration of the network.

Tasks for independent completion

1. Carry out the diagnostics of a network failure with the use of the system utilities.
2. Restore the network connection after a failure.
3. Compile an algorithm for the diagnostics and the restoration of a network connection.

Conclusion: As a result of the work, the student has formed the skills of diagnosing and restoring the operation of network systems and connections after failures.

Control Questions

1. What may cause the failures of a network connection?
2. What utilities are used for the diagnostics of a network?
3. How is the physical connection of a network checked?
4. What does the ping utility show, and what does the tracert utility show?
5. How is a network connection restored after a failure?

Practical work 1.5.19 Development of scenarios for the restoration of the operability of a computer system

Objective: To form practical skills in developing scenarios for the restoration of the operability of a computer system for various types of failures.

Theoretical background

A scenario of the restoration is a documented sequence of the actions for restoring the operability of a system after a specific type of the failure. The well-developed scenarios reduce the time of the restoration and exclude the errors.

The scenarios are developed for the typical failures: the failure of a drive, the corruption of the operating system, the malware, the loss of data, the network failure. Each scenario contains the procedure of the actions and the necessary resources.

Equipment and materials

- A description of a computer system.
- Templates of the scenarios of the restoration.
- Information about the means of the restoration.
- A report form.

Procedure

1. Determine the typical types of the failures for the given computer system.
2. For each type of the failure, determine the procedure of the restoration of the operability.
3. Develop a detailed scenario of the restoration for one of the types of the failures.
4. Determine the resources necessary for the restoration (the backups, the means, the time).
5. Assess the time of the restoration (RTO) and the volume of the possible loss of data (RPO).
6. Make a conclusion about the developed scenarios of the restoration.

Tasks for independent completion

1. Develop a scenario of the restoration of the operability for a given type of the failure.
2. Determine the resources and the time necessary for the restoration.
3. Compile a list of the typical failures with the brief scenarios of the restoration.

Conclusion: As a result of the work, the student has formed the skills of developing scenarios for the restoration of the operability of a computer system for various types of failures.

Control Questions

1. What is a scenario of the restoration?
2. For what types of the failures are the scenarios developed?
3. What does a scenario of the restoration contain?
4. How are the resources necessary for the restoration determined?
5. How are the RTO and the RPO assessed for a scenario?

Practical work 1.5.21 Comprehensive analysis of the state of the operating system and the hardware components of a PC

Objective: To consolidate the skills of the comprehensive analysis of the state of the operating system and the hardware components with the use of WMI and the diagnostic utilities.

Theoretical background

The comprehensive analysis of the state of a computer with the use of WMI allows one to obtain the data about the equipment and the operating system in a unified form and to automate their collection. This complements the analysis with the diagnostic utilities.

The combination of WMI, the diagnostic utilities and the analysis of the event logs gives the most complete picture of the state of the computer and allows one to detect the problems and to plan the maintenance.

Equipment and materials

- A computer with the Windows operating system.
- PowerShell with the access to WMI.
- The diagnostic and the monitoring utilities.
- A report form.

Procedure

1. Obtain the data about the hardware components through WMI with the help of PowerShell.

2. Obtain the data about the operating system and its state through WMI.
3. Supplement the data with the results of the diagnostic and the monitoring utilities.
4. Analyse the event logs of the system for the warnings and the errors.
5. Systematize all the obtained data into a comprehensive report.
6. Make a conclusion about the state of the computer and the recommendations for the maintenance.

Tasks for independent completion

1. Prepare a comprehensive report on the state of the computer with the use of WMI.
2. Combine the data of WMI, the utilities and the event logs.
3. Formulate the recommendations for the maintenance on the basis of the analysis.

Conclusion: As a result of the work, the student has consolidated the skills of the comprehensive analysis of the state of the operating system and the hardware components with the use of WMI and the diagnostic utilities.

Control Questions

1. How does WMI complement the analysis of the state of a computer?
2. What data is obtained through WMI?
3. How are the data of WMI, the utilities and the logs combined?
4. What gives the most complete picture of the state of a computer?
5. How are the recommendations for the maintenance formulated?

Practical work 1.5.23 Creation of a script for the automatic diagnostics of a computer in Python

Objective: To form practical skills in creating a script for the automatic diagnostics of a computer in the Python language with the use of the system libraries.

Theoretical background

Python is a high-level programming language convenient for the automation of the tasks of the administration and the diagnostics. For obtaining the information about the system, the libraries psutil (the processes and the resources), WMI (the access to WMI) and platform (the information about the system) are used.

A script for the automatic diagnostics collects the data about the state of the computer (the processor, the memory, the disks, the network), analyses them and forms a report. This automates the routine diagnostics and increases its speed and accuracy.

Equipment and materials

- A computer with the installed Python.
- The libraries psutil, WMI, platform.
- A code editor.
- A report form.

Procedure

1. Study the capabilities of the psutil library for obtaining the information about the resources of the system.
2. Write a fragment of the script for obtaining the information about the processor and its load with the help of psutil.
3. Add to the script the obtaining of the information about the memory, the disks and the network.
4. Add the analysis of the obtained data and the formation of a report on the state of the system.
5. Run the script, check its operation and analyse the report.
6. Make a conclusion about the capabilities of Python for the automation of the diagnostics.

Tasks for independent completion

1. Write a Python script for obtaining the information about the processor, the memory and the disks.

2. Add the formation of a report on the state of the system to the script.

3. Describe the capabilities of the psutil library for the diagnostics.

Conclusion: As a result of the work, the student has formed the skills of creating a script for the automatic diagnostics of a computer in Python with the use of the system libraries.

Control Questions

1. Why is Python convenient for the automation of the diagnostics?
2. What libraries are used for obtaining the information about the system?
3. What does the psutil library allow one to obtain?
4. How is a report on the state of the system formed in a script?
5. What are the advantages of the automatic diagnostics?

Practical work 1.5.24 Investigation of the capabilities of PowerShell for the diagnostics and the restoration of network connections

Objective: To form practical skills in using PowerShell for the diagnostics and the restoration of network connections.

Theoretical background

PowerShell is a powerful framework for the automation of the administration of Windows. It contains the cmdlets for the work with the network: Test-Connection (the analogue of ping), Get-NetAdapter, Get-NetIPConfiguration, Restart-NetAdapter and others.

The cmdlets of PowerShell allow one to diagnose and to restore the network connections in an automated way, which is convenient for the mass administration and the creation of the scripts of the restoration.

Equipment and materials

- A computer with the Windows operating system and a network connection.
- PowerShell.
- Examples of the network cmdlets.
- A report form.

Procedure

1. Study the cmdlets of PowerShell for the work with the network and their parameters.
2. Check the availability of the network nodes with the cmdlet Test-Connection.
3. Obtain the information about the network adapters and the settings with the cmdlets Get-NetAdapter and Get-NetIPConfiguration.
4. Perform the restart of the network adapter with the cmdlet Restart-NetAdapter.
5. Write a fragment of the script for the automatic diagnostics of the network connection.
6. Make a conclusion about the capabilities of PowerShell for the diagnostics and the restoration of the network.

Tasks for independent completion

1. Compile a list of the cmdlets of PowerShell for the work with the network.
2. Write a fragment of the script for the diagnostics of a network connection.
3. Describe the procedure for the restoration of a network connection with PowerShell.

Conclusion: As a result of the work, the student has formed the skills of using PowerShell for the diagnostics and the restoration of network connections.

Control Questions

1. What is PowerShell, and what is it for?
2. What cmdlets are used for the work with the network?

3. What does the cmdlet Test-Connection do?
4. How is the restart of a network adapter performed?
5. What are the advantages of PowerShell for the administration of the network?

Practical work 1.5.25 Performing the diagnostics and the restoration of network connections with the use of PowerShell

Objective: To consolidate the practical skills of the diagnostics and the restoration of network connections with the use of PowerShell scripts.

Theoretical background

The automation of the diagnostics and the restoration of the network with the help of PowerShell scripts allows one to quickly and reliably eliminate the typical network problems. A script can check the connection, the settings and restart the adapter automatically.

The script of the restoration of the network combines the diagnostics (the check of the connection and the settings) and the actions of the restoration (the reset of the settings, the restart of the adapter, the renewal of the address).

Equipment and materials

- A computer with PowerShell and a network connection.
- Examples of the scripts of the diagnostics.
- Administrator rights.
- A report form.

Procedure

1. Write a PowerShell script for the diagnostics of a network connection (the check of the adapter, the settings, the availability of the nodes).
2. Add to the script the actions of the restoration (the restart of the adapter, the renewal of the IP address).
3. Test the script on a working and a model-broken network connection.
4. Analyse the operation of the script and its messages.
5. Improve the script by adding the handling of the errors and the logging.
6. Make a conclusion about the automation of the diagnostics and the restoration of the network.

Tasks for independent completion

1. Write a PowerShell script for the diagnostics and the restoration of a network connection.
2. Test the script and describe the result.
3. Add the logging of the actions to the script.

Conclusion: As a result of the work, the student has consolidated the skills of the diagnostics and the restoration of network connections with the use of PowerShell scripts.

Control Questions

1. What does a PowerShell script for the diagnostics of a network include?
2. What actions of the restoration can a script perform?
3. How is a network-diagnostics script tested?
4. Why is the handling of the errors necessary in a script?
5. What does the logging of the actions of a script give?

Practical work 1.5.27 Development of a script for the automated check of the state of a computer system in Python

Objective: To consolidate, in a comprehensive form, the skills of the automation of the diagnostics by developing a complete script for the automated check of the state of a computer system in Python.

Theoretical background

A script for the automated check of the state of a computer system collects the data about all the main components (the processor, the memory, the disks, the network), analyses them, compares them with the norms and forms a report indicating the problems.

The complete script combines the obtaining of the data with the help of psutil and WMI, the analysis, the formation of the report and the logging. This is a comprehensive task that demonstrates the mastery of the automation of the diagnostics.

Equipment and materials

- A computer with the installed Python and the libraries.
- A code editor.
- Reference data on the norms of the parameters.
- A final-assessment form.

Procedure

1. Design the structure of the script: the modules of the collection of the data, the analysis and the report.
2. Implement the collection of the data about the processor, the memory, the disks and the network with the help of psutil.
3. Implement the analysis of the data: the comparison with the norms and the detection of the problems.
4. Implement the formation of the report on the state of the system and the logging.
5. Test the script and check the correctness of its operation and the report.
6. Make a conclusion about the developed script and the capabilities of its application.

Tasks for independent completion

1. Develop a complete Python script for the check of the state of a computer system.
2. Implement the analysis of the data and the formation of the report in the script.
3. Test the script and describe the result of its operation.

Conclusion: As a result of the work, the student has consolidated the comprehensive skills of the automation of the diagnostics by developing a complete script for the automated check of the state of a computer system in Python.

Control Questions

1. What does a script for the check of the state of a computer system include?
2. How is the collection of the data about the components implemented?
3. How is the analysis of the data and the detection of the problems implemented?
4. How is the report on the state of the system formed?
5. Why is the logging necessary in a diagnostics script?

Practical work 1.5.28 Introduction to the diagnostics of computers and the software methods of restoration

Objective: To form a holistic understanding of the software methods of the diagnostics and the restoration of computers and to consolidate the skills of their application.

Theoretical background

The software methods of the diagnostics and the restoration use the programs and the scripts to detect and to eliminate the problems of computer systems. They complement the hardware methods and allow one to automate the maintenance.

The software methods include the diagnostic utilities, the system tools of the restoration, the scripts of the automation and the monitoring systems. The combination of these means provides the comprehensive software maintenance of the computers.

Equipment and materials

- A computer for the diagnostics.
- A set of the software means of the diagnostics and the restoration.
- Documentation on the means.
- A report form.

Procedure

1. Systematize the software methods of the diagnostics and the restoration of computers.
2. Apply the diagnostic utilities to assess the state of the computer.
3. Apply the system tools of the restoration to eliminate a model problem.
4. Apply a script of the automation for the diagnostics or the maintenance.
5. Compare the effectiveness of the various software methods.
6. Make a conclusion about the software methods of the diagnostics and the restoration.

Tasks for independent completion

1. Compile a classification of the software methods of the diagnostics and the restoration.
2. Apply at least three different software methods on practice.
3. Compare the effectiveness of the applied methods.

Conclusion: As a result of the work, the student has formed a holistic understanding of the software methods of the diagnostics and the restoration of computers and has consolidated the skills of their application.

Control Questions

1. What are the software methods of the diagnostics and the restoration?
2. How do the software methods complement the hardware ones?
3. What software means of the restoration exist?
4. What are the advantages of the software methods?
5. How is the effectiveness of the various methods compared?

Practical work 1.5.29 Analysis of the system logs: principles, formats, main files

Objective: To form practical skills in analysing the system logs of an operating system for the diagnostics of the problems.

Theoretical background

The system logs (the event logs) record the events of the operation of the system, the applications and the security. The analysis of the logs is an important method of the diagnostics that allows one to detect the causes of the problems and to forecast the failures.

In Windows, the logs are stored in the Event Viewer and are divided into the logs of the system, the applications and the security. In Linux, the logs are stored in the directory /var/log in the form of the text files. Each event has the level (the information, the warning, the error).

Equipment and materials

- A computer with the Windows and (or) Linux operating system.
- The Event Viewer (Windows) and the access to /var/log (Linux).
- Tools for the analysis of the logs.
- A report form.

Procedure

1. Study the structure of the system logs in Windows (the Event Viewer) and the levels of the events.

2. View the log of the system and detect the warnings and the errors.
3. Analyse the detected errors and determine their possible causes.
4. Study the structure of the logs in Linux (/var/log) and the main files.
5. Find in the logs the events connected with the problems of the equipment or the software.
6. Make a conclusion about the application of the analysis of the logs for the diagnostics.

Tasks for independent completion

1. Describe the structure of the system logs in Windows and Linux.
2. Analyse the log and detect the events connected with the problems.
3. Compile a list of the typical events that indicate the problems.

Conclusion: As a result of the work, the student has formed the skills of analysing the system logs of an operating system for the diagnostics of the problems.

Control Questions

1. What do the system logs record?
2. How are the logs organized in Windows and Linux?
3. What levels of the events exist?
4. How does the analysis of the logs help in the diagnostics?
5. What events indicate the problems?

Practical work 1.5.30 Use of Python for reading and processing logs

Objective: To form practical skills in using Python for the automated reading, processing and analysis of the logs.

Theoretical background

Python is convenient for the automated processing of the logs: the reading of the files, the search for the events by the patterns, the filtering, the counting of the statistics and the formation of the reports. For this, the standard means of the language and the modules re (the regular expressions), and for Windows the modules for the work with the Event Log are used.

The automated analysis of the logs allows one to quickly process the large volumes of the data, to detect the patterns and the anomalies and to form the reports, which is impossible to do manually.

Equipment and materials

- A computer with the installed Python.
- Files of the logs for the analysis.
- A code editor.
- A report form.

Procedure

1. Study the means of Python for the reading and the processing of the text files.
2. Write a fragment of the script for the reading of a log file.
3. Add the search for the events by the pattern with the help of the regular expressions.
4. Add the filtering of the events by the level (the errors, the warnings) and the counting of the statistics.
5. Add the formation of a report on the detected events.
6. Make a conclusion about the capabilities of Python for the analysis of the logs.

Tasks for independent completion

1. Write a Python script for the reading and the filtering of a log file.
2. Add the search for the events by the pattern and the counting of the statistics.
3. Form a report on the detected events.

Conclusion: As a result of the work, the student has formed the skills of using Python for the automated reading, processing and analysis of the logs.

Control Questions

1. Why is Python convenient for the processing of the logs?
2. What means are used for the reading of the files and the search for the events?
3. What are the regular expressions, and how are they used?
4. How is the filtering of the events by the level performed?
5. What are the advantages of the automated analysis of the logs?

Practical work 1.5.31 The fundamentals of the system monitoring: parameters, methods, tools

Objective: To form practical skills in organizing the system monitoring of a computer: the choice of the parameters, the methods and the tools.

Theoretical background

The system monitoring is the continuous tracking of the parameters of the state of a computer system: the load of the processor, the use of the memory and the disks, the network activity, the temperatures and the availability of the services. The monitoring allows one to detect the problems in a timely manner.

The methods of the monitoring include the periodic polling of the parameters, the tracking of the events and the setting of the thresholds with the notification of their excess. The tools include the system utilities, the scripts and the specialized monitoring systems.

Equipment and materials

- A computer for the monitoring.
- The system tools and the monitoring utilities.
- Python with the psutil library.
- A report form.

Procedure

1. Determine the parameters of the system that are subject to the monitoring.
2. Study the system tools of the monitoring (the task manager, the resource monitor).
3. Configure the monitoring of the parameters with the help of the utilities.
4. Write a fragment of the script for the periodic collection of the parameters with the help of psutil.
5. Configure the thresholds of the parameters and the notification of their excess.
6. Make a conclusion about the methods and the tools of the system monitoring.

Tasks for independent completion

1. Compile a list of the parameters for the monitoring of a computer system.
2. Configure the monitoring of the parameters with the help of the tools.
3. Write a fragment of the script for the collection of the parameters.

Conclusion: As a result of the work, the student has formed the skills of organizing the system monitoring of a computer: the choice of the parameters, the methods and the tools.

Control Questions

1. What is the system monitoring?
2. What parameters are subject to the monitoring?
3. What methods of the monitoring exist?
4. What tools are used for the monitoring?
5. How are the thresholds of the parameters and the notifications configured?

Practical work 1.5.32 PowerShell for the diagnostics of the equipment: basic commands

Objective: To form practical skills in using the basic commands of PowerShell for the diagnostics of the equipment of a computer.

Theoretical background

PowerShell provides the cmdlets for obtaining the information about the equipment through WMI/CIM: Get-CimInstance with the various classes, Get-ComputerInfo, Get-PhysicalDisk, Get-Volume and others. These cmdlets allow one to diagnose the equipment in an automated way.

The basic commands of PowerShell for the diagnostics form the foundation for the creation of the scripts of the automated diagnostics and the monitoring of the equipment.

Equipment and materials

- A computer with the Windows operating system.
- PowerShell.
- Examples of the diagnostics cmdlets.
- A report form.

Procedure

1. Study the basic cmdlets of PowerShell for obtaining the information about the equipment.
2. Obtain the information about the processor and the memory with the help of Get-CimInstance.
3. Obtain the information about the disks with the help of Get-PhysicalDisk and Get-Volume.
4. Obtain the general information about the computer with the help of Get-ComputerInfo.
5. Write a fragment of the script that combines several commands of the diagnostics.
6. Make a conclusion about the capabilities of PowerShell for the diagnostics of the equipment.

Tasks for independent completion

1. Compile a list of the basic cmdlets of PowerShell for the diagnostics of the equipment.
2. Obtain the information about the main components with the help of PowerShell.
3. Write a fragment of the script of the diagnostics of the equipment.

Conclusion: As a result of the work, the student has formed the skills of using the basic commands of PowerShell for the diagnostics of the equipment of a computer.

Control Questions

1. What cmdlets of PowerShell are used for the diagnostics of the equipment?
2. How is the information about the processor and the memory obtained?
3. How is the information about the disks obtained?
4. What does the cmdlet Get-ComputerInfo provide?
5. How are several commands of the diagnostics combined in a script?

Practical work 1.5.33 Creation of the first monitoring scripts in Python

Objective: To form practical skills in creating the first scripts of the monitoring of the state of a computer in Python.

Theoretical background

A monitoring script in Python periodically collects the parameters of the state of the system (the load of the processor, the use of the memory and the disks, the network activity) with the help of the psutil library and displays or saves them.

The basis of a monitoring script is a cycle with a delay, in which the parameters are collected at the specified interval. The collected data can be displayed on the screen, saved to a file or analysed.

Equipment and materials

- A computer with the installed Python and the psutil library.
- A code editor.
- Examples of the monitoring scripts.
- A report form.

Procedure

1. Study the functions of the psutil library for obtaining the parameters of the system.
2. Write a fragment of the script for obtaining the load of the processor and the use of the memory.
3. Add a cycle with a delay for the periodic collection of the parameters.
4. Add the output of the parameters on the screen and the saving to a file.
5. Run the monitoring script and observe its operation.
6. Make a conclusion about the creation of the monitoring scripts in Python.

Tasks for independent completion

1. Write a Python script for the periodic monitoring of the load of the processor and the memory.
2. Add the saving of the parameters to a file.
3. Describe the structure of a monitoring script.

Conclusion: As a result of the work, the student has formed the skills of creating the scripts of the monitoring of the state of a computer in Python.

Control Questions

1. What does a monitoring script in Python collect?
2. What functions of psutil are used for obtaining the parameters?
3. How is the periodic collection of the parameters organized?
4. How are the collected parameters saved?
5. What is the structure of a monitoring script?

Practical work 1.5.34 Work with the Windows API by means of Python (WMI)

Objective: To form practical skills in working with the Windows API through WMI by means of Python for obtaining the information about the system and managing it.

Theoretical background

Python can interact with the Windows API through WMI with the help of the WMI library (a wrapper over the COM interface of WMI). This allows one to obtain the information about the equipment and the system and to perform the operations of the management directly from a Python script.

The WMI library provides a convenient access to the WMI classes from Python: the obtaining of the instances, the reading of the properties and the call of the methods. This combines the power of WMI and the convenience of Python.

Equipment and materials

- A computer with the Windows operating system and the installed Python.
- The WMI library for Python.
- A code editor.
- A report form.

Procedure

1. Install the WMI library for Python and study its capabilities.
2. Write a fragment of the script for the connection to WMI and obtaining the information about the processor.
3. Obtain the information about the memory, the disks and the operating system through WMI.
4. Obtain the information about the services and the processes through WMI.

5. Form a report on the system on the basis of the data of WMI.
6. Make a conclusion about the capabilities of the work with WMI from Python.

Tasks for independent completion

1. Write a Python script for obtaining the information about the system through WMI.
2. Obtain the information about several components through WMI.
3. Form a report on the system on the basis of the data of WMI.

Conclusion: As a result of the work, the student has formed the skills of working with the Windows API through WMI by means of Python.

Control Questions

1. How does Python interact with WMI?
2. What does the WMI library provide?
3. How is the connection to WMI from Python performed?
4. What information can be obtained through WMI from Python?
5. What are the advantages of the combination of WMI and Python?

Practical work 1.5.35 Reading and changing the parameters of the system through winreg

Objective: To form practical skills in reading and changing the parameters of the system through the Windows registry with the help of the winreg module of Python.

Theoretical background

The Windows registry is a hierarchical database of the settings of the operating system and the programs. The winreg module of Python allows one to read and to change the parameters of the registry, which is used for the configuration and the automation.

The work with the registry requires caution: the incorrect changes may disrupt the operation of the system. Before the changes, it is recommended to make a backup of the registry.

Equipment and materials

- A computer with the Windows operating system and the installed Python.
- The winreg module (built into Python).
- A code editor.
- A report form.

Procedure

1. Study the structure of the Windows registry and the main sections.
2. Write a fragment of the script for the reading of a parameter of the registry with the help of winreg.
3. Read several system parameters from the registry.
4. Write a fragment of the script for the change of a parameter of the registry (on a test key).
5. Observe the safety measures: make a backup before the change.
6. Make a conclusion about the capabilities of the work with the registry through winreg.

Tasks for independent completion

1. Write a Python script for the reading of the parameters of the registry.
2. Demonstrate the change of a parameter of the registry on a test key.
3. Describe the safety measures when working with the registry.

Conclusion: As a result of the work, the student has formed the skills of reading and changing the parameters of the system through the registry with the help of the winreg module of Python.

Control Questions

1. What is the Windows registry?
2. What does the winreg module allow one to do?

3. How is a parameter of the registry read?
4. What safety measures must be observed when changing the registry?
5. Why is it recommended to make a backup before the changes?

Practical work 1.5.36 Management of the processes and the services through subprocess

Objective: To form practical skills in managing the processes and the services of an operating system with the help of the subprocess module of Python.

Theoretical background

The subprocess module of Python allows one to launch the external programs and the system commands from a script and to obtain their result. This is used for the automation of the management of the processes, the services and the execution of the system commands.

Through subprocess, one can launch the system utilities (tasklist, sc, net), obtain their output and process it. This combines the capabilities of the system commands and the logic of Python.

Equipment and materials

- A computer with the Windows operating system and the installed Python.
- The subprocess module (built into Python).
- A code editor.
- A report form.

Procedure

1. Study the capabilities of the subprocess module for the launch of the system commands.
2. Write a fragment of the script for obtaining the list of the processes with the help of subprocess.
3. Write a fragment of the script for the management of a service (the start, the stop) through the system commands.
4. Process the result of the execution of the commands in the script.
5. Add the handling of the errors of the execution of the commands.
6. Make a conclusion about the management of the processes and the services through subprocess.

Tasks for independent completion

1. Write a Python script for obtaining the list of the processes through subprocess.
2. Demonstrate the management of a service from a script.
3. Add the handling of the errors to the script.

Conclusion: As a result of the work, the student has formed the skills of managing the processes and the services of an operating system with the help of the subprocess module of Python.

Control Questions

1. What does the subprocess module allow one to do?
2. How is a system command launched from a script?
3. How is the result of the execution of a command obtained?
4. How are the processes and the services managed through subprocess?
5. Why is the handling of the errors of the execution of the commands necessary?

Practical work 1.5.37 Scripts for the cleaning and the optimization of the state of the system

Objective: To form practical skills in creating the scripts for the cleaning and the optimization of the state of a computer system.

Theoretical background

The scripts for the cleaning and the optimization automate the routine tasks of the maintenance: the removal of the temporary files, the cleaning of the caches, the optimization of the autostart and the freeing of the disk space. This maintains the performance of the system.

The cleaning scripts must be careful and remove only the files that are knowingly unnecessary, in order not to damage the system. Before the mass removal, it is recommended to check the list of the files.

Equipment and materials

- A computer for the cleaning and the optimization.
- Python or PowerShell.
- A code editor.
- A report form.

Procedure

1. Determine the objects of the cleaning: the temporary files, the caches, the logs, the recycle bin.
2. Write a fragment of the script for the search and the removal of the temporary files.
3. Add to the script the cleaning of the caches and the freeing of the disk space.
4. Add the safety measures: the check of the files before the removal and the logging.
5. Test the script and assess the freed space.
6. Make a conclusion about the automation of the cleaning and the optimization of the system.

Tasks for independent completion

1. Write a script for the cleaning of the temporary files of the system.
2. Add the logging of the removed files to the script.
3. Assess the volume of the freed disk space.

Conclusion: As a result of the work, the student has formed the skills of creating the scripts for the cleaning and the optimization of the state of a computer system.

Control Questions

1. What tasks do the cleaning and optimization scripts automate?
2. What objects are subject to the cleaning?
3. Why must the cleaning scripts be careful?
4. Why is the logging of the removed files necessary?
5. How is the result of the cleaning assessed?

Practical work 1.5.38 Automatic diagnostics of network problems

Objective: To form practical skills in creating the scripts for the automatic diagnostics of network problems.

Theoretical background

A script for the automatic diagnostics of the network checks the main aspects of the network connection: the availability of the gateway and the external nodes, the correctness of the settings, the operation of the DNS. This allows one to quickly determine the cause of a network problem.

The diagnostics is built sequentially: from the check of the local connection to the check of the access to the Internet, which allows one to localize the problem at a specific level.

Equipment and materials

- A computer with a network connection.
- Python or PowerShell.
- A code editor.
- A report form.

Procedure

1. Determine the sequence of the checks for the diagnostics of the network (the adapter, the settings, the gateway, the DNS, the Internet).
2. Write a fragment of the script for the check of the availability of the gateway and the external nodes.
3. Add the check of the operation of the DNS (the resolution of the names).
4. Add the analysis of the results and the determination of the level of the problem.
5. Test the script on a working and a model-broken network.
6. Make a conclusion about the automatic diagnostics of the network problems.

Tasks for independent completion

1. Write a script for the automatic diagnostics of a network connection.
2. Add the determination of the level of the problem to the script.
3. Test the script and describe the result.

Conclusion: As a result of the work, the student has formed the skills of creating the scripts for the automatic diagnostics of network problems.

Control Questions

1. What does a script for the diagnostics of the network check?
2. In what sequence is the diagnostics of the network built?
3. How is the operation of the DNS checked?
4. How is the level of a network problem determined?
5. What are the advantages of the automatic diagnostics of the network?

Practical work 1.5.39 Methods of the restoration of network connections through PowerShell

Objective: To form practical skills in applying the methods of the restoration of network connections with the help of PowerShell.

Theoretical background

PowerShell provides the cmdlets for the restoration of the network: Restart-NetAdapter (the restart of the adapter), the renewal of the IP address, the reset of the DNS cache (Clear-DnsClientCache), the reset of the network stack. These methods allow one to eliminate the typical network problems.

The script of the restoration of the network combines the diagnostics and the consistent application of the methods of the restoration until the connection is restored.

Equipment and materials

- A computer with the Windows operating system and a network connection.
- PowerShell with administrator rights.
- Examples of the cmdlets of the restoration.
- A report form.

Procedure

1. Study the cmdlets of PowerShell for the restoration of the network.
2. Perform the restart of the network adapter with the cmdlet Restart-NetAdapter.
3. Perform the renewal of the IP address and the reset of the DNS cache.
4. Write a fragment of the script that consistently applies the methods of the restoration.
5. Test the script on a model-broken network connection.
6. Make a conclusion about the methods of the restoration of the network through PowerShell.

Tasks for independent completion

1. Compile a list of the methods of the restoration of the network in PowerShell.
2. Write a script for the consistent restoration of a network connection.
3. Test the script and describe its operation.

Conclusion: As a result of the work, the student has formed the skills of applying the methods of the restoration of network connections with the help of PowerShell.

Control Questions

1. What cmdlets are used for the restoration of the network?
2. How is the restart of a network adapter performed?
3. What does the reset of the DNS cache give?
4. How does a script of the restoration of the network work?
5. In what sequence are the methods of the restoration applied?

Practical work 1.5.40 Utilities for the check of the file system (Python + system tools)

Objective: To form practical skills in creating the utilities for the check of the file system with the combination of Python and the system tools.

Theoretical background

The check of the file system reveals and corrects the errors that arise as a result of the incorrect shutdowns and the faults. The system tools (chkdsk in Windows, fsck in Linux) perform the check, and Python automates their launch and the analysis of the results.

The combination of Python and the system tools allows one to create the utilities that automatically check the file system, analyse the result and form a report.

Equipment and materials

- A computer with a file system.
- Python and the system tools of the check.
- A code editor.
- A report form.

Procedure

1. Study the system tools of the check of the file system (chkdsk, fsck).
2. Write a fragment of the script for the launch of the check of the file system through subprocess.
3. Add the analysis of the result of the check and the extraction of the information about the errors.
4. Add the formation of a report on the state of the file system.
5. Test the utility and analyse the report.
6. Make a conclusion about the automation of the check of the file system.

Tasks for independent completion

1. Write a utility for the launch of the check of the file system and the analysis of the result.
2. Form a report on the state of the file system.
3. Describe the system tools of the check of the file system.

Conclusion: As a result of the work, the student has formed the skills of creating the utilities for the check of the file system with the combination of Python and the system tools.

Control Questions

1. What does the check of the file system reveal?
2. What system tools are used for the check?
3. How does Python automate the launch of the check?
4. How is the result of the check analysed?
5. What does a report on the state of the file system contain?

Practical work 1.5.41 Diagnostics and fixing of the problems of the drivers

Objective: To form practical skills in diagnosing and fixing the problems of the drivers of the devices.

Theoretical background

The drivers ensure the interaction of the operating system with the devices. The problems of the drivers (the absence, the outdated version, the conflict, the corruption) lead to the incorrect operation or the inoperability of the devices.

The diagnostics of the drivers is performed through the device manager, which shows the devices with the problems, and through the system tools. The fixing consists in the installation, the update or the rollback of the drivers.

Equipment and materials

- A computer with the devices and the drivers.
- The device manager and the system tools.
- PowerShell for the automation.
- A report form.

Procedure

1. Study the device manager and detect the devices with the problems of the drivers.
2. Determine the type of the problem of the driver (the absence, the outdated version, the conflict).
3. Perform the update of the driver of the device.
4. Perform the rollback of the driver to the previous version (model the situation).
5. Obtain the information about the drivers with the help of PowerShell (Get-PnpDevice).
6. Make a conclusion about the diagnostics and the fixing of the problems of the drivers.

Tasks for independent completion

1. Detect the devices with the problems of the drivers in the device manager.
2. Perform the update or the rollback of a driver.
3. Obtain the information about the drivers with the help of PowerShell.

Conclusion: As a result of the work, the student has formed the skills of diagnosing and fixing the problems of the drivers of the devices.

Control Questions

1. What do the drivers ensure?
2. What problems of the drivers exist?
3. How are the problems of the drivers detected?
4. How are the problems of the drivers fixed?
5. How is the information about the drivers obtained with PowerShell?

Practical work 1.5.42 Creation of a Python utility for the comprehensive check of the system

Objective: To form practical skills in creating a Python utility for the comprehensive check of the state of a computer system.

Theoretical background

A utility for the comprehensive check combines the checks of the various aspects of the system: the equipment, the file system, the network, the drivers, the security. It collects the data, analyses them and forms a comprehensive report on the state of the system.

The combination of several checks in one utility requires the modular structure: each check is implemented as a separate function, and the main program calls them and combines the results.

Equipment and materials

- A computer with the installed Python and the libraries.
- A code editor.

- Reference data on the norms.
- A report form.

Procedure

1. Design the modular structure of the utility: the functions of the checks and the main program.
2. Implement the function of the check of the equipment (the processor, the memory, the disks) with psutil.
3. Implement the function of the check of the file system and the network.
4. Implement the formation of the comprehensive report combining all the checks.
5. Test the utility and analyse the report.
6. Make a conclusion about the creation of the comprehensive utility of the check.

Tasks for independent completion

1. Create a Python utility for the comprehensive check of the system.
2. Implement at least three different checks in the utility.
3. Form a comprehensive report on the state of the system.

Conclusion: As a result of the work, the student has formed the skills of creating a Python utility for the comprehensive check of the state of a computer system.

Control Questions

1. What checks does a utility for the comprehensive check combine?
2. Why is the modular structure of the utility necessary?
3. How is each check implemented?
4. How is the comprehensive report formed?
5. What are the advantages of the comprehensive check?

Practical work 1.5.43 Integration of Python and PowerShell in one maintenance process

Objective: To form practical skills in integrating Python and PowerShell in one process of the maintenance of a computer system.

Theoretical background

Python and PowerShell complement each other: PowerShell provides a convenient access to the Windows components, and Python provides the rich capabilities of the processing of the data and the logic. Their integration allows one to combine the strengths of both languages.

The integration is performed by the launch of the PowerShell commands from Python (through subprocess) or the Python scripts from PowerShell. This allows one to use the optimal tool for each task.

Equipment and materials

- A computer with the installed Python and PowerShell.
- A code editor.
- Examples of the integration.
- A report form.

Procedure

1. Study the methods of the integration of Python and PowerShell.
2. Write a fragment of the Python script that launches a PowerShell command and obtains the result.
3. Process the result of the PowerShell command in the Python script.
4. Create a maintenance process that combines the capabilities of both languages.
5. Test the integrated process and analyse its operation.
6. Make a conclusion about the integration of Python and PowerShell.

Tasks for independent completion

1. Write a Python script that launches a PowerShell command.
2. Process the result of the PowerShell command in Python.
3. Create a maintenance process combining both languages.

Conclusion: As a result of the work, the student has formed the skills of integrating Python and PowerShell in one process of the maintenance of a computer system.

Control Questions

1. How do Python and PowerShell complement each other?
2. How is the integration of Python and PowerShell performed?
3. How is a PowerShell command launched from Python?
4. How is the result of a PowerShell command processed in Python?
5. What are the advantages of the integration of the two languages?

Practical work 1.5.44 Creation of an automated recovery scenario (a mini-project)

Objective: To form practical skills in creating an automated scenario of the restoration of the operability of a computer system as a mini-project.

Theoretical background

An automated scenario of the restoration combines the diagnostics, the decision-making and the actions of the restoration in a single script. It detects a problem and automatically performs the actions for its elimination, minimizing the participation of the person.

The mini-project of the automated restoration integrates the studied skills: the collection of the data, the analysis, the actions of the restoration, the logging and the formation of the report. This is a comprehensive task that demonstrates the mastery of the automation.

Equipment and materials

- A computer with the installed Python and (or) PowerShell.
- A code editor.
- A description of the scenario of the restoration.
- A report form.

Procedure

1. Choose a typical problem for the automated restoration (for example, the restoration of the network or the cleaning of the system).
2. Design the scenario of the restoration: the diagnostics, the conditions, the actions, the check of the result.
3. Implement the diagnostics of the problem in the script.
4. Implement the actions of the restoration and the check of the result.
5. Add the logging and the formation of the report and test the scenario.
6. Make a conclusion about the developed automated scenario of the restoration.

Tasks for independent completion

1. Develop an automated scenario of the restoration for a chosen problem.
2. Implement the diagnostics, the restoration and the check of the result in the script.
3. Test the scenario and describe its operation.

Conclusion: As a result of the work, the student has formed the skills of creating an automated scenario of the restoration of the operability of a computer system as a mini-project.

Control Questions

1. What does an automated scenario of the restoration combine?
2. How is the scenario of the restoration designed?

3. How is the diagnostics of the problem implemented in the script?
4. How is the check of the result of the restoration implemented?
5. Why are the logging and the report necessary in the scenario?

Practical work 1.5.45 Testing and demonstration of the operability of the developed utility

Objective: To consolidate and check, in a comprehensive form, the skills of the development of the means of the automation by testing and demonstrating the operability of the developed utility, summing up the learning outcome.

Theoretical background

The testing of the developed utility confirms its operability and reveals the errors. The testing is performed on the various scenarios, including the boundary and the erroneous situations. The demonstration shows the operation of the utility in the real conditions.

This lesson generalizes the material of the learning outcome on the restoration and the prevention of the failures and the automation of the maintenance. The student demonstrates the developed utility and assesses its operability.

Equipment and materials

- The utility developed in the previous lessons.
- A computer for the testing.
- Test scenarios.
- A final-assessment form.

Procedure

1. Prepare the developed utility for the testing and the demonstration.
2. Compile a plan of the testing covering the main scenarios and the boundary situations.
3. Perform the testing of the utility on the various scenarios.
4. Detect and fix the errors revealed during the testing.
5. Demonstrate the operation of the utility and assess its operability.
6. Make a final conclusion about the developed utility and its application.

Tasks for independent completion

1. Compile a plan of the testing of the developed utility.
2. Perform the testing and fix the revealed errors.
3. Demonstrate the operation of the utility and prepare a report.

Conclusion: As a result of the work, the student has consolidated the comprehensive skills of the development, the testing and the demonstration of the means of the automation of the maintenance of computer systems.

Control Questions

1. What does the testing of the developed utility confirm?
2. How is the plan of the testing compiled?
3. What scenarios must be covered by the testing?
4. How are the errors revealed during the testing fixed?
5. What does the demonstration of the operability of the utility show?

5. LIST OF SOFTWARE AND MULTIMEDIA SUPPORT FOR TRAINING SESSIONS

- Multimedia projector and interactive (smart) board; digital podium; white marker board.

- PowerPoint presentations for every topic of the module.
- Personal computers with Internet access and access to local networks.
- Disassembled training PC stands; sets of motherboards, power supplies, RAM modules, storage drives and graphics adapters.
- Peripheral and office equipment: printers, scanners, multifunction devices (MFPs), plotters.
- Measuring instruments: multimeters, POST-cards, power-supply and cable testers.
- Anti-static wrist straps and mats; PC assembly and repair tool kits; soldering stations.
- Hardware-diagnostic software: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo, HWMonitor, OCCT.
- Programming and scripting tools: Python 3 (with psutil, WMI, pywin32), PowerShell, Windows Subsystem for Linux.
- Consumables of office equipment (cartridges, toner, ink, ribbons) for industrial training.
- Learning Management System (LMS) for hosting materials and assignments; video lectures on PC maintenance and repair.

6. LIST OF LITERATURE

Core (fundamental) literature:

1. Phil Hughes MBE, Ed Ferrett. Introduction to Health and Safety at Work. ISBN: 978-0-415-72308-4.
2. Mark A. Friend, James P. Kohn. Fundamentals of Occupational Safety and Health. ISBN-13: 978-0-86587-171-7.
3. Charles K. Alexander, Matthew N. O. Sadiku. Fundamentals of Electric Circuits. ISBN: 978-0-07-352955-4.
4. John L. Hennessy, David A. Patterson. Computer Architecture: A Quantitative Approach (5th Edition). ISBN-13: 978-8178672663.
5. Jim Ledin. Modern Computer Architecture and Organization (2nd Edition). ISBN-13: 978-1803234519.

Additional literature:

1. Saurabh Shrivastava, Neelanjali Srivastav. Solutions Architect's Handbook (2nd Edition). ISBN-13: 978-1801816618.
2. Joseph Ingeno. Software Architect's Handbook. ISBN-13: 978-1788624060.
3. Robert C. Martin. Clean Architecture: A Craftsman's Guide to Software Structure and Design. ISBN-13: 978-0134494166.

Internet resources:

1. Computer Architecture Topics — https://en.wikipedia.org/wiki/Computer_architecture
2. Fire Protection and Prevention (OSHA) — <https://www.osha.gov/sites/default/files/2019-03/fireprotection.pdf>
3. <https://learn.microsoft.com> — official Microsoft documentation on Windows and administration.
4. <https://support.microsoft.com> — technical support and instructions for configuring hardware and software.
5. <https://docs.python.org> — documentation on automation and system administration in Python.

VII. LO 1.6 Replace the consumables used in computing and office equipment

1. ACADEMIC CALENDAR

LO 1.6 Replace the consumables used in computing and office equipment

Instructor: D.M. Yermekov

Specialty: 06130100 – «Software Engineering (by types)»

Total hours: 36 hours (18 lessons of 2 academic hours each)

№	Name of topics	Date of lesson	Type of activity
1	The system unit. The hard disk drive and its characteristics. Memory modules. Processors and their types		Study of new material
2	Maintenance and replacement of the batteries in uninterruptible power supplies (UPS)		Combined lesson
3	The role of consumables in computing and office equipment		Application and consolidation of knowledge
4	Replacement and care of the cartridges and the toner in printers and copiers		Skills-improvement lesson
5	Replacement and care of the print heads, thermal films, thermal ribbons and thermal heads in printers		Study of new material
6	Assembly of a computer system		Combined lesson
7	Replacement and care of keyboards, mice, batteries and accumulators of laptops and mobile devices		Application and consolidation of knowledge
8	Replacement and care of monitors: cleaning and adjustment		Skills-improvement lesson
9	Replacement and care of the filters and the fans in computers		Study of new material
10	Replacement and care of the battery elements in office equipment, buffer batteries in servers and network devices		Skills-improvement lesson
11	Replacement and care of the lamps and the LEDs in scanners and printers		Application and consolidation of knowledge
12	Replacement and care of the rubber rollers, the rollers and the thermal sensors in printing devices		Skills-improvement lesson
13	Replacement and care of the cleaning blades and elements in plotters		Combined lesson
14	Replacement and care of the toner drums, the filters and the separators in laser printers		Skills-improvement lesson

15	Replacement and care of the anodes, cathodes, corona wires and rollers in scanners, fax machines and laser printers		Application and consolidation of knowledge
16	Organization of the stock of consumables and their efficient use		Application and consolidation of knowledge
17	Analysis of the methods of the preventive maintenance of computer and office equipment		Application and consolidation of knowledge
18	Analysis of the methods of the maintenance of the consumables of office equipment		Comprehensive assessment of knowledge

2. GLOSSARY

The glossary contains the main definitions, terms and concepts that reflect the core content of this learning outcome. The terms are arranged in alphabetical order.

Term	Definition
Cartridge	A replaceable container with toner or ink, used in printers and copiers for forming an image on paper.
Consumable	A material that is used up in the course of the operation of equipment and is subject to periodic replacement (toner, ink, paper, batteries, rollers).
Drum (photoconductor)	A photosensitive cylinder of a laser printer onto which the image is formed by a laser before being transferred to paper.
Fuser	The unit of a laser printer that fixes the toner on the paper by heating and pressure (thermal film, rollers, thermal head).
Ink	A liquid colouring material used in inkjet printers.
Thermal head	A printing element of a thermal printer that forms an image by heating the points of a thermal-sensitive material.
Toner	A fine powder used in laser printers and copiers for forming an image.
UPS (Uninterruptible Power Supply)	A device that provides the power supply of equipment from batteries during a power outage and protects against voltage surges.
Roller (platen/pickup)	A rubber or polymer cylinder of a printing device that feeds and moves the paper.
Consumables inventory	The stock of consumables of an enterprise and the system of its accounting and replenishment for the uninterrupted operation of the equipment.

3. LECTURE MATERIALS

LO 1.6 Replace the consumables used in computing and office equipment

Lecture 1.6.1 The system unit. The hard disk drive and its characteristics. Memory modules. Processors and their types

Content

- The composition of the system unit.
- The hard disk drive and its characteristics.
- The memory modules and their types.
- The processors and their types.

The system unit unites the main components of a computer. The understanding of its composition and the characteristics of the components is necessary for the assembly, the maintenance, the modernization and the replacement of the consumables and the wearing elements of the computing equipment.

The hard disk drive (HDD) stores the data on the rotating magnetic plates. Its main characteristics are the capacity (the volume of the stored data), the rotation speed of the plates (which affects the speed of the access), the volume of the cache memory, and the interface of the connection (SATA).

The capacity of modern hard disk drives reaches several terabytes, which makes them suitable for storing large volumes of data. The rotation speed (5400 or 7200 revolutions per minute) affects the performance: the higher the speed, the faster the access to the data.

Along with the HDD, the solid-state drives (SSD) are widely used; they have no moving parts and provide a much higher speed. In modern computers, the SSD is used for the system and the programs, and the HDD for the storage of large volumes of data.

The memory modules (RAM) store the data and the code of the running programs. Their characteristics are the type (DDR4, DDR5), the volume, the frequency and the timings. The volume and the frequency of the memory affect the performance of the computer.

The memory modules are made in the form of the DIMM (for the desktop computers) and the SO-DIMM (for the laptops). The number and the volume of the modules determine the total volume of the random-access memory. The modules must be compatible with the motherboard and the processor.

The processor is the main computing component. Its characteristics are the number of cores and threads, the clock frequency, the volume of the cache memory and the heat-release power. The more cores and the higher the frequency, the higher the performance of the processor.

The processors differ by the manufacturers and the product lines, oriented to the various tasks: from the economical ones for the office computers to the powerful ones for the workstations and the gaming computers. The processor must correspond to the socket of the motherboard.

When choosing and replacing the components, their compatibility is taken into account: the processor must correspond to the socket, the memory to the supported standard, the drives to the available interfaces. The incompatibility of the components leads to the inoperability of the system.

The understanding of the characteristics of the components allows one to competently select them for the assembly and the modernization of a computer, to assess the performance and to determine the components that need to be replaced or upgraded.

The maintenance of the components includes the cleaning of the dust, the replacement of the thermal paste, the check of the state and the timely replacement of the wearing elements. This maintains the operability and prolongs the service life of the equipment.

Thus, the knowledge of the composition of the system unit and the characteristics of the hard disk drive, the memory modules and the processors is the basis for the assembly, the maintenance, the modernization and the replacement of the components of the computing equipment.

Control Questions

1. What components are included in the system unit?
2. What are the main characteristics of a hard disk drive?
3. What characteristics do the memory modules have?
4. What are the main characteristics of a processor?

5. Why is the compatibility of the components important when replacing them?

Lecture 1.6.2 Maintenance and replacement of the batteries in uninterruptible power supplies (UPS)

Content

The purpose and the operating principle of a UPS.

The types of UPS.

The batteries of a UPS and their service life.

The maintenance and the replacement of the batteries.

An uninterruptible power supply (UPS) provides the power supply of the equipment from the batteries during a power outage and protects against the voltage surges. The UPS is critically important for the servers, the workstations and the network equipment, since it prevents the loss of data and the damage of the equipment.

The operating principle of a UPS consists in the constant charging of the batteries from the mains and the switching to the power supply from the batteries when the mains voltage disappears or goes beyond the permissible limits. This provides the equipment with the time for the correct shutdown.

There are several types of UPS. The off-line (standby) UPS switches to the batteries only when the mains disappears. The line-interactive UPS additionally stabilizes the voltage. The on-line (double-conversion) UPS constantly powers the equipment from the converted voltage, providing the best protection.

The choice of the type of UPS depends on the requirements for the protection and the budget. For the office computers, the line-interactive UPS is sufficient; for the servers and the critical equipment, the on-line UPS is used, which provides the constant clean voltage.

The batteries are the heart of a UPS and its most wearing element. In most UPS, the sealed lead-acid batteries are used. Their service life is limited (usually 3-5 years) and depends on the operating conditions, especially the temperature.

The signs of the wear of the batteries are the reduction of the operating time from the batteries, the frequent indication of the need for the replacement, and the failure of the self-test. The worn batteries cannot provide the equipment with the energy for the necessary time.

The high temperature accelerates the degradation of the batteries, so the UPS must be installed in a cool, ventilated place. The deep discharges and the long storage without the charging also reduce the service life of the batteries.

The maintenance of a UPS includes the periodic check of the state of the batteries, the test of the operation from the batteries, the cleaning of the dust and the check of the connections. The modern UPS have the means of the self-diagnostics and notify of the need for the replacement of the batteries.

The replacement of the batteries is performed when they are worn out. Before the replacement, the UPS is turned off and de-energized (where possible), and the safety measures are observed, since the batteries store a charge. The new batteries must correspond to the original ones in their parameters.

After the replacement of the batteries, the UPS is tested: the charging of the new batteries and the check of the operation from them. The correct replacement restores the protective functions of the UPS and ensures the reliable operation of the equipment.

The utilization of the old lead-acid batteries is performed in accordance with the ecological requirements, since they contain harmful substances. The batteries are handed over to the specialized points of the reception and the recycling.

Thus, the maintenance and the replacement of the batteries of a UPS ensure the reliable protection of the equipment against the power outages and the voltage surges, which prevents the loss of data and the damage of the equipment.

Control Questions

1. What is a UPS for, and how does it work?
2. What types of UPS exist, and what is the difference between them?
3. What batteries are used in a UPS, and what is their service life?
4. What are the signs of the wear of the batteries of a UPS?
5. How is the replacement of the batteries of a UPS performed?

Lecture 1.6.5 Replacement and care of the print heads, thermal films, thermal ribbons and thermal heads in printers

Content

The printing elements of various types of printers.

The print heads of inkjet printers.

The thermal films and the fuser of laser printers.

The thermal heads and the thermal ribbons of thermal printers.

Various types of printers use various printing elements that form the image on the paper. The understanding of their design, the operating principle and the methods of the replacement and the care is necessary for the maintenance of the printing equipment.

The print head of an inkjet printer forms the image by spraying the microscopic drops of the ink onto the paper. The head contains the nozzles through which the ink is supplied. The clogging of the nozzles is the main problem of the inkjet printers.

The print heads are of two types: built into the cartridge (replaced together with it) and permanent (installed in the printer). The permanent heads require the careful care, since their replacement is expensive.

The care of the print heads includes the regular printing (to prevent the drying of the ink), the cleaning of the nozzles by the means of the printer, and the deep cleaning when clogged. The timely care prevents the clogging of the nozzles.

The fuser of a laser printer fixes the toner on the paper by the heating and the pressure. It includes the thermal film or the thermal roller and the pressure roller. The thermal film is a consumable that wears out over time.

The thermal film of the fuser is a thin heat-resistant film that transfers the heat to the toner. Its wear or damage leads to the deterioration of the fixing of the toner (the printing is smeared) and requires the replacement.

The thermal printers form the image by the heating of the points of the thermal-sensitive material. They are used in the cash registers, the label printers and the fax machines. The printing element of a thermal printer is the thermal head.

The thermal head of a thermal printer contains the heating elements that form the image. Its care includes the regular cleaning of the deposits, since the contamination worsens the quality of the printing. The thermal head wears out and requires the replacement over time.

The thermal-transfer printers use the thermal ribbon — a film with a colouring layer that is transferred to the paper by the heating. The thermal ribbon is a consumable that is replaced as it is used up.

The replacement of the printing elements requires the observance of the technology and the safety measures (the fuser is heated to a high temperature). The replacement is performed with the suitable elements, observing the correct installation.

The care of the printing elements prolongs their service life and maintains the quality of the printing. The regular cleaning, the correct operation and the timely replacement of the worn elements are the basis of the maintenance of the printing equipment.

Thus, the various types of printers use various printing elements — the print heads, the thermal films, the thermal heads and the thermal ribbons — the replacement and the care of which require the knowledge of their design and the technology of the maintenance.

Control Questions

1. What printing elements do various types of printers use?
2. How does the print head of an inkjet printer work, and what is its main problem?
3. What is the fuser of a laser printer for, and what does it include?
4. How do thermal printers form the image?
5. What does the care of the printing elements include?

Lecture 1.6.6 Assembly of a computer system

Content

The preparation for the assembly of a computer.

The sequence of the assembly.

The connection of the components and the cables.

The check of the assembled system.

The assembly of a computer system is the process of installing and connecting the components into a single working device. The correct assembly ensures the operability, the reliability and the convenience of the maintenance of the computer.

The preparation for the assembly includes the selection of the compatible components, the preparation of the tools and the workplace, and the observance of the anti-static measures. The components must be compatible with each other in their interfaces and parameters.

The anti-static measures are mandatory when assembling a computer, since the static electricity can damage the electronic components. An anti-static wrist strap, a mat and the discharge of the static from oneself before the work are used.

The sequence of the assembly is usually as follows: the installation of the processor into the socket of the motherboard, the installation of the cooling system, the installation of the memory modules, the installation of the motherboard into the case, the installation of the drives, the power supply and the expansion cards.

The installation of the processor requires caution: the processor is installed into the socket by the key, without the application of force, after which it is fixed by the lever. The contacts of the processor and the socket must not be touched, in order not to damage them.

After the installation of the processor, the cooling system is installed: the thermal paste is applied to the processor, and the radiator with the fan is fixed. The quality of the thermal contact is critically important for the heat removal and the reliable operation of the processor.

The memory modules are installed into the slots until the latches click, observing the correct orientation by the key. The drives are installed into the bays of the case and are connected to the motherboard and the power supply by the cables.

The connection of the cables requires attentiveness: the power cables of the motherboard, the processor and the components; the data cables of the drives; the cables of the front panel of the case (the power button, the indicators, the USB). The incorrect connection may lead to the inoperability.

The management of the cables (the neat laying and the fixing) improves the air circulation inside the case, simplifies the maintenance and improves the appearance. The neat assembly is a sign of the professional approach.

After the assembly, the check is performed: the correctness of all the connections, the absence of the foreign objects, the reliability of the fixing of the components. Only after this the computer is connected to the mains and is turned on for the first time.

The first start checks the operability of the assembled system: the passage of the POST, the appearance of the image, the recognition of the components. After the successful start, the operating system and the drivers are installed, and the testing of the stability is performed.

Thus, the assembly of a computer system requires the selection of the compatible components, the observance of the anti-static measures and the sequence of the assembly, the correct connection of the components and the cables, and the check of the operability of the assembled system.

Control Questions

1. What does the preparation for the assembly of a computer include?
2. Why are the anti-static measures necessary when assembling a computer?
3. What is the sequence of the assembly of a computer?
4. Why is the quality of the thermal contact of the processor important?
5. What is checked after the assembly before the first start?

Lecture 1.6.9 Replacement and care of the filters and the fans in computers

Content

The role of the filters and the fans in the cooling system.

The contamination of the filters and the fans.

The cleaning and the replacement of the filters.

The maintenance and the replacement of the fans.

The filters and the fans are the important elements of the cooling system of a computer. The fans provide the air circulation that removes the heat, and the dust filters protect the internal components from the contamination. Their maintenance is necessary for the efficient cooling.

The fans create the air flow that removes the heat from the components. In a computer, the fans of the case (for the through ventilation), the processor (on the radiator), the video card and the power supply are used. The malfunction of a fan leads to the overheating.

The dust filters are installed on the air intakes of the case and trap the dust, preventing its penetration inside. This reduces the contamination of the components but requires the regular cleaning of the filters themselves, since the clogged filter worsens the ventilation.

The contamination of the filters and the fans with the dust is the main problem of the cooling system. The dust accumulates on the blades of the fans, the radiators and the filters, worsening the air flow and the heat removal, which leads to the overheating of the components.

The signs of the contamination of the cooling system are the increase of the temperature of the components, the increased noise of the fans (which spin faster), and the dust visible inside the case. The regular check allows one to detect the contamination in a timely manner.

The cleaning of the dust filters is performed regularly: the filters are removed, cleaned of the dust (washed or blown out) and installed back. The regularity of the cleaning depends on the conditions: in the dusty premises, the filters are cleaned more often.

The cleaning of the fans is performed with compressed air, a brush and a vacuum cleaner. The dust is removed from the blades and the radiators. When cleaning, the blades of the fan are held to prevent its spinning, which may damage the bearing.

The fans wear out over time: the bearing wears out, which leads to the increase of the noise, the reduction of the rotation speed and, eventually, the stop of the fan. The worn fans are replaced to ensure the efficient cooling.

The signs of the wear of a fan are the extraneous noise (the rattling, the humming), the reduction of the rotation speed and the periodic stops. The worn fan is replaced with a new one of the corresponding size and parameters.

The replacement of a fan is performed with the observance of the correct orientation of the air flow (the direction of the blowing) and the connection to the corresponding connector of the motherboard. After the replacement, the operation of the fan and the temperature of the components are checked.

The regular maintenance of the filters and the fans (the cleaning and the timely replacement) ensures the efficient cooling, prevents the overheating and prolongs the service life of the components of the computer. This is an important part of the preventive maintenance.

Thus, the filters and the fans are the important elements of the cooling system, the maintenance of which — the regular cleaning and the timely replacement — ensures the efficient cooling and prevents the overheating of the components.

Control Questions

1. What role do the filters and the fans play in the cooling system?
2. What is the main problem of the cooling system connected with the contamination?
3. How is the cleaning of the dust filters performed?
4. What are the signs of the wear of a fan?
5. What is taken into account when replacing a fan?

Lecture 1.6.13 Replacement and care of the cleaning blades and elements in plotters

Content

The purpose and the types of plotters.

The consumables and the wearing elements of plotters.

The cleaning blades and the maintenance elements.

The maintenance of plotters.

A plotter is a device for the printing of the large-format images (the drawings, the posters, the diagrams). The plotters are used in the design, the engineering and the advertising. Their maintenance has its own features connected with the large format and the special consumables.

There are several types of plotters. The inkjet plotters print by the spraying of the ink and are the most widespread. The cutting plotters cut the material along the contour. The pen plotters (the older type) draw with the pens. Each type has its own consumables.

The consumables of the inkjet plotters are the ink and the print heads, similar to the inkjet printers, but of the larger volume. In addition, the large-format paper or the other materials are used. The replacement of these consumables is performed as they are used up.

The cleaning blades and the cleaning elements maintain the cleanliness of the print heads and the other working elements of the plotter. They remove the excess ink and the deposits, ensuring the quality of the printing. Over time, they wear out and require the replacement.

The cleaning system of the inkjet plotters includes the cleaning blades, the absorbers and the parking stations of the heads. These elements prevent the drying of the ink in the nozzles and maintain the cleanliness of the heads, which is critical for the quality of the printing.

The wear of the cleaning blades leads to the deterioration of the cleaning of the heads and, as a consequence, to the deterioration of the quality of the printing (the stripes, the gaps). The timely replacement of the cleaning elements maintains the quality of the printing.

The maintenance of the plotters includes the regular cleaning of the print heads, the replacement of the cleaning elements, the cleaning of the mechanism of the movement, and the check of the feeding of the material. The regularity of the maintenance ensures the reliable operation.

The cleaning of the print heads of a plotter is performed by the means of the device (the automatic cleaning) and, when necessary, manually. The regular cleaning prevents the clogging of the nozzles, which is the main problem of the inkjet devices.

The maintenance of the mechanism of the movement of the carriage and the feeding of the material includes the cleaning of the guides, the lubrication (where required) and the check of the tension of the belts. The smooth movement of the carriage ensures the quality of the printing.

The replacement of the wearing elements (the cleaning blades, the absorbers, the rollers) is performed according to the regulations of the manufacturer or as they wear out. The use of the suitable elements maintains the quality and the reliability of the operation of the plotter.

The observance of the regulations of the maintenance, the use of the suitable consumables and the timely replacement of the wearing elements ensure the long and the reliable operation of the plotter and the high quality of the printing.

Thus, the maintenance of the plotters includes the replacement and the care of the consumables and the wearing elements, including the cleaning blades, which ensures the quality of the printing and the reliable operation of the device.

Control Questions

1. What is a plotter for, and what types of plotters exist?
2. What consumables do the inkjet plotters use?
3. What are the cleaning blades and the cleaning elements for?
4. What does the wear of the cleaning blades lead to?
5. What does the maintenance of the plotters include?

4. INDUSTRIAL TRAINING MATERIALS

LO 1.6 Replace the consumables used in computing and office equipment

Industrial training task 1.6.3 The role of consumables in computing and office equipment

Objective: To form practical skills in determining the types of consumables of computing and office equipment, their purpose and the rules of handling them.

Theoretical background

Consumables are the materials that are used up in the course of the operation of the equipment and are subject to the periodic replacement: the toner, the ink, the paper, the cartridges, the ribbons, the batteries, the rollers and the thermal films. The timely replacement of the consumables ensures the quality of the work of the equipment.

Each type of the equipment uses its own consumables. The laser printers use the toner and the drums; the inkjet printers use the ink; the matrix printers use the ribbons; the UPS use the batteries. The correct choice of the consumables is important for the quality and the reliability.

Equipment and materials

- A set of consumables of various types.
- Office equipment (printers, copiers, scanners).
- Documentation on the consumables.
- A report form.

Procedure

1. Examine the proposed consumables and determine the type and the purpose of each of them.
2. Determine for which equipment each consumable is intended.
3. Study the rules of the storage and the handling of the consumables.
4. Determine the signs of the need for the replacement of the consumables.
5. Compile a table of the consumables with an indication of their purpose and the equipment.
6. Make a conclusion about the role of the consumables in the operation of the equipment.

Tasks for independent completion

1. Compile a table of the consumables indicating their type, purpose and the equipment.
2. Determine the signs of the need for the replacement of each consumable.
3. Describe the rules of the storage and the handling of the consumables.

Conclusion: As a result of the work, the student has formed the skills of determining the types of consumables of computing and office equipment, their purpose and the rules of handling them.

Control Questions

1. What are consumables?
2. What types of consumables are used in office equipment?
3. What consumables does a laser printer use?
4. What are the signs of the need for the replacement of the consumables?
5. What rules of the storage of the consumables must be observed?

Industrial training task 1.6.4 Replacement and care of the cartridges and the toner in printers and copiers

Objective: To form practical skills in replacing and caring for the cartridges and the toner in laser printers and copiers.

Theoretical background

The cartridge of a laser printer contains the toner and, in many models, the photoconductor drum and the developing unit. The replacement of the cartridge is the most frequent operation of the maintenance of laser printers. The toner is a fine powder that is dangerous to inhale.

The care of the cartridges includes the correct storage, the careful handling, the shaking for the even distribution of the toner, and the timely replacement. The refilling of the cartridges is possible but requires the observance of the technology.

Equipment and materials

- A laser printer (or a copier).
- Cartridges for the replacement.
- Means of protection (gloves, a mask).
- A report form.

Procedure

1. Study the design of the cartridge of a laser printer and its elements.
2. Determine the signs of the need for the replacement of the cartridge (the pale printing, the messages of the printer).
3. Extract the old cartridge from the printer, observing the safety measures.
4. Prepare and install the new cartridge (remove the protective elements, shake it).

5. Check the quality of the printing after the replacement of the cartridge.
6. Make a conclusion about the replacement and the care of the cartridges.

Tasks for independent completion

1. Perform the replacement of the cartridge of a laser printer.
2. Describe the signs of the need for the replacement of a cartridge.
3. Compile a memo on the care of the cartridges and the toner.

Conclusion: As a result of the work, the student has formed the skills of replacing and caring for the cartridges and the toner in laser printers and copiers.

Control Questions

1. What does the cartridge of a laser printer contain?
2. What are the signs of the need for the replacement of a cartridge?
3. What safety measures must be observed when working with the toner?
4. How is the new cartridge prepared for the installation?
5. How is the care of the cartridges performed?

Industrial training task 1.6.7 Replacement and care of keyboards, mice, batteries and accumulators of laptops and mobile devices

Objective: To form practical skills in replacing and caring for keyboards, mice, batteries and accumulators of laptops and mobile devices.

Theoretical background

Keyboards and mice are the most used input devices, subject to the contamination and the wear. The batteries and the accumulators of laptops and mobile devices have a limited service life and require the replacement as they are depleted.

The care of the keyboards and the mice includes the regular cleaning of the dust and the dirt; the care of the accumulators includes the observance of the rules of the charging and the operation, which prolongs their service life.

Equipment and materials

- Keyboards, mice, laptop accumulators.
- Means for the cleaning.
- Tools for the disassembly.
- A report form.

Procedure

1. Perform the cleaning of the keyboard from the dust and the dirt, observing the technology.
2. Perform the cleaning and the maintenance of the mouse.
3. Determine the signs of the wear of the accumulator of a laptop (the reduction of the operating time).
4. Study the procedure for the replacement of the accumulator of a laptop or a mobile device.
5. Describe the rules of the operation of the accumulators for the prolongation of their service life.
6. Make a conclusion about the replacement and the care of the input devices and the accumulators.

Tasks for independent completion

1. Perform the cleaning and the maintenance of a keyboard and a mouse.
2. Describe the signs of the wear and the procedure for the replacement of an accumulator.
3. Compile a memo on the operation of the accumulators.

Conclusion: As a result of the work, the student has formed the skills of replacing and caring for keyboards, mice, batteries and accumulators of laptops and mobile devices.

Control Questions

1. What does the care of the keyboards and the mice include?
2. What are the signs of the wear of an accumulator of a laptop?
3. How is the replacement of an accumulator performed?
4. What rules of the operation prolong the service life of an accumulator?
5. Why do the accumulators require the replacement over time?

Industrial training task 1.6.8 Replacement and care of monitors: cleaning and adjustment

Objective: To form practical skills in caring for, cleaning and adjusting monitors.

Theoretical background

Monitors require the regular care: the cleaning of the screen from the dust and the prints, and the correct adjustment of the parameters of the image (the brightness, the contrast, the colour rendering). The correct care prolongs the service life and maintains the quality of the image.

The cleaning of the screen is performed with the soft means that do not damage the coating. The adjustment of the parameters ensures the comfortable and the correct display of the image.

Equipment and materials

- Monitors of various types.
- Means for the cleaning of the screens.
- Tools for the adjustment.
- A report form.

Procedure

1. Study the rules of the cleaning of the screens of various types of monitors.
2. Perform the cleaning of the screen of the monitor with the suitable means.
3. Adjust the parameters of the image: the brightness, the contrast, the colour temperature.
4. Check the absence of the defects of the image (the dead pixels, the spots, the stripes).
5. Describe the rules of the operation of the monitors for the prolongation of their service life.
6. Make a conclusion about the care and the adjustment of the monitors.

Tasks for independent completion

1. Perform the cleaning and the adjustment of a monitor.
2. Check the monitor for the defects of the image.
3. Compile a memo on the care of the monitors.

Conclusion: As a result of the work, the student has formed the skills of caring for, cleaning and adjusting monitors.

Control Questions

1. How is the cleaning of the screen of a monitor performed correctly?
2. What parameters of the image are adjusted in a monitor?
3. What defects of the image can a monitor have?
4. Why must one not use aggressive means for the cleaning of the screen?
5. What rules of the operation prolong the service life of a monitor?

Industrial training task 1.6.10 Replacement and care of the battery elements in office equipment, buffer batteries in servers and network devices

Objective: To form practical skills in replacing and caring for the battery elements in office equipment, the buffer batteries in servers and network devices.

Theoretical background

The battery elements are used in various equipment: the CMOS batteries of motherboards, the buffer batteries of the RAID controllers of servers, the batteries of the UPS, and the elements of the backup power of network devices. They have a limited service life and require the replacement.

The CMOS battery of a motherboard maintains the settings of the BIOS and the clock when the power is off. Its discharge leads to the loss of the settings and the time. The buffer batteries of the RAID controllers protect the data in the cache during a power outage.

Equipment and materials

- A motherboard, a server (or its components), network devices.
- Battery elements for the replacement.
- A multimeter for the check.
- A report form.

Procedure

1. Determine the location and the type of the battery elements in the equipment.
2. Check the voltage of the CMOS battery of the motherboard with a multimeter.
3. Perform the replacement of the CMOS battery, observing the polarity.
4. Study the buffer batteries of the RAID controllers and the procedure for their replacement.
5. Describe the signs of the discharge of the battery elements and the consequences.
6. Make a conclusion about the replacement and the care of the battery elements.

Tasks for independent completion

1. Perform the replacement of the CMOS battery of a motherboard.
2. Describe the signs of the discharge of the battery elements.
3. Compile a list of the equipment that uses the battery elements.

Conclusion: As a result of the work, the student has formed the skills of replacing and caring for the battery elements in office equipment, the buffer batteries in servers and network devices.

Control Questions

1. What battery elements are used in the equipment?
2. What is the CMOS battery of a motherboard for?
3. What does the discharge of the CMOS battery lead to?
4. How is the replacement of the CMOS battery performed?
5. What are the buffer batteries of the RAID controllers for?

Industrial training task 1.6.11 Replacement and care of the lamps and the LEDs in scanners and printers

Objective: To form practical skills in replacing and caring for the lamps and the LEDs in scanners and printers.

Theoretical background

The lamps and the LEDs are used in scanners (for the illumination of the originals during the scanning) and in some printing devices. Their wear or failure worsens the quality of the scanning or the operation of the device.

The modern scanners use the LED illumination, which is durable and economical. The older models use the fluorescent lamps. The care includes the cleaning and the timely replacement of the failed elements.

Equipment and materials

- A scanner (or a multifunction device).
- Lamps or LEDs for the replacement (if available).
- Tools for the disassembly.

- A report form.

Procedure

1. Study the design of the illumination system of a scanner.
2. Determine the type of the illumination (a lamp or LEDs) and its state.
3. Determine the signs of the wear of the illumination (the deterioration of the quality of the scanning).
4. Study the procedure for the replacement of the illumination element.
5. Perform the cleaning of the optical system and the glass of the scanner.
6. Make a conclusion about the replacement and the care of the lamps and the LEDs.

Tasks for independent completion

1. Determine the type and the state of the illumination of a scanner.
2. Describe the signs of the wear of the illumination.
3. Perform the cleaning of the optical system of a scanner.

Conclusion: As a result of the work, the student has formed the skills of replacing and caring for the lamps and the LEDs in scanners and printers.

Control Questions

1. What are the lamps and the LEDs used for in scanners?
2. What types of illumination are used in scanners?
3. What are the signs of the wear of the illumination?
4. How is the cleaning of the optical system of a scanner performed?
5. What are the advantages of the LED illumination?

Industrial training task 1.6.12 Replacement and care of the rubber rollers, the rollers and the thermal sensors in printing devices

Objective: To form practical skills in replacing and caring for the rubber rollers, the rollers and the thermal sensors in printing devices.

Theoretical background

The rubber rollers and the rollers feed and move the paper in the printing devices. They wear out over time, which leads to the problems of the feeding of the paper (the jamming, the skewing, the multi-feeding). The thermal sensors control the temperature of the fuser.

The care of the rollers includes the cleaning of the dust and the paper deposits, and the timely replacement of the worn ones. The contamination of the rollers worsens the grip with the paper and causes the problems of the feeding.

Equipment and materials

- A printer (or a copier).
- Rollers and rubber elements for the replacement.
- Means for the cleaning.
- A report form.

Procedure

1. Study the design of the paper-feeding mechanism of a printer.
2. Determine the signs of the wear of the rollers (the problems of the feeding of the paper).
3. Perform the cleaning of the rollers from the dust and the deposits.
4. Study the procedure for the replacement of the worn rollers.
5. Study the role of the thermal sensors in the control of the temperature of the fuser.
6. Make a conclusion about the replacement and the care of the rollers and the thermal sensors.

Tasks for independent completion

1. Perform the cleaning of the rollers of the paper-feeding mechanism.

2. Describe the signs of the wear of the rollers and the procedure for their replacement.

3. Describe the role of the thermal sensors in a printer.

Conclusion: As a result of the work, the student has formed the skills of replacing and caring for the rubber rollers, the rollers and the thermal sensors in printing devices.

Control Questions

1. What do the rubber rollers and the rollers do in a printer?

2. What are the signs of the wear of the rollers?

3. How is the cleaning of the rollers performed?

4. What do the problems of the feeding of the paper lead to?

5. What is the role of the thermal sensors in a printer?

Industrial training task 1.6.14 Replacement and care of the toner drums, the filters and the separators in laser printers

Objective: To form practical skills in replacing and caring for the toner drums, the filters and the separators in laser printers.

Theoretical background

The photoconductor drum is a photosensitive cylinder onto which the image is formed by the laser before the transfer to the paper. The drum wears out and requires the replacement. The filters trap the toner dust, and the separators separate the paper from the drum.

The care of the drum requires the careful handling (the protection from the light and the touches of the surface), since the damage of the photosensitive layer worsens the quality of the printing.

Equipment and materials

- A laser printer.
- A photoconductor drum and the filters for the replacement.
- Means of protection.
- A report form.

Procedure

1. Study the design of a laser printer and the location of the drum, the filters and the separators.
2. Determine the signs of the wear of the drum (the defects of the printing, the repeating marks).
3. Study the procedure for the replacement of the photoconductor drum, observing the careful handling.
4. Perform the cleaning or the replacement of the toner filters.
5. Study the role of the separators in the separation of the paper from the drum.
6. Make a conclusion about the replacement and the care of the drums, the filters and the separators.

Tasks for independent completion

1. Describe the signs of the wear of the photoconductor drum.
2. Describe the procedure for the replacement of the drum, observing the careful handling.
3. Describe the role of the filters and the separators in a laser printer.

Conclusion: As a result of the work, the student has formed the skills of replacing and caring for the toner drums, the filters and the separators in laser printers.

Control Questions

1. What is the photoconductor drum, and what is its role?
2. What are the signs of the wear of the drum?
3. Why does the drum require the careful handling?
4. What are the toner filters for?
5. What do the separators do in a laser printer?

Industrial training task 1.6.15 Replacement and care of the anodes, cathodes, corona wires and rollers in scanners, fax machines and laser printers

Objective: To form practical skills in replacing and caring for the corona wires, the rollers and the other wearing elements in scanners, fax machines and laser printers.

Theoretical background

The corona wire (the corotron) creates the electric charge on the drum or the paper in a laser printer, which is necessary for the transfer of the toner. The contamination or the wear of the corona wire worsens the quality of the printing. The rollers feed and move the paper.

The care of these elements includes the cleaning of the deposits and the timely replacement. The contamination of the corona wire leads to the defects of the printing (the stripes, the uneven density).

Equipment and materials

- A laser printer (or a fax machine).
- Wearing elements for the replacement.
- Means for the cleaning.
- A report form.

Procedure

1. Study the role of the corona wires (the corotrons) in a laser printer.
2. Determine the signs of the contamination or the wear of the corona wire.
3. Perform the cleaning of the corona wire, observing the careful handling.
4. Study the procedure for the replacement of the worn rollers and the elements.
5. Describe the consequences of the wear of these elements for the quality of the printing.
6. Make a conclusion about the replacement and the care of the corona wires and the rollers.

Tasks for independent completion

1. Describe the role of the corona wires in a laser printer.
2. Describe the signs of the contamination of the corona wire and the method of its cleaning.
3. Compile a list of the wearing elements of a laser printer.

Conclusion: As a result of the work, the student has formed the skills of replacing and caring for the corona wires, the rollers and the other wearing elements in printing devices.

Control Questions

1. What is the role of the corona wire in a laser printer?
2. What are the signs of the contamination of the corona wire?
3. How is the cleaning of the corona wire performed?
4. What do the defects of the printing connected with the corona wire look like?
5. What wearing elements does a laser printer have?

Industrial training task 1.6.16 Organization of the stock of consumables and their efficient use

Objective: To form practical skills in organizing the stock of consumables and the planning of their efficient use.

Theoretical background

The organization of the stock of consumables ensures the uninterrupted operation of the equipment. The stock must be sufficient to avoid the downtime, but not excessive, in order not to freeze the funds and not to allow the expiration of the shelf life.

The planning of the stock is based on the analysis of the consumption of the consumables, the determination of the optimal level of the stock and the order point. The accounting of the consumables allows one to control the consumption and to plan the purchases.

Equipment and materials

- Data on the consumption of the consumables.
- A computer for the accounting.
- Templates of the accounting of the stock.
- A report form.

Procedure

1. Compile a list of the consumables necessary for the operation of the equipment.
2. Analyse the consumption of the consumables over a period and determine the average consumption.
3. Determine the optimal level of the stock and the order point for each consumable.
4. Organize the accounting of the consumables (the receipt, the consumption, the balance).
5. Develop a plan of the purchases of the consumables.
6. Make a conclusion about the organization of the stock and the efficient use of the consumables.

Tasks for independent completion

1. Compile a system of the accounting of the consumables.
2. Determine the optimal level of the stock and the order point.
3. Develop a plan of the purchases of the consumables.

Conclusion: As a result of the work, the student has formed the skills of organizing the stock of consumables and the planning of their efficient use.

Control Questions

1. Why is the organization of the stock of consumables important?
2. How is the optimal level of the stock determined?
3. What is the order point?
4. How is the accounting of the consumables organized?
5. How does the planning of the stock help to avoid the downtime?

Industrial training task 1.6.17 Analysis of the methods of the preventive maintenance of computer and office equipment

Objective: To form practical skills in analysing and applying the methods of the preventive maintenance of computer and office equipment.

Theoretical background

The preventive maintenance of computer and office equipment includes the regular cleaning, the check of the state, the replacement of the consumables and the wearing elements, and the testing. The regular maintenance prevents the faults and prolongs the service life.

The methods of the preventive maintenance differ for the various types of equipment but have the common principles: the regularity, the systematicity, the documenting and the use of the suitable means and consumables.

Equipment and materials

- Computer and office equipment.
- Tools and means for the maintenance.
- Schedules and journals of the maintenance.
- A report form.

Procedure

1. Study the methods of the preventive maintenance for the various types of equipment.

2. Compile a schedule of the preventive maintenance for a given set of equipment.
3. Perform the preventive maintenance of one of the devices (the cleaning, the check, the replacement of the consumables).
4. Document the carried-out maintenance in the journal.
5. Compare the methods of the maintenance of the various types of equipment.
6. Make a conclusion about the methods of the preventive maintenance.

Tasks for independent completion

1. Compile a schedule of the preventive maintenance for a set of equipment.
2. Perform and document the preventive maintenance of a device.
3. Compile a comparative table of the methods of the maintenance of the various types of equipment.

Conclusion: As a result of the work, the student has formed the skills of analysing and applying the methods of the preventive maintenance of computer and office equipment.

Control Questions

1. What does the preventive maintenance of equipment include?
2. What are the common principles of the preventive maintenance?
3. How is a schedule of the preventive maintenance compiled?
4. Why is the documenting of the maintenance necessary?
5. How do the methods of the maintenance of the various types of equipment differ?

Industrial training task 1.6.18 Analysis of the methods of the maintenance of the consumables of office equipment

Objective: To check and consolidate, in a comprehensive form, the skills of the replacement and the maintenance of the consumables of computing and office equipment, summing up the learning outcome.

Theoretical background

This lesson generalizes the material of the learning outcome on the replacement of the consumables of computing and office equipment. The student must demonstrate the comprehensive skills of the maintenance: from the determination of the need for the replacement to its execution and the check of the result.

The maintenance of the consumables includes their accounting, the timely replacement, the correct handling and the utilization. The comprehensive approach ensures the uninterrupted operation of the equipment.

Equipment and materials

- A set of computing and office equipment.
- Various consumables.
- Tools and means of protection.
- A final-assessment form.

Procedure

1. Determine the consumables of the proposed equipment and their state.
2. Determine the need for the replacement of the consumables by their signs.
3. Perform the replacement of one of the consumables, observing the technology and the safety measures.
4. Check the quality of the operation of the equipment after the replacement.
5. Compile a plan of the maintenance of the consumables of the equipment.
6. Make a final conclusion about the maintenance of the consumables.

Tasks for independent completion

1. Perform the replacement of a consumable and check the result.
2. Compile a comprehensive plan of the maintenance of the consumables of the equipment.
3. Prepare a memo on the maintenance of the consumables of office equipment.

Conclusion: As a result of the work, the student has demonstrated comprehensive skills of the replacement and the maintenance of the consumables of computing and office equipment.

Control Questions

1. What consumables does office equipment use?
2. How is the need for the replacement of a consumable determined?
3. What is included in the maintenance of the consumables?
4. How is the quality of the operation of the equipment checked after the replacement?
5. How is the utilization of the used consumables performed?

5. LIST OF SOFTWARE AND MULTIMEDIA SUPPORT FOR TRAINING SESSIONS

- Multimedia projector and interactive (smart) board; digital podium; white marker board.
- PowerPoint presentations for every topic of the module.
- Personal computers with Internet access and access to local networks.
- Disassembled training PC stands; sets of motherboards, power supplies, RAM modules, storage drives and graphics adapters.
- Peripheral and office equipment: printers, scanners, multifunction devices (MFPs), plotters.
- Measuring instruments: multimeters, POST-cards, power-supply and cable testers.
- Anti-static wrist straps and mats; PC assembly and repair tool kits; soldering stations.
- Hardware-diagnostic software: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo, HWMonitor, OCCT.
- Programming and scripting tools: Python 3 (with psutil, WMI, pywin32), PowerShell, Windows Subsystem for Linux.
- Consumables of office equipment (cartridges, toner, ink, ribbons) for industrial training.
- Learning Management System (LMS) for hosting materials and assignments; video lectures on PC maintenance and repair.

6. LIST OF LITERATURE

Core (fundamental) literature:

1. Phil Hughes MBE, Ed Ferrett. Introduction to Health and Safety at Work. ISBN: 978-0-415-72308-4.
2. Mark A. Friend, James P. Kohn. Fundamentals of Occupational Safety and Health. ISBN-13: 978-0-86587-171-7.
3. Charles K. Alexander, Matthew N. O. Sadiku. Fundamentals of Electric Circuits. ISBN: 978-0-07-352955-4.
4. John L. Hennessy, David A. Patterson. Computer Architecture: A Quantitative Approach (5th Edition). ISBN-13: 978-8178672663.
5. Jim Ledin. Modern Computer Architecture and Organization (2nd Edition). ISBN-13: 978-1803234519.

Additional literature:

1. Saurabh Shrivastava, Neelanjali Srivastav. Solutions Architect's Handbook (2nd Edition). ISBN-13: 978-1801816618.

2. Joseph Ingeno. Software Architect's Handbook. ISBN-13: 978-1788624060.
3. Robert C. Martin. Clean Architecture: A Craftsman's Guide to Software Structure and Design. ISBN-13: 978-0134494166.

Internet resources:

1. Computer Architecture Topics — https://en.wikipedia.org/wiki/Computer_architecture
2. Fire Protection and Prevention (OSHA) — <https://www.osha.gov/sites/default/files/2019-03/fireprotection.pdf>
3. <https://learn.microsoft.com> — official Microsoft documentation on Windows and administration.
4. <https://support.microsoft.com> — technical support and instructions for configuring hardware and software.
5. <https://docs.python.org> — documentation on automation and system administration in Python.

VIII. LO 1.7 Manage the internal and external devices of a personal computer

1. ACADEMIC CALENDAR

LO 1.7 Manage the internal and external devices of a personal computer

Instructor: D.M. Yermekov

Specialty: 06130100 – «Software Engineering (by types)»

Total hours: 36 hours (18 lessons of 2 academic hours each)

№	Name of topics	Date of lesson	Type of activity
1	Introduction to the management of devices through programming		Skills-improvement lesson
2	Overview of the system interfaces of Windows (WMI, PowerShell, Device Manager API)		Skills-improvement lesson
3	Work with the Python library psutil: basic capabilities		Skills-improvement lesson
4	Obtaining the information about the equipment through WMI		Skills-improvement lesson
5	Management of the services and the drivers through PowerShell		Skills-improvement lesson
6	Management of the processes and the resources with the help of Python		Skills-improvement lesson
7	Connection and configuration of the external devices through the system commands		Skills-improvement lesson
8	Work with USB devices through the programming interfaces		Skills-improvement lesson
9	Determination of the status of the devices and the diagnostics of the errors		Skills-improvement lesson
10	Monitoring of the temperature of the processor through the OpenHardwareMonitor API		Skills-improvement lesson
11	Monitoring of the load of CPU, RAM, Disk with psutil		Skills-improvement lesson
12	Automation of the update of the drivers with the help of scripts		Skills-improvement lesson

13	Writing of the utilities for the management of the network devices		Skills-improvement lesson
14	Management of the configuration of the system through Python scripts		Skills-improvement lesson
15	Configuration of the parameters of the devices: sound, network, input		Skills-improvement lesson
16	Execution of the system commands through subprocess		Skills-improvement lesson
17	Collection of the data about the equipment into log files		Skills-improvement lesson
18	Final mini-project: «Automated device-management system»		Consolidation and assessment of knowledge

2. GLOSSARY

The glossary contains the main definitions, terms and concepts that reflect the core content of this learning outcome. The terms are arranged in alphabetical order.

Term	Definition
API (Application Programming Interface)	A set of functions and procedures through which a program interacts with the operating system or another program.
Device Manager	A Windows component for viewing and managing the devices installed in a computer and their drivers.
OpenHardwareMonitor	An open program and library for monitoring the temperatures, the voltages and the load of the hardware components of a computer.
psutil	A cross-platform Python library for obtaining information about the processes and the use of the system resources.
PnP (Plug and Play)	A technology for the automatic detection and configuration of the devices connected to a computer.
Service	A program that runs in the background of the operating system and performs system functions without the direct participation of the user.
subprocess	A Python module for launching external programs and system commands from a script.
USB device	A device connected over the universal serial bus and managed through the corresponding controllers and drivers.
WMI	Windows Management Instrumentation — an infrastructure for managing data and operations and obtaining information about the equipment in Windows.
Script automation	The automation of repetitive tasks of management and diagnostics with the help of programs in scripting languages (Python, PowerShell, Bash).

4. PRACTICAL CLASS MATERIALS

LO 1.7 Manage the internal and external devices of a personal computer

Practical work 1.7.1 Introduction to the management of devices through programming

Objective: To form a holistic understanding of the management of the devices of a computer through programming and to consolidate the skills of using the system tools.

Theoretical background

The management of the devices of a computer can be performed not only manually, through the graphical interface, but also programmatically — with the help of the scripts and the programs that turn to the system interfaces. The programmatic management allows one to automate the routine tasks and to manage many devices.

For the programmatic management of the devices in Windows, the interfaces WMI, the registry, the Device Manager API and the PowerShell cmdlets are used. The scripting languages Python and PowerShell provide a convenient access to these interfaces.

Equipment and materials

- A computer with the Windows operating system.
- Python and PowerShell.
- The Device Manager.
- A report form.

Procedure

1. Study the methods of the management of the devices: the graphical interface and the programmatic management.
2. Examine the Device Manager and the information about the devices that it provides.
3. Study the system interfaces for the programmatic management of the devices (WMI, PowerShell).
4. Obtain the list of the devices with the help of PowerShell (Get-PnpDevice).
5. Compare the manual and the programmatic management of the devices.
6. Make a conclusion about the capabilities of the programmatic management of the devices.

Tasks for independent completion

1. Compile a list of the system interfaces for the management of the devices.
2. Obtain the list of the devices with the help of PowerShell.
3. Compare the advantages of the manual and the programmatic management.

Conclusion: As a result of the work, the student has formed a holistic understanding of the management of the devices of a computer through programming.

Control Questions

1. What methods of the management of the devices exist?
2. What interfaces are used for the programmatic management of the devices?
3. What does the Device Manager provide?
4. What are the advantages of the programmatic management?
5. What languages are used for the management of the devices?

Practical work 1.7.2 Overview of the system interfaces of Windows (WMI, PowerShell, Device Manager API)

Objective: To form practical skills in using the system interfaces of Windows for obtaining the information about the devices and managing them.

Theoretical background

Windows provides several system interfaces for the management of the devices and the system. WMI provides a unified access to the information about the equipment and the operations of the management. PowerShell provides the cmdlets for the administration. The Device Manager API provides the management of the PnP devices.

The understanding of these interfaces and their capabilities allows one to choose the optimal tool for a specific task of the management of the devices and to combine them in the scripts.

Equipment and materials

- A computer with the Windows operating system.
- PowerShell with the access to WMI.
- Documentation on the interfaces.
- A report form.

Procedure

1. Study the WMI interface and its classes for the work with the devices.
2. Study the cmdlets of PowerShell for the management of the devices (Get-PnpDevice, Enable/Disable-PnpDevice).
3. Obtain the information about the devices through WMI with the help of PowerShell.
4. Obtain the information about the devices through the PnP cmdlets.
5. Compare the capabilities of the various interfaces for the management of the devices.
6. Make a conclusion about the system interfaces of Windows for the management of the devices.

Tasks for independent completion

1. Compile a comparative table of the system interfaces of Windows.
2. Obtain the information about the devices through WMI and the PnP cmdlets.
3. Determine the optimal interface for various tasks of the management.

Conclusion: As a result of the work, the student has formed the skills of using the system interfaces of Windows for obtaining the information about the devices and managing them.

Control Questions

1. What system interfaces of Windows are used for the management of the devices?
2. What does the WMI interface provide?
3. What cmdlets of PowerShell are used for the management of the devices?
4. What is the Device Manager API for?
5. How is the optimal interface for a task chosen?

Practical work 1.7.3 Work with the Python library psutil: basic capabilities

Objective: To form practical skills in using the basic capabilities of the psutil library for obtaining the information about the resources and the devices of a computer.

Theoretical background

psutil (process and system utilities) is a cross-platform Python library for obtaining the information about the use of the system resources (the processor, the memory, the disks, the network) and the management of the processes. It is convenient for the creation of the utilities of the monitoring and the diagnostics.

psutil provides the functions for obtaining the load of the processor, the use of the memory, the information about the disks and the partitions, the network connections and the running processes. These data are used for the monitoring and the management.

Equipment and materials

- A computer with the installed Python and the psutil library.
- A code editor.
- Documentation on psutil.
- A report form.

Procedure

1. Install the psutil library and study its main functions.
2. Obtain the information about the load of the processor with the help of psutil.
3. Obtain the information about the use of the memory and the disks.
4. Obtain the information about the network connections and the processes.
5. Write a fragment of the script that displays the main parameters of the system.
6. Make a conclusion about the capabilities of the psutil library.

Tasks for independent completion

1. Write a Python script for obtaining the information about the resources of the system with psutil.
2. Display the load of the processor, the memory and the disks.
3. Describe the main functions of the psutil library.

Conclusion: As a result of the work, the student has formed the skills of using the basic capabilities of the psutil library for obtaining the information about the resources and the devices of a computer.

Control Questions

1. What is the psutil library?
2. What information can be obtained with the help of psutil?
3. How is the load of the processor obtained?
4. How is the information about the disks and the network obtained?
5. For what utilities is psutil convenient?

Practical work 1.7.4 Obtaining the information about the equipment through WMI

Objective: To form practical skills in obtaining the detailed information about the equipment of a computer through WMI.

Theoretical background

WMI provides the detailed information about all the components of the equipment through the corresponding classes: Win32_Processor, Win32_PhysicalMemory, Win32_DiskDrive, Win32_VideoController and others. This information is used for the inventory, the diagnostics and the management.

The access to WMI is possible from PowerShell (Get-CimInstance) and from Python (the WMI library). The obtaining of the information about the equipment through WMI is the basis for the automation of the administration.

Equipment and materials

- A computer with the Windows operating system.
- PowerShell and (or) Python with the WMI library.
- A list of the WMI classes.
- A report form.

Procedure

1. Study the WMI classes for obtaining the information about the equipment.

2. Obtain the information about the processor and the memory through WMI.
3. Obtain the information about the disks and the video controller through WMI.
4. Obtain the information about the motherboard and the BIOS through WMI.
5. Systematize the obtained information into a report on the equipment.
6. Make a conclusion about obtaining the information about the equipment through WMI.

Tasks for independent completion

1. Obtain the information about the main components of the equipment through WMI.
 2. Compile a report on the equipment of the computer.
 3. Compile a list of the WMI classes for obtaining the information about the equipment.
- Conclusion:** As a result of the work, the student has formed the skills of obtaining the detailed information about the equipment of a computer through WMI.

Control Questions

1. What WMI classes are used for obtaining the information about the equipment?
2. How is the information about the processor obtained through WMI?
3. How is the information about the disks obtained through WMI?
4. From what tools is the access to WMI possible?
5. How is WMI used for the inventory of the equipment?

Practical work 1.7.5 Management of the services and the drivers through PowerShell

Objective: To form practical skills in managing the services and the drivers of an operating system with the help of PowerShell.

Theoretical background

The services are the programs that run in the background and perform the system functions. PowerShell provides the cmdlets for the management of the services: Get-Service, Start-Service, Stop-Service, Restart-Service. The drivers are managed through the PnP cmdlets and pnputil.

The programmatic management of the services and the drivers allows one to automate the administration, to mass-manage the systems and to create the scripts of the maintenance and the restoration.

Equipment and materials

- A computer with the Windows operating system.
- PowerShell with administrator rights.
- A list of the cmdlets for the management of the services.
- A report form.

Procedure

1. Obtain the list of the services with the help of the cmdlet Get-Service.
2. Study the state of the services (running, stopped) and their types of start.
3. Perform the start, the stop and the restart of a service with the corresponding cmdlets.
4. Obtain the information about the drivers with the help of the PnP cmdlets.
5. Write a fragment of the script for the management of a service.
6. Make a conclusion about the management of the services and the drivers through PowerShell.

Tasks for independent completion

1. Obtain the list of the services and analyse their state.
2. Perform the management of a service (the start, the stop) from PowerShell.
3. Write a fragment of the script for the management of a service.

Conclusion: As a result of the work, the student has formed the skills of managing the services and the drivers of an operating system with the help of PowerShell.

Control Questions

1. What are the services, and what are they for?
2. What cmdlets are used for the management of the services?
3. How is the start and the stop of a service performed?
4. How are the drivers managed through PowerShell?
5. What are the advantages of the programmatic management of the services?

Practical work 1.7.6 Management of the processes and the resources with the help of Python

Objective: To form practical skills in managing the processes and the resources of a computer with the help of Python and the psutil library.

Theoretical background

psutil allows one not only to obtain the information about the processes but also to manage them: to obtain the list of the processes, the information about each of them (the use of the processor and the memory), and to terminate the processes. This is used for the monitoring and the management of the resources.

The management of the processes from a Python script allows one to automate the tracking of the resource-intensive processes, the termination of the hung processes and the analysis of the load of the system.

Equipment and materials

- A computer with the installed Python and the psutil library.
- A code editor.
- Documentation on psutil.
- A report form.

Procedure

1. Obtain the list of the running processes with the help of psutil.
2. Obtain the information about each process: the name, the identifier, the use of the processor and the memory.
3. Sort the processes by the use of the resources and detect the resource-intensive ones.
4. Write a fragment of the script for the termination of a process by its identifier.
5. Add the monitoring of the load of the resources by the processes.
6. Make a conclusion about the management of the processes and the resources with Python.

Tasks for independent completion

1. Write a Python script for obtaining and analysing the list of the processes.
2. Detect the most resource-intensive processes.
3. Demonstrate the termination of a process from a script.

Conclusion: As a result of the work, the student has formed the skills of managing the processes and the resources of a computer with the help of Python and the psutil library.

Control Questions

1. What does psutil allow one to do with the processes?
2. How is the list of the processes obtained?
3. How are the resource-intensive processes detected?
4. How is a process terminated from a script?
5. What is the management of the processes used for?

Practical work 1.7.7 Connection and configuration of the external devices through the system commands

Objective: To form practical skills in connecting and configuring the external devices with the help of the system commands and the scripts.

Theoretical background

The external devices can be configured not only through the graphical interface but also through the system commands and the scripts. This allows one to automate the configuration, to apply the uniform settings to many computers and to manage the devices remotely.

For the configuration of the devices through the commands, the utilities pnputil, the PowerShell cmdlets and the WMI are used. They allow one to enable, disable and configure the devices programmatically.

Equipment and materials

- A computer with the connected external devices.
- PowerShell and the system utilities.
- External devices for the experiments.
- A report form.

Procedure

1. Connect an external device and observe its automatic detection (PnP).
2. Obtain the information about the connected device through the system commands.
3. Configure the parameters of the device through the commands or the scripts.
4. Demonstrate the enabling and the disabling of the device with the PnP cmdlets.
5. Write a fragment of the script for the configuration of a device.
6. Make a conclusion about the configuration of the external devices through the system commands.

Tasks for independent completion

1. Obtain the information about a connected device through the commands.
2. Demonstrate the enabling and the disabling of a device from PowerShell.
3. Write a fragment of the script for the configuration of a device.

Conclusion: As a result of the work, the student has formed the skills of connecting and configuring the external devices with the help of the system commands and the scripts.

Control Questions

1. How can the external devices be configured besides the graphical interface?
2. What utilities are used for the configuration of the devices through the commands?
3. What is the technology PnP?
4. How is a device enabled and disabled programmatically?
5. What are the advantages of the configuration of the devices through the scripts?

Practical work 1.7.8 Work with USB devices through the programming interfaces

Objective: To form practical skills in obtaining the information about the USB devices and managing them through the programming interfaces.

Theoretical background

The USB devices are connected over the universal serial bus and are managed through the corresponding controllers and drivers. The information about the USB devices can be obtained through WMI (the class Win32_USBControllerDevice) and the PnP cmdlets.

The programmatic work with the USB devices allows one to obtain the list of the connected devices, their parameters, and to track the connection and the disconnection of the devices, which is used for the monitoring and the security.

Equipment and materials

- A computer with the USB ports.
- USB devices for the experiments.
- PowerShell and (or) Python.
- A report form.

Procedure

1. Obtain the list of the connected USB devices through WMI or the PnP cmdlets.
2. Obtain the information about each USB device: the name, the identifier, the type.
3. Connect and disconnect a USB device and observe the change of the list.
4. Write a fragment of the script for obtaining the list of the USB devices.
5. Study the possibilities of tracking the connection of the USB devices.
6. Make a conclusion about the work with the USB devices through the programming interfaces.

Tasks for independent completion

1. Write a script for obtaining the list of the connected USB devices.
2. Obtain the information about the parameters of the USB devices.
3. Describe the possibilities of tracking the connection of the USB devices.

Conclusion: As a result of the work, the student has formed the skills of obtaining the information about the USB devices and managing them through the programming interfaces.

Control Questions

1. How are the USB devices managed?
2. Through what interfaces is the information about the USB devices obtained?
3. How is the list of the connected USB devices obtained?
4. How can the connection of the USB devices be tracked?
5. What is the tracking of the USB devices used for?

Practical work 1.7.9 Determination of the status of the devices and the diagnostics of the errors

Objective: To form practical skills in determining the status of the devices and diagnosing the errors of their operation programmatically.

Theoretical background

Each device in the system has a status that reflects its state: working correctly, having a problem, disabled. The status and the error codes of the devices can be obtained through the PnP cmdlets and WMI, which allows one to automate the diagnostics.

The error codes of the devices (the Device Manager codes) indicate the nature of the problem: the absence of the driver, the conflict of the resources, the disabling of the device. The analysis of these codes helps to determine the cause of the problem.

Equipment and materials

- A computer with the connected devices.
- PowerShell with the PnP cmdlets.
- A reference of the error codes of the devices.
- A report form.

Procedure

1. Obtain the list of the devices and their status with the help of the cmdlet Get-PnpDevice.
2. Detect the devices with the problems (the status differs from OK).
3. Obtain the error codes of the problem devices.
4. Analyse the error codes and determine the cause of the problems.
5. Write a fragment of the script for the detection of the problem devices.

6. Make a conclusion about the determination of the status and the diagnostics of the errors of the devices.

Tasks for independent completion

1. Write a script for obtaining the status of the devices and the detection of the problems.
2. Analyse the error codes of the problem devices.
3. Compile a table of the typical error codes and their meaning.

Conclusion: As a result of the work, the student has formed the skills of determining the status of the devices and diagnosing the errors of their operation programmatically.

Control Questions

1. What is the status of a device?
2. How is the status of the devices obtained?
3. What do the error codes of the devices indicate?
4. How are the problem devices detected?
5. How does the analysis of the error codes help in the diagnostics?

Practical work 1.7.10 Monitoring of the temperature of the processor through the OpenHardwareMonitor API

Objective: To form practical skills in monitoring the temperature of the processor and the other components through the OpenHardwareMonitor API.

Theoretical background

OpenHardwareMonitor is an open program and library for monitoring the temperatures, the voltages, the speed of the fans and the load of the components. It provides the access to the data of the sensors through WMI, which allows one to use it in the scripts.

The monitoring of the temperature of the processor and the components allows one to track the thermal mode, to detect the overheating and to take the measures in a timely manner. This is an important part of the maintenance and the management of the equipment.

Equipment and materials

- A computer with the installed OpenHardwareMonitor.
- PowerShell or Python for the access to the data.
- A code editor.
- A report form.

Procedure

1. Install and launch OpenHardwareMonitor and study the data of the sensors.
2. Study the access to the data of OpenHardwareMonitor through WMI.
3. Obtain the temperature of the processor through the WMI namespace of OpenHardwareMonitor.
4. Obtain the temperatures of the other components and the speed of the fans.
5. Write a fragment of the script for the monitoring of the temperature.
6. Make a conclusion about the monitoring of the temperature through the OpenHardwareMonitor API.

Tasks for independent completion

1. Obtain the temperature of the processor through OpenHardwareMonitor.
2. Write a script for the monitoring of the temperatures of the components.
3. Describe the access to the data of OpenHardwareMonitor through WMI.

Conclusion: As a result of the work, the student has formed the skills of monitoring the temperature of the processor and the other components through the OpenHardwareMonitor API.

Control Questions

1. What is OpenHardwareMonitor?
2. What data does OpenHardwareMonitor provide?
3. How is the access to the data of OpenHardwareMonitor performed?
4. How is the temperature of the processor obtained programmatically?
5. What is the monitoring of the temperature used for?

Practical work 1.7.11 Monitoring of the load of CPU, RAM, Disk with psutil

Objective: To form practical skills in monitoring the load of the processor, the memory and the disks with the help of the psutil library.

Theoretical background

psutil provides the functions for obtaining the load of the processor (`cpu_percent`), the use of the memory (`virtual_memory`), and the load of the disks (`disk_usage`, `disk_io_counters`). On their basis, the monitoring of the load of the system is built.

The monitoring of the load allows one to detect the bottlenecks of the system, the resource-intensive processes and the moments of the peak load, which is used for the optimization and the diagnostics.

Equipment and materials

- A computer with the installed Python and the psutil library.
- A code editor.
- A tool for creating the load.
- A report form.

Procedure

1. Obtain the load of the processor with the help of psutil.
2. Obtain the use of the memory and the load of the disks.
3. Organize the periodic collection of the parameters in a cycle.
4. Add the output of the load on the screen and the saving to a file.
5. Create the load on the system and observe the change of the parameters.
6. Make a conclusion about the monitoring of the load with psutil.

Tasks for independent completion

1. Write a Python script for the monitoring of the load of the CPU, the RAM and the disks.
2. Save the data of the monitoring to a file.
3. Analyse the dynamics of the load under the load on the system.

Conclusion: As a result of the work, the student has formed the skills of monitoring the load of the processor, the memory and the disks with the help of the psutil library.

Control Questions

1. What functions of psutil are used for the monitoring of the load?
2. How is the load of the processor obtained?
3. How is the use of the memory and the disks obtained?
4. How is the periodic collection of the parameters organized?
5. What does the monitoring of the load allow one to detect?

Practical work 1.7.12 Automation of the update of the drivers with the help of scripts

Objective: To form practical skills in automating the update of the drivers of the devices with the help of scripts.

Theoretical background

The update of the drivers maintains the correct operation of the devices and eliminates the errors. The automation of the update with the help of the scripts (pnputil, PowerShell) allows one to update the drivers on many computers and to apply the uniform drivers.

The utility pnputil allows one to add, to delete and to manage the drivers from the command line, which is convenient for the automation. The PowerShell cmdlets provide the additional capabilities of the management of the drivers.

Equipment and materials

- A computer with the Windows operating system.
- PowerShell and the utility pnputil.
- Driver files for the experiments.
- A report form.

Procedure

1. Obtain the list of the installed drivers with the help of pnputil.
2. Study the commands of pnputil for the management of the drivers.
3. Demonstrate the addition of a driver to the system through pnputil.
4. Write a fragment of the script for the automation of the update of the drivers.
5. Add the logging of the actions to the script.
6. Make a conclusion about the automation of the update of the drivers.

Tasks for independent completion

1. Obtain the list of the installed drivers with the help of pnputil.
2. Write a fragment of the script for the management of the drivers.
3. Describe the commands of pnputil for the work with the drivers.

Conclusion: As a result of the work, the student has formed the skills of automating the update of the drivers of the devices with the help of scripts.

Control Questions

1. Why is the update of the drivers necessary?
2. What utility is used for the management of the drivers from the command line?
3. What commands does pnputil provide?
4. How is the addition of a driver automated?
5. What are the advantages of the automation of the update of the drivers?

Practical work 1.7.13 Writing of the utilities for the management of the network devices

Objective: To form practical skills in writing the utilities for the management of the network devices and the network settings.

Theoretical background

The network devices (the network adapters) can be managed programmatically through the PowerShell cmdlets (Get-NetAdapter, Set-NetIPAddress) and through WMI. This allows one to automate the configuration of the network and to manage the network settings of many computers.

The utilities for the management of the network devices automate the typical tasks: the configuration of the IP addresses, the enabling and the disabling of the adapters, the configuration of the DNS, and the diagnostics of the network.

Equipment and materials

- A computer with the network adapters.
- PowerShell with administrator rights.
- A code editor.

- A report form.

Procedure

1. Obtain the list of the network adapters with the help of Get-NetAdapter.
2. Obtain and analyse the network settings of the adapters.
3. Demonstrate the enabling and the disabling of a network adapter.
4. Write a fragment of the script for the configuration of the network settings.
5. Add the diagnostics of the network connection to the utility.
6. Make a conclusion about the management of the network devices through the scripts.

Tasks for independent completion

1. Write a utility for obtaining and analysing the network settings.
2. Demonstrate the management of a network adapter from a script.
3. Add the configuration of the network settings to the utility.

Conclusion: As a result of the work, the student has formed the skills of writing the utilities for the management of the network devices and the network settings.

Control Questions

1. How are the network devices managed programmatically?
2. What cmdlets are used for the management of the network adapters?
3. How are the network settings configured through the scripts?
4. How is the enabling and the disabling of an adapter performed?
5. What tasks do the utilities for the management of the network automate?

Practical work 1.7.14 Management of the configuration of the system through Python scripts

Objective: To form practical skills in managing the configuration of an operating system through Python scripts.

Theoretical background

The management of the configuration of the system includes the change of the settings of the operating system, the registry, the services and the parameters of the devices. Python allows one to automate these tasks through the modules winreg, subprocess and WMI.

The automation of the management of the configuration allows one to apply the uniform settings to many computers, to quickly configure the new systems and to maintain the configuration in the required state.

Equipment and materials

- A computer with the installed Python.
- The modules winreg, subprocess, WMI.
- A code editor.
- A report form.

Procedure

1. Determine the parameters of the configuration of the system that are to be managed.
2. Write a fragment of the script for the reading of the parameters of the configuration through winreg.
3. Write a fragment of the script for the change of the settings of the system.
4. Add the management of the services through the script.
5. Add the safety measures and the logging to the script.
6. Make a conclusion about the management of the configuration of the system through Python.

Tasks for independent completion

1. Write a Python script for the management of the parameters of the configuration of the system.
2. Demonstrate the change of a setting of the system from a script.

3. Add the logging of the changes to the script.

Conclusion: As a result of the work, the student has formed the skills of managing the configuration of an operating system through Python scripts.

Control Questions

1. What does the management of the configuration of the system include?
2. What modules of Python are used for the management of the configuration?
3. How are the parameters of the configuration read and changed?
4. How are the services managed from a script?
5. What are the advantages of the automation of the management of the configuration?

Practical work 1.7.15 Configuration of the parameters of the devices: sound, network, input

Objective: To form practical skills in configuring the parameters of the various devices (sound, network, input) programmatically.

Theoretical background

The parameters of the various devices — the sound (the volume, the playback device), the network (the IP, the DNS), the input (the keyboard, the mouse) — can be configured programmatically through the system interfaces and the scripts. This allows one to automate the configuration of the workplaces.

For the configuration of the various devices, the corresponding PowerShell cmdlets, the WMI and the system utilities are used. The combination of these means allows one to configure the devices comprehensively.

Equipment and materials

- A computer with the sound, network and input devices.
- PowerShell and the system utilities.
- A code editor.
- A report form.

Procedure

1. Study the methods of the programmatic configuration of the sound devices.
2. Configure the network parameters of the device through the scripts.
3. Study the configuration of the parameters of the input devices.
4. Write a fragment of the script for the comprehensive configuration of the devices.
5. Test the script and check the applied settings.
6. Make a conclusion about the configuration of the parameters of the devices programmatically.

Tasks for independent completion

1. Configure the network parameters of a device through a script.
2. Describe the methods of the configuration of the sound and the input devices.
3. Write a fragment of the script for the configuration of the devices.

Conclusion: As a result of the work, the student has formed the skills of configuring the parameters of the various devices programmatically.

Control Questions

1. What parameters of the devices can be configured programmatically?
2. How are the sound devices configured?
3. How are the network parameters configured through the scripts?
4. How are the parameters of the input devices configured?
5. What are the advantages of the programmatic configuration of the devices?

Practical work 1.7.16 Execution of the system commands through subprocess

Objective: To form practical skills in executing the system commands and managing the devices through the subprocess module of Python.

Theoretical background

The subprocess module of Python allows one to launch the system commands and the utilities from a script, to obtain their output and to process it. This is used for the management of the devices through the system utilities that do not have a direct API.

Through subprocess, one can launch the utilities `pnputil`, `netsh`, `sc` and others, combining their capabilities with the logic of Python. This makes Python a universal tool for the automation.

Equipment and materials

- A computer with the installed Python.
- The subprocess module.
- A code editor.
- A report form.

Procedure

1. Study the capabilities of the subprocess module for the launch of the commands.
2. Write a fragment of the script for the launch of a system command and obtaining its output.
3. Process the result of the execution of the command in the script.
4. Launch the device-management utilities through subprocess.
5. Add the handling of the errors of the execution of the commands.
6. Make a conclusion about the execution of the system commands through subprocess.

Tasks for independent completion

1. Write a Python script for the launch of the system commands.
2. Process the result of the execution of a command.
3. Add the handling of the errors to the script.

Conclusion: As a result of the work, the student has formed the skills of executing the system commands and managing the devices through the subprocess module of Python.

Control Questions

1. What does the subprocess module allow one to do?
2. How is a system command launched from a script?
3. How is the output of a command obtained and processed?
4. What utilities can be launched through subprocess?
5. Why is the handling of the errors necessary?

Practical work 1.7.17 Collection of the data about the equipment into log files

Objective: To form practical skills in collecting the data about the equipment and the state of the system into log files for the subsequent analysis.

Theoretical background

The collection of the data about the equipment and the state of the system into log files allows one to track the dynamics of the parameters, to detect the trends and to diagnose the problems. The logs are written in a structured form for the convenience of the analysis.

For the collection of the data, the scripts that periodically obtain the parameters of the system and write them to a file with a time stamp are used. The format of the log (text, CSV, JSON) is chosen for the convenience of the subsequent analysis.

Equipment and materials

- A computer with the installed Python.
- The libraries psutil and WMI.
- A code editor.
- A report form.

Procedure

1. Determine the parameters of the equipment and the system to be logged.
2. Write a fragment of the script for obtaining the parameters and writing them to a file.
3. Add the time stamp and the structured format (CSV) to the log.
4. Organize the periodic collection of the data in a cycle.
5. Analyse the collected log and the dynamics of the parameters.
6. Make a conclusion about the collection of the data about the equipment into the logs.

Tasks for independent completion

1. Write a script for the collection of the data about the equipment into a log file.
2. Organize the structured format of the log with the time stamps.
3. Analyse the collected log.

Conclusion: As a result of the work, the student has formed the skills of collecting the data about the equipment and the state of the system into log files for the subsequent analysis.

Control Questions

1. What does the collection of the data into the logs allow one to do?
2. In what format are the logs written?
3. How is the periodic collection of the data organized?
4. Why is the time stamp necessary in a log?
5. How is the collected log analysed?

Practical work 1.7.18 Final mini-project: «Automated device-management system»

Objective: To consolidate and check, in a comprehensive form, the skills of the management of the devices through programming by developing a final mini-project — an automated device-management system, summing up the learning outcome and the module as a whole.

Theoretical background

The final mini-project integrates all the skills acquired in the learning outcome: obtaining the information about the devices, monitoring their state, managing the devices and the services, and automating the typical tasks. The result is a complete automated device-management system.

The automated device-management system collects the information about the devices, monitors their state, detects the problems and performs the actions of the management or the notification. This is a comprehensive task that demonstrates the mastery of the management of the devices through programming.

Equipment and materials

- A computer with the installed Python and (or) PowerShell.
- The libraries psutil, WMI and the system interfaces.
- A code editor.
- A final-assessment form.

Procedure

1. Design the structure of the device-management system: the modules of the collection of the information, the monitoring and the management.
2. Implement the module of obtaining the information about the devices and their state.

3. Implement the module of the monitoring of the parameters and the detection of the problems.
4. Implement the module of the management of the devices or the notification of the problems.
5. Integrate the modules into a single system and add the logging and the report.
6. Test the system, demonstrate its operation and make a conclusion about the project.

Tasks for independent completion

1. Develop an automated device-management system as a mini-project.
2. Implement the collection of the information, the monitoring and the management in the system.
3. Test and demonstrate the operation of the system and prepare a report.

Conclusion: As a result of the work, the student has consolidated the comprehensive skills of the management of the devices of a computer through programming and has developed a complete automated device-management system, summing up the learning outcome and the module.

Control Questions

1. What skills does the final mini-project integrate?
2. What does an automated device-management system include?
3. How is the structure of the system designed?
4. How is the monitoring and the detection of the problems implemented?
5. How is the operation of the developed system tested and demonstrated?

5. LIST OF SOFTWARE AND MULTIMEDIA SUPPORT FOR TRAINING SESSIONS

- Multimedia projector and interactive (smart) board; digital podium; white marker board.
- PowerPoint presentations for every topic of the module.
- Personal computers with Internet access and access to local networks.
- Disassembled training PC stands; sets of motherboards, power supplies, RAM modules, storage drives and graphics adapters.
- Peripheral and office equipment: printers, scanners, multifunction devices (MFPs), plotters.
- Measuring instruments: multimeters, POST-cards, power-supply and cable testers.
- Anti-static wrist straps and mats; PC assembly and repair tool kits; soldering stations.
- Hardware-diagnostic software: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo, HWMonitor, OCCT.
- Programming and scripting tools: Python 3 (with psutil, WMI, pywin32), PowerShell, Windows Subsystem for Linux.
- Consumables of office equipment (cartridges, toner, ink, ribbons) for industrial training.
- Learning Management System (LMS) for hosting materials and assignments; video lectures on PC maintenance and repair.

6. LIST OF LITERATURE

Core (fundamental) literature:

1. Phil Hughes MBE, Ed Ferrett. Introduction to Health and Safety at Work. ISBN: 978-0-415-72308-4.
2. Mark A. Friend, James P. Kohn. Fundamentals of Occupational Safety and Health. ISBN-13: 978-0-86587-171-7.
3. Charles K. Alexander, Matthew N. O. Sadiku. Fundamentals of Electric Circuits. ISBN: 978-0-07-352955-4.

4. John L. Hennessy, David A. Patterson. Computer Architecture: A Quantitative Approach (5th Edition). ISBN-13: 978-8178672663.
5. Jim Ledin. Modern Computer Architecture and Organization (2nd Edition). ISBN-13: 978-1803234519.

Additional literature:

1. Saurabh Shrivastava, Neelanjali Srivastav. Solutions Architect's Handbook (2nd Edition). ISBN-13: 978-1801816618.
2. Joseph Ingeno. Software Architect's Handbook. ISBN-13: 978-1788624060.
3. Robert C. Martin. Clean Architecture: A Craftsman's Guide to Software Structure and Design. ISBN-13: 978-0134494166.

Internet resources:

1. Computer Architecture Topics — https://en.wikipedia.org/wiki/Computer_architecture
2. Fire Protection and Prevention (OSHA) — <https://www.osha.gov/sites/default/files/2019-03/fireprotection.pdf>
3. <https://learn.microsoft.com> — official Microsoft documentation on Windows and administration.
4. <https://support.microsoft.com> — technical support and instructions for configuring hardware and software.
5. <https://docs.python.org> — documentation on automation and system administration in Python.

IX. SELF-STUDY TOPICS FOR THE MODULE

Independent (self-study) work is an obligatory part of the module. It is aimed at deepening the theoretical knowledge obtained in lectures, developing practical skills, and forming the ability to work independently with technical documentation and information sources. The following topics are recommended for independent study and for the preparation of reports, presentations and abstracts.

1. Modern requirements for occupational safety and ergonomics at the workplace of an IT specialist.
2. International electrical-safety standards and their application in information-service enterprises.
3. Fundamentals of circuitry: analysis of passive and active components of computer devices.
4. Methods of diagnosing power supplies of computer systems with measuring instruments.
5. Modern peripheral devices: classification, characteristics and areas of application.
6. Data-storage technologies: HDD, SSD, NVMe, external drives and NAS.
7. Software tools for hardware diagnostics: AIDA64, HWiNFO, MemTest86, Victoria, CrystalDiskInfo.
8. BIOS/UEFI: purpose, settings, update and diagnostics.
9. Backup and data-recovery strategies after hardware and software failures.
10. Automation of computer-system diagnostics with Python and PowerShell.
11. Analysis of system event logs to detect and prevent failures.
12. Replacement and maintenance of consumables in printers, copiers, scanners and plotters.
13. Maintenance of uninterruptible power supplies (UPS) and replacement of their batteries.
14. Managing internal and external PC devices through the Windows API (WMI, winreg, psutil).
15. Development of a comprehensive utility for monitoring and diagnosing the state of a computer system.

X. EXAM QUESTIONS

The intermediate certification of the module is conducted in the form of an examination. The exam checks the student's mastery of the learning outcomes of the module — both theoretical knowledge and practical skills in maintaining and configuring computer hardware and software. Below is the list of questions for preparation for the examination.

1. Occupational-safety, electrical-safety and fire-safety requirements at information-service enterprises.
2. Organization of a safe workplace for an IT specialist; risk assessment and risk-reduction measures.
3. Individual electrical-protective equipment: classification, design and rules of use.
4. Fundamentals of circuitry; passive components (resistors, capacitors) and their role in computer devices.
5. Conductors and dielectrics in an electric field; the basics of electromagnetism in electronic circuits.
6. Diagnosing faults of computer power supplies; methods of measuring electrical-circuit parameters.
7. Testing and diagnostics of the CPU, RAM and motherboard at the circuit level.
8. The role of BIOS/UEFI and methods of its diagnostics.
9. Peripheral devices of a PC: purpose, classification and characteristics.
10. Data-storage devices: SSD, NVMe, NAS; input/output interfaces.
11. Operation of computers: power-on, power-off, OS initialization; software management in Windows and Linux.
12. Composition and types of hardware of computer systems; the system unit and its components.
13. Hardware diagnostics with diagnostic programs and utilities; configuring BIOS/UEFI.
14. Diagnostics and repair of the CPU, RAM, motherboard, graphics card and power supply.
15. Preventive maintenance of computer systems and extension of equipment service life.
16. Recovery of equipment after hardware and software failures; backup and data recovery.
17. Protection against malware and viruses; recovery of operating systems after failures.
18. Automation of diagnostics and recovery of computer systems with Python and PowerShell.
19. Replacement of consumables in computing and office equipment; maintenance of UPS units.
20. Maintenance of printers, copiers, scanners and plotters; replacement of cartridges, toner and rollers.
21. Managing internal and external PC devices through the Windows API; psutil and WMI.
22. Monitoring of CPU, RAM and disk load; collecting hardware data into log files.
23. Automation of driver updates and management of system configuration with scripts.
24. Development of an automated device-management and monitoring utility (final mini-project).